

Know Your Users via Image Analytics Before Developing Posts: Data-Driven Optimization Framework to Enhance Social Media Engagement

Mayukh Majumdar,^a Subodha Kumar,^b and Chelliah Sriskandarajah^c

^a Knauss School of Business, University of San Diego, San Diego, California 92110;

^b Fox School of Business, Temple University, Philadelphia, Pennsylvania 19122;

^c Mays Business School, Texas A&M University, College Station, Texas 77843

mmajumdar@sandiego.edu, subodha@temple.edu, chelliah@mays.tamu.edu

Social media platforms are popular for advertisers to promote products to their audience via posts. Developing these posts with the appropriate number of image features is essential as these features enhance informativeness and visual complexity, impacting how users engage with posts. However, the existing literature offers a limited investigation into this topic, primarily examining low-level imagery information while overlooking high-level image features, cross-platform differences arising from different user expectations, and the trade-offs firms must make in tailoring their strategies. To address this critical gap, we develop an optimization framework for analyzing and publishing social media image posts for different platforms within a firm's limited budget. This optimization framework is grounded in empirical modeling of the association between image features and social media user engagement, with primary and secondary features identified using deep learning algorithms. We focus on secondary features because they add visual complexity and informativeness, are controllable by designers, and are costly to extract, underscoring the need to assess their value for engagement. We find a nonlinear association between secondary features and engagement that varies across the two platforms. The optimization framework, which models the nonlinear relationship and is tested on realistic scenarios, considers and compares our approach against commonly used strategies that allocate budgets solely based on the user base of platforms, those that ignore secondary features, and those that do not use duplicate features that can enhance informational richness and context. We present key implications for firms seeking to maximize user engagement on social media platforms.

Key words: data-driven optimization, user engagement, social media, discrete optimization, image analytics, deep learning

1. Introduction

Social media platforms enable firms to promote their products to large audiences. Social media users spend a substantial amount of time online (Statista 2025). The global social media advertising market is projected to increase from \$228 billion in 2024 to approximately \$256.5 billion in 2025, representing a substantial growth in advertising (Johnson 2025). However, this increase in social media advertising represents a significant challenge for firms, which demands that they develop posts that can maximize user engagement. Maximizing user engagement on social media platforms has major implications for firms, such as brand reputation and future purchases (Lang 2025). Thus,

failing to understand what aspects of social media posts drive higher engagement has implications for firms' marketing goals and profitability (Ma et al. 2024).

Therefore, given the rising trend of social media marketing campaigns, producing posts that achieve high user engagement is crucial from an industry standpoint, as understanding the drivers of higher engagement is important for firms (Qian and Jain 2024). Posts with images are considered more effective for advertising on social media platforms, as image content plays a central role in driving user engagement (Cashyap 2024). These image posts play a crucial role in social media advertising by disseminating relevant information to social media users across platforms (He et al. 2024). Within the category of image posts, firm-generated content (FGC) posts are important and often considered better than paid advertisement posts for long-term potential, as reported in Quin (2023). While social media marketers' experience and intuition are invaluable, practical data-driven decisions to maximize engagement on FGC posts are important in the sense that they often overcome the bias associated with marketers' own assumptions (Barqar 2024).

Despite substantial research on the drivers of social media user engagement, there are important gaps in understanding the specific dimensions of images that influence user engagement that warrant further exploration. The multitude of factors in an image that can drive engagement presents exciting research opportunities, aligning with the problems practitioners face (Koshy 2024). It also underscores the importance of resource allocation, turning the focus from merely maximizing outcomes to optimizing results within available resources, such as a limited marketing budget. Moreover, differences across platforms may indicate that users may interact with posts differently, highlighting the need to analyze image features in relation to engagement on different platforms to gain deeper insights. However, the academic literature features a limited investigation of this topic. To bridge this critical gap, we examine how image posts with certain characteristics are associated with engagement across different platforms and generate insights. We propose an optimization framework that utilizes those insights and provides guidelines for efficiently utilizing social media analytics to maximize user engagement.

1.1. Motivation

The rise of social media platforms has changed the role of customers from passive observers of content to active participants or users (Shahbaznezhad et al. 2021). Most users are active on multiple platforms, which motivates firms to deploy a multi-platform strategy to reach a wider audience (Voorveld et al. 2018). On these platforms, users engage with social media posts through various actions, including liking posts, commenting on posts, and sharing posts with friends

(Peng and Van den Bulte 2024). Most social media platforms offer a combination of these different engagement types. Higher engagement, such as a high number of likes on a post, indicates greater resonance with social media users. By monitoring which image posts with what characteristics and on which platform receive higher engagement levels, firms can gain important insights.

Social media FGC posts by firms contain information; this informativeness of FGCs is associated with user engagement (e.g., likes and comments) (Ren et al. 2023). Following the Elaboration Likelihood Model (ELM) in the seminal work of Petty et al. (1986), this informativeness is conveyed to users via the central route, which facilitates diligent consumption of information that motivates these users to become more engaged with the ads. In traditional advertising (e.g. Cox and Cox 2002), it has been shown that the objects (human/nonhuman) or features present in an image deliver information to the users about the product(s) advertised. Studies show that users perceive the objects present in these images (e.g., Oliva 2005, Zhou and Xue 2021) and sometimes grasp the overall gist of a scene, recognizing key elements without necessarily identifying or objectively judging the features (Shin et al. 2020). This emphasizes the importance of considering an image as an important artifact, where the number of features serves as an important measure.

Previous research suggests that visual elements in the central route, such as image features in advertisements, aid in creating narratives and enabling consumers to form mental imagery (Babin and Burns 1997). Thus, as the narrative framing concept proposed by Baumgartner (2002) suggests, posts with varying levels of features may receive different levels of engagement. Images usually have two broad feature classes: (i) primary features that display the product or group of products advertised and (ii) additional secondary features that help create a dynamic environment. These secondary features instill a sense-making process, stimulating user engagement, and increasing brand popularity (Wulf et al. 2019). This categorization of features is common in images, where these secondary features offer support to the primary features (Tanaka 2019). These secondary features also amplify the complexity of an image, which is extremely important in terms of image design (Pieters et al. 2010). Understanding these features is crucial for practitioners, as they may help explain engagement patterns on posts, particularly since many users tend to skip prominent brand elements, such as tags and captions (Social Intelligence Lab 2023). By exploring these features—an approach increasingly discussed among practitioners—brands can gain a richer understanding of audience preferences and uncover subtler drivers of engagement (Brandwatch 2025).

The academic literature has so far offered limited investigation of this important topic. Some insights have emerged with regard to the importance of image characteristics. However, most of

these insights are focused on low-level imagery information or pixel-level measurements, such as color properties and image quality, which are part of the peripheral route or features. Analysis of these low-level image features captures only a narrow aspect of what makes image posts engaging on social media. Additionally, existing research, mostly in traditional advertising, treats image features as undifferentiated, ignoring the differing effects of secondary features, which is important in advertising theory. The theoretical ambiguity surrounding the impact of image features in traditional media, where fewer features (or more features) are associated with higher engagement, raises several other important questions that are empirically unexplored and have implications for social media platforms. In other words, are these secondary features associated nonlinearly with user engagement on social media platforms, recognizing that effects may vary at low vs. high levels of features? Prior research also highlights the importance of “content context” or the differences in engagement based on the nature of a social media platform (e.g., Voorveld et al. 2018, Shahbaznezhad et al. 2021). Yet, it has not been explored whether the effects of image features on engagement are consistent or vary across platforms. Thus, we explore the impact of image features on engagement on different platforms, assuming the differential roles these social media platforms play in a user’s portfolio. This approach addresses the issue where firms can efficiently adapt their image posts’ design and development strategies for different platforms in response to those differences.

We expand the existing body of knowledge in this domain by integrating the three theoretical lenses - ELM framework (Petty et al. 1986), narrative framing (Baumgartner 2002), and visual complexity (Pieters et al. 2010) - to analyze the association between image features, with a particular emphasis on the role of secondary features. Using this set of alternative theoretical lenses allows us to address the limits of a single-framework approach, which may have been insufficient in explaining the varied effects of image features on user engagement. This approach brings into focus secondary features—elements that are central to informativeness and complexity yet remain within the control of post designers—thus enhancing both theoretical understanding and actionable implications. We examine the nonlinear relationship of secondary features and user engagement—a dynamic that, to the best of our knowledge, has not been previously documented in the literature. This approach also addresses the limitations of linear models, which may produce oversimplified insights and lead to suboptimal social media strategies.

Although many prior studies use human annotators to code image attributes, more recent studies have leveraged deep learning algorithms to extract image features (Shin et al. 2020). For instance, Hartmann et al. (2021) study the impact of brand selfies by leveraging deep learning methods. Zhang

et al. (2022) use deep learning algorithms to identify high- and low-quality pictures from Airbnb. We leverage advancements in computer vision algorithms and operationalize image features with the use of two deep learning algorithms, the Single Shot Detector (SSD) and the Faster Region-based Convolutional Neural Network (R-CNN). Primary features are the primary advertising objects, which can be easily extracted using simple algorithms (e.g., the SSD). Besides primary features, Faster R-CNN also extracts secondary features, albeit at a higher effort cost to the firm. Using different algorithms enables us to quantitatively examine the costs of social media analytics for user engagement: understanding the impact of these features on engagement is costly or resource-intensive, requiring significant time, computational power, and analytical effort via social media analytics. Thus, our exploration delves into understanding simplicity—images with minimalistic features—and complexity—images with intricate details and more features—to understand their impact on engagement and associated varying costs.

We utilize the empirical insights in an optimization framework that generates important practical insights and actionable guidance. This framework allows us to incorporate important yet often overlooked dimensions together in social media advertising: firms' budget constraints, social media costs, and engagement intensity levels. While such constraints are difficult to capture in empirical models, which is not surprising as most empirical analyses on image features are constrained to deep learning model comparisons or descriptive studies, they substantially influence engagement outcomes, especially in multi-platform settings, where the relationship between features and engagement varies across platforms due to differences among users.

Our optimization framework models the costs of analyzing and developing posts. Social media analytics enable firms to influence user engagement with FGC by identifying, creating, and publishing posts that incorporate appropriate image features. This gives the manager control over the informational content or complexity in an image by increasing or decreasing the number of secondary features. However, firms also incur high costs for analytics based on business needs and utilization. For example, utilizing advanced methods, such as Faster R-CNN, requires more computing power, data, and time (Li 2021). Meanwhile, even if a firm decides to outsource analytics operations, it may need an in-house team to coordinate with vendors. Unmetric, a well-known analytics firm, offers services to firms wanting to use social media insights. Pricing for their basic analytics options starts at \$1,000 and increases according to the depth of the service (Unmetric 2024). Beyond analytics, developing social media posts also involves a cost to firms.

Notably, despite its significance for firms, developing engaging image posts for different social media platforms has not been rigorously examined, especially given the multitude of factors, such as the features within an image that can influence user engagement. In addition, to have a complete understanding of this problem, it is important to include resource allocation perspectives, such as budget and costs associated with analytics and development, which can influence the use of features, thereby affecting overall engagement. These aspects remain largely unexplored and warrant further investigation. In this study, we answer these questions by examining the engagement-features association across two different platforms, numerically analyzing it under important test cases, and presenting insights to managerially relevant questions. By integrating deep learning for feature extraction, econometric modeling, and scenario testing, our proposed optimization framework systematically links data-driven prediction and prescriptive optimization, yielding deeper insights and more effective decision support than either could provide in isolation.

1.2. Method and Algorithms

We focus on organic or free-to-publish FGC across different platforms. Within FGC, we focus on image posts because posts with images are seen to enhance engagement (Cashyap 2024). Motivated by the differences across platforms and to model the relationship between features, especially secondary features, and their impact on user engagement, we use deep learning and econometric methods on data obtained from two companies owned by Meta, namely, Instagram and Facebook. Although it may seem that engagement should be proportional to image features and should be different across platforms, the exact nature of this relationship—and how it differs between platforms—is far less straightforward. A key finding of our paper is the identification of the nonlinear association between secondary image features and user engagement and the concave and convex nature of this association on Instagram and Facebook, respectively, in our context. Our investigation has parallels with marketing studies in traditional and social media advertising on image complexity as a key attribute of user engagement. Beyond image features, we also consider several important attributes of social media image posts, such as post-interval, textual content, and image color level, that support and contribute to the literature.

We apply the feature-engagement relationships derived from these empirical analyses in our optimization framework. In practice, firms allocate a budget to explore the drivers of user engagement and develop posts accordingly (Otachl 2025). Budget allocation becomes even more crucial when a firm operates on multiple platforms. Budget constraints lead to the following main features of our

framework: (i) we model social media user engagement as a Mixed Integer Nonlinear Programming (MINLP) formulation that incorporates the feature-engagement relationship derived from the empirical analysis in conjunction with costs and engagement levels; (ii) because MINLP does not produce analytically tractable solutions, we transform the models into equivalent solvable linear formulations for both single and multiple platforms; (iii) we develop polynomially solvable and easy-to-implement algorithms based on the structural insights from our models for practical use.

We consider several other attributes of social media platforms, such as weights for platforms' user capacity, engagement types, and days of posts, indicating users' engagement intensity. Budget limitations make selecting or prioritizing platforms essential for achieving the maximum possible engagement (Gain 2024). It is essential to prioritize the days to post depending on when users are more active. As a possible application of our multiple platform framework, we extend our analysis to a four-platform scenario (in the e-companion) and discuss a budget allocation mechanism based on i) a firm's conversion rate on platforms, ii) each platform's potential in terms of user reach, and iii) appropriate image features. Overall, our study contributes to the literature on social media user engagement and image analysis, which does not systematically study the impact of image features or often ignores what delivers the most value given resource limitations, bridging empirical insights with analytical modeling and showing that maximizing engagement is a multi-factor, trade-off-driven problem that varies by platforms.

1.3. Results and Insights Relevant to Managers

Our study presents important structural results for single and multiple platforms, which lead to practically relevant and efficient algorithms. These efficient algorithms provide reasonable solutions to general problems. We also analyze several interesting questions using the structural insights and algorithms and generate insights based on computational experiments, producing the following consequent insights, with the first two insights based on our empirical results:

- ***Q1: How do post-interval, textual content, and color affect engagement?*** Our empirical analyses reveal how post-interval, textual content, and image color level are associated with engagement. First, infrequent posting reduces engagement across platforms. Second, firms should carefully use the descriptions on their social media posts. For example, Instagram posts show a higher level of engagement on posts with more textual content. Conversely, on Facebook, more text correlates with lower engagement. Finally, high image saturation lowers engagement

- ***Q2: How do engagement patterns on image posts with specific features vary across social media platforms?*** Our analysis reveals distinct engagement patterns between Instagram likes and Facebook shares on specific features: Instagram likes are negatively associated with the presence of people in images, while Facebook shares exhibit a negative association with images featuring plants. These findings, along with our primary empirical result, support and extend the work of Li and Xie (2020), where the authors explore the impact of image characteristics, such as color composition, on user engagement across different platforms.
- ***Q3: Is the user base of a social media platform sufficient for maximum engagement?*** We further analyze a user-base-focused setting, since firms typically tailor strategies solely based on the number of users (Cucu 2022). We show that focusing completely on a platform's user base when allocating a budget for social media analytics and activities associated with developing posts may produce sub-optimal results under certain settings. Under varying budgets, engagement intensity, user base, and analytics costs, our approach delivers up to 14% higher overall engagement compared to an approach that focuses solely on the user base. This means that it is important for firms to make informed decisions on budget allocation.
- ***Q4: Why are secondary features important for social media engagement?*** The features that an image contains in social media posts crucially contribute to its success (Overgoor et al. 2022). The narrative framing theory discusses the role of features as a broader sense-making process. Based on that theory, we explore the relationship between secondary features and user engagement in our optimization framework, which also varies across platforms. We find the addition of secondary features boosts engagement by 5% and 16% on Instagram and Facebook, respectively, compared to posts without such features. This increase is substantial from an industry point of view.
- ***Q5: Is removing duplicate features enhance overall engagement?*** Social media advertising image posts often utilize duplicate or multiple features of the same type to convey important advertising information. By repeating key objects, such as the features (e.g., multiple dresses with different colors) within the same image, they aim to improve brand recognition and enhance the likelihood of capturing the audience's attention (Mialki 2023). Overall, using duplicated features in social media posts is reasonable in the digital landscape. As an important extension, we study whether user engagement is affected when removing duplicates. We find that for a majority of the scenarios, posts with duplicate features exhibit a higher level of engagement compared to the case where we removed the duplicates. Also, eliminating duplicate

features may not always reduce the costs of social media analytics, as removing them does not meaningfully reduce computational or analytic effort.

2. Literature Review

Social media image posts and user engagement analysis are areas of substantial interest in the operations management (OM) literature. Our work draws primarily from two major research streams: (i) investigation of the impact of social media image posts on user engagement and (ii) design and development of social media posts and platform management. We discuss the contributions of our study in this section and demonstrate the positioning of our work in Figure EC.1 and Table EC.1.

2.1. Investigation of the Impact of Social Media Image Posts on User Engagement

This stream of research concerns the value of images in advertising. To fully understand it, we explore the traditional advertising literature where studies have consistently shown the positive impact of images (e.g., Edell and Staelin 1983, Wu et al. 2017), indicating its preference over textual content in conveying product information more effectively to consumers. Despite the literature on image content in traditional literature, researchers have mostly focused on analyzing textual content in social media posts (e.g., Lee et al. 2018, Chung et al. 2020). This exclusive focus on textual data, which is more commonly related to the functional attributes of products (Liu et al. 2020), may lead to the potential neglect of the informativeness of images. Nevertheless, studies on textual content greatly contribute to understanding social media posts and their impact on user engagement. The analysis of images has emerged as a recent trend in the literature (e.g., Jaakonmäki et al. 2017, Ma et al. 2018, Wulf et al. 2019, Li and Xie 2020, Zhang et al. 2022). However, although it seems intuitive that images impact engagement (Li and Xie 2020), more analysis is required to understand what image characteristics do it.

In their seminal work, Petty et al. (1986) discuss that images influence consumers in two ways: (i) the central route, which influences a consumer's information processing and is more robust and (ii) the peripheral route, via which consumers receive positive or negative cues and is more temporary. The authors argue that the central (peripheral) route facilitates high (low) involvement under which consumers exert more (less) cognitive effort to evaluate an advertisement. Studies have extensively shown the impact of peripheral features on user engagement in social media, such as colorfulness (Li and Xie 2020), quality of images (Zhang et al. 2022), pixel-level complexity and celebrity endorsements (Shin et al. 2020), and brand logos (Rietveld et al. 2020). How user engagement on social media platforms is impacted via the central route, a traditional but much-debated topic in practice (Argyris et al. 2020), is still under-explored in social media advertising.

In the traditional advertising literature, we find the discussion on central routes in which image features constitute an important aspect (Pieters et al. 2010). Using an experiment, Cox and Cox (2002) find that consumers prefer a moderate number of features in advertising images of fashion products. On the other hand, Bialkova et al. (2013) report that consumers' attention decreases as more features are presented on front-of-pack labels on packages. Deng and Poole (2010) find that consumers prefer images with more features. Zhang et al. (2021) use conjoint analysis to understand users' preference for souvenirs from images. In line with the importance of image features and mixed findings in traditional advertising literature, we expect that these image features may also be associated with user engagement on social media platforms. The availability of advanced deep learning algorithms has opened up opportunities to estimate more granular measures, such as image features. Shin et al. (2020) and Overgoor et al. (2022) study the central route by analyzing image complexity (pixel variations, objects). This is essential from the practitioners' standpoints as features or objects constitute the composition of an image, which is a crucial aspect in advertising (Indeed 2025). Using an eye-tracking experiment, Zhou and Xue (2021) show that social media users pay attention to advertisements where different objects or products appear in images, in line with the insights from Loftus and Mackworth (1978). Our study has parallels with the existing literature on considering central and peripheral features as key image attributes that affect user engagement, although there are substantial differences.

There are three major gaps in the literature, which are also important for practice. First, the existing literature is mostly constrained to either developing deep learning models and making comparisons or descriptive studies to identify the antecedents of user engagement, overlooking the value of developing image posts with features for maximum user engagement. As outlined in the literature, these features directly or indirectly via the conceptual gist (Oliva 2005) affect users. There are conflicting findings on whether simplicity with fewer features (Aitchison 2012) or complexity with more intricate details (Putrevu et al. 2004) better influences user engagement. This opens up the need for empirical investigation of the differing impact of features in different settings, such as different engagement types and platforms. These considerations are critical from the industry point of view, where practitioners face challenges developing posts for different platforms with distinct user experiences (Markerly 2022). Second, most studies in the literature focus primarily on UGC (e.g., Li and Xie 2020, Zhang et al. 2022). There is a substantial gap in analyzing FGC in the social media context. Examining FGC is important, as it sheds more light on consumer preferences, which can result in potential future sales (Ren et al. 2023). Third, the evolving stream of literature on the

impact of image characteristics on user engagement focuses on the analytics aspect by examining the mere presence of images with (central and peripheral) features of an image by using deep learning algorithms. Research on the use and effectiveness of analytics is missing in the literature, while there is an increase in the ongoing discourse surrounding the strategic deployment of such methodologies in practice (Boesch 2024).

We address these important gaps in the literature through an optimization framework aimed at maximizing user engagement. First, we expand the current knowledge in the literature on social media user engagement of image posts. We combine three diverse theoretical perspectives — ELM framework (Petty et al. 1986), visual complexity and user attention (Pieters et al. 2010), and narrative framing (Baumgartner 2002)— to better understand how these secondary image features impact user engagement on social media platforms. We show then that there are nonlinear relationships between secondary features and user engagement and that this nonlinear relationship also differs across platforms. Thus, simplicity and intricacy must also be carefully considered when designing posts to achieve higher engagement on different social media platforms. Moreover, our empirical analysis also reveals interesting insights on how post interval, saturation of an image, wordiness in post captions, and specific features, such as people and plants, that are pretty standard in furniture posts, are associated with user engagement and differ across platforms.

Second, our framework provides a novel mechanism to develop social media image posts (FGC) by tailoring image features for different platforms to optimize engagement. The analysis and development of FGC is crucial for understanding user preferences and gaining a competitive edge (Tyrväinen et al. 2023). Thus, our study can help social media managers develop more engaging FGC posts for their users on multiple platforms. Additionally, our optimization framework is also methodologically novel: it allows for more rigorous scenario testing while answering managerially relevant questions. Moreover, this optimization framework with budget allocation for FGCs is also relevant as the existing literature focuses mostly on targeted ads (e.g., Luzon et al. 2022).

Third, we demonstrate how to utilize analytics to develop engaging posts on social media platforms. By juxtaposing two canonical deep learning algorithms with different efficiency levels, we aim to elucidate the trade-offs between resource intensity and cost-effectiveness and the resulting business value of analytics—elements that are broadly representative of real-world social media optimization problems. As social media platforms, user preferences on these platforms, and post-related costs—including analytical efforts—continue to evolve, our engagement-maximizing framework and the insights will continue to remain valuable.

2.2. Design and Development of Social Media Posts and Platform Management

Our study shares some similarities with the research into resource allocation for design and development within the OM literature (e.g., Klingebiel and Rammer 2014, Shi et al. 2025). Specifically, our study partly relates to the emerging field of new product development (NPD) under budget constraints, with the social media analytics process for designing posts considered as a design phase. Meanwhile, the manufacturing phase corresponds to the development of posts. Existing NPD studies explore domains such as funding (Santiago and Vakili 2005), resource allocation (Hutchison-Krupat and Kavadias 2015), development of an advertisement pipeline (Ding and Eliashberg 2002), and most importantly, the NPD phases of design and manufacturing (Bajaj et al. 2004). Despite substantial interest in social media's role in the NPD process (Giannakis et al. 2022), there is little research concerning social media within the purview of NPD, despite its importance in practice (Newberry 2024). Some recent studies explore design or manufacturing phases in social media advertising. For example, Mallipeddi et al. (2021) analyze social media posts on Twitter and investigate the relationship between posts generated by users and their user engagement. Kumar et al. (2020) develop an optimization framework to determine advertisement timing and length to influence user engagement. Mallipeddi et al. (2022) study the selection of social media influencers and scheduling of their posts for a planning horizon.

There are two major gaps in the literature discussed above. First, none of the studies combined the design and development phases, where user insights are analyzed in the context of developing engaging posts. This is important for firms as engaging posts often result in potential future sales (Lang 2025). Second, the consideration of multiple social media platforms is missing in the literature, given its importance when firms operate under a limited budget (Sprout Social 2021). In our study, we address these two important gaps in the literature. First, we consider both the design (of image posts via feature analysis using analytics) and the development (developing engaging posts) phases together in our study. Given the importance of social media analytics to developing compelling posts (Barkakoty 2025), our study examines this common social media optimization problem, which has practical implications. Second, our framework includes the more common setting in practice where firms operate and prioritize different social media platforms under a limited budget (Influencity 2025). We develop different algorithms for different platforms and propose a budget allocation for multiple platforms.

3. Problem Setting

We examine a setting where a firm wants to maximize user engagement on its social media platform(s) by using analytics to develop engaging posts. This involves closely following industry examples and proposing an optimization framework for analyzing and publishing social media posts across multiple platforms. Our framework incorporates the trade-off between the competing costs associated with social media analytics and the development of posts under a limited budget. We describe the social media marketing ecosystem where firms develop and publish posts featuring their products on platforms and users engage with those posts.

3.1. Social Media Marketing Ecosystem

Firms develop and publish image posts with features on one or multiple social media platforms. Upon seeing a post on their social media account, an Instagram user may engage by liking, commenting, or both, and a Facebook user may share, comment, or both. User engagement—or the extent to which a user interacts with a service and frequently wants to use that service—represents a vital outcome metric for the firm. To analyze user engagement, we analyze three engagement types: liking and commenting on Instagram and sharing and commenting on Facebook. Users react to posts that firms develop and publish, making the following activities paramount to the development of a social media marketing strategy: (i) measuring and quantifying social media user engagement, (ii) establishing the relationship between the image features of posts and user engagement on social media platforms, and (iii) considering the costs associated with a marketing strategy. We discuss the variables and feature-engagement relationship in Sections 3.2 and EC.1.4. in detail.

3.1.1. Social Media User Engagement. To capture user engagement on a given post, we let x_{lmk} be the total engagement of type m on post k on platform l . For example, the engagement type like (m) in post (k) published on Instagram (l) by IKEA (see Figure 1) is $x_{lmk} = 6,701$. The firm derives some benefits by maximizing user engagement: $\sum_{l=1}^L \sum_{m=1}^M \sum_{k=1}^K x_{lmk}$. Firms usually publish one post each day, as revealed by our data and reported by industry-based examples (Preloznik 2024). Thus, in our study, each k also represents one day, meaning that the planning horizon comprises K days. As the engagement level on a post in the first 24 hours is usually considered very critical (Minter 2021), we analyze engagement levels on 100 posts from each Facebook and Instagram. We observe that, on average, most engagement occurs within 24 hours from publication, which is consistent with the findings in the literature (Vassio et al. 2022). Figures EC.2 and EC.3 in the e-companion present these results. Thus, we assume one social media post for each platform on each day. Table 1 presents the key parameters and variables.

Figure 1 A Post from IKEA USA Official Instagram Page



Table 1 Key Parameters and Variables

Parameters	
a	Unit cost of adding one feature to a post k , $k = 1, 2, \dots, K$
B	Total allocated budget for a planning horizon
u_{1l}, u_{2l}	Upper bounds for primary features and secondary features extracted for each platform l , $l = 1, 2, \dots, L$
b_{lm}	Impact factor of post measures on user engagement of k posts, obtained from empirical analysis for each platform l , $l = 1, 2, \dots, L$ and engagement type m , $m = 1, 2, \dots, M$
β_{lm}	Impact factor of primary features on user engagement of k posts, derived from empirical analysis for each platform l , $l = 1, 2, \dots, L$ and engagement type m , $m = 1, 2, \dots, M$
ζ_{lm}	Linear impact factor of secondary features on user engagement of k posts, estimated by empirical analysis for each platform l , $l = 1, 2, \dots, L$ and engagement type m , $m = 1, 2, \dots, M$
η_{lm}	Quadratic impact factor of secondary features on user engagement of k posts, obtained from empirical analysis for each platform l , $l = 1, 2, \dots, L$ and engagement type m , $m = 1, 2, \dots, M$
z_{lm}	Weight factor for each platform l , $l = 1, 2, \dots, L$ and engagement type m , $m = 1, 2, \dots, M$
w_k	Weight of post k on user engagement, where $\sum_{k=1}^K w_k = 1$
C_f	Fixed cost incurred for social media analytics activities for a planning horizon
e	Unit cost of the effort of extracting/analyzing secondary features using the advanced deep learning method Faster R-CNN
q_{lk}	Minimum number of primary features required by posts k ($q_k \leq s_k$), where k , $k = 1, 2, \dots, K$ and l , $l = 1, 2, \dots, L$
Variables	
U_l	Maximum number of secondary features to be included during the planning horizon for each platform l , $l = 1, 2, \dots, L$
x_{lmk}	User engagement for post k , where l , $l = 1, 2, \dots, L$, engagement type m , $m = 1, 2, \dots, M$, and k , $k = 1, 2, \dots, K$
s_{lk}	Number of primary features to be included in the image of post k , where l , $l = 1, 2, \dots, L$, engagement type m , $m = 1, 2, \dots, M$, and k , $k = 1, 2, \dots, K$
f_{lk}	Number of secondary features to be included in the image of post k , where l , $l = 1, 2, \dots, L$, engagement type m , $m = 1, 2, \dots, M$, and k , $k = 1, 2, \dots, K$

Social media platforms have important differences, which motivate us to develop an optimization framework that can be adjusted and used for different platforms. Here, we describe some key differences between Instagram and Facebook. Socialbee (2025) comments that Instagram has more users below the age of 24 years and fewer users over 34 than Facebook. According to BackLinco (2025), both Instagram and Facebook have more female users than male users. These differences in demographic variables in social media platforms' users have appeared in the literature (Sashittal et al. 2016). Besides these key demographic differences, studies also show differences across usage patterns of users across these platforms. For instance, Alhabash and Ma (2017) show that users show greater time engagement on Facebook than on Instagram. Pearce et al. (2020)

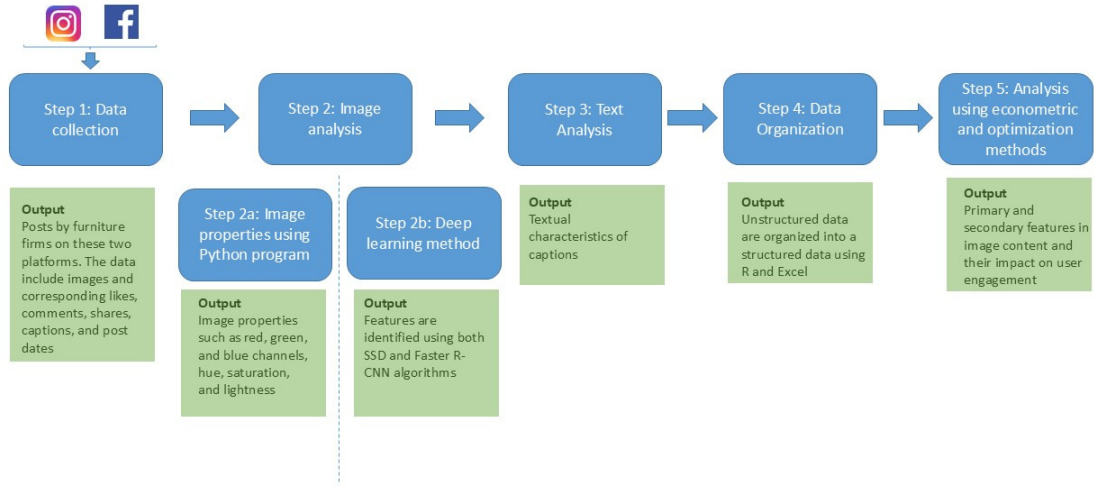
find that Instagram users prefer more aesthetically pleasing content than Facebook users. Due to these differences across platforms, we conjecture that users may evaluate content differently when deciding which post to engage with on each platform.

Together, these findings in the literature highlight the importance of developing tailored social media advertising strategies for different platforms to obtain maximum user engagement, as highlighted by Alhabash and Ma (2017). However, firms are also faced with the challenge of developing posts for different platforms under a limited budget. Thus, they need guidance to make their social media strategies by analyzing the distinctive features of social media posts. We develop our framework for individual platforms and extend it to multi-platform settings, following the literature and industry practices to offer insights. We assign weights of z_{lm} to each engagement type m for each platform l , indicating its importance.

3.1.2. Features of Posts and User Engagement on Social Media Platforms. According to practitioners, firms must tailor their social media marketing strategy to create and customize posts that attract and engage users by competing effectively with the numerous other posts published every minute (Abimbola 2025). Recall that a firm may emphasize the primary features in an image and introduce secondary features that provide users with an impression of the broader advertising narrative. Because primary features convey the main information or represent the primary product(s), it is difficult for firms to modify them. However, they can change the number of secondary features to achieve the desired level of engagement. Our optimization model defines primary and secondary features as s_{lk} and f_{lk} for each post k on each platform l . Therefore, user engagement x_{lmk} can be expressed as $x_{lmk} = f(s_{lk}, f_{lk}, \tau)$, where the vector τ represents additional parameters derived from the empirical analysis. We use the empirical analysis to estimate the engagement-feature association $x_{lmk} = f(s_{lk}, f_{lk}, \tau)$.

3.1.3. Costs Associated with Social Media Marketing. Wu et al. (2020) find that firms allocate resources to perform in-house market research and create advertisements for social media platforms. This entails two major costs: social media analytics and the development of posts. We denote U_l as the social media analytics effort for every platform l . Social media analytics involves the analysis and insights gained from structured and unstructured data (Sprout Social 2021). Gaining more insights from these data requires greater analytics efforts and incurs higher costs. We include this cost in our optimization framework and analyze the impact of changing the cost on user engagement.

Developing posts for social media includes the costs of creating images, which are associated with photographers, human actors, objects, renting space, and several miscellaneous items. We

Figure 2 Social Media Analytics Process

capture a substantial part of this cost by assuming a unit development cost per feature for each s_{lk} and f_{lk} . For example, Otachl (2025) mentions that organizations' spending can cross \$50,000 per project on social media activities, with the cost varying according to the frequency of posting, the number of platforms, and the graphics. Meanwhile, a study by UP Reports (2025) finds that each graphic or object in a social media post can cost around \$20-50. Our model accounts for these costs by including the number of platforms and posts. We also study user engagement levels across various values for a and consider a fixed cost C_f that represents a technical team performing social media analytics, regardless of the number of posts developed. We present the budget constraint as $a \sum_{l=1}^L \sum_{k=1}^K (s_{lk} + f_{lk}) + e \sum_{l=1}^L U_l + LC_f \leq B$, where B denotes the total budget for social media marketing and e represents the per unit effort cost of including features in a post.

3.2. Data Collection and Empirical Analysis

To study the impact of image features on user engagement, we conduct an empirical analysis using data from Instagram and Facebook, the two most popular social media platforms for advertising. To establish the relationship between engagement and features, we follow two steps: (i) extracting primary and secondary features by running deep learning algorithms (SSD and Faster R-CNN) on social media posts' images and (ii) estimating the association between user engagement and these extracted image features (both primary features and secondary features) using econometric methods. We illustrate our social media analytics process in developing the framework in Figure 2 and discuss it in the e-companion (Section EC.1).

Our empirical analysis is motivated by the process model developed by O'Brien and Toms (2008) and the narrative framing theory of Baumgartner (2002). User engagement occurs via a process in which users show their engagement with a system—a social media post in our research context. Users engage with a post by liking, commenting, or sharing it with others. The information captured in an image helps users to form an idea of the product(s) advertised on social media platforms. An image comprises not only primary features, but also secondary features that offer a holistic visual experience. Deep learning methods have facilitated opportunities to extract image features from social media posts at a substantially low cost and without the human biases associated with data-intensive projects (Wu et al. 2015). This has enabled the analysis of engagement levels on image posts with varying features. In line with the marketing literature on user reactions (Yin et al. 2017), we argue that user engagement on image posts may vary with secondary features non-linearly.

3.2.1. Deep Learning Methods. Social media analytics utilizes structured and unstructured data and helps firms identify the antecedents to user engagement (IBM 2021). As a part of social media analytics, we apply deep learning algorithms to the images collected from social media data. These algorithms facilitate a process of gathering information from images and discerning how much of that information contributes to user engagement. Consistent with practice, we use pre-trained robust deep learning algorithms on our social media data to extract features from images (Shin et al. 2020). The algorithms offer reasonably high accuracy in our context because the objects or features are well-defined and easily identifiable. Additionally, we manually analyze the extracted features after applying the pre-trained algorithms; the outcomes are similar to those delivered by the deep learning algorithms. Section EC.1.2 (in the e-companion) provides further insight.

We define the number of features extracted by SSD as primary features. Secondary features are the additional features extracted by the Faster R-CNN besides extracting the primary features. Thus, primary and secondary features together represent the total number of features in an image extracted by Faster R-CNN. For instance, a post may have 1 couch (primary feature), 2 vases, and 1 chair (secondary features). As a simple algorithm with one layer, SSD is designed to extract an image's most prominent feature(s). This aligns with the content marketing strategies of firms focusing on clarity or the absence of ambiguity in an advertisement where the primary objects are prominently highlighted (Erdem and Swait 1998).

3.2.2. Empirical Findings. To formulate and solve our optimization model to maximize user engagement (next section), we empirically analyze the association between a post's user engagement and its corresponding image features based on the data we collected. Further details of the

empirical results appear in the e-companion. Table EC.3 presents the variables used in the empirical analysis of Instagram and Facebook data. The summary statistics for the variables used for the Instagram and Facebook analyses are in Tables EC.4 and EC.5. Since our dependent variables are count variables and show high dispersion, we utilize the Negative Binomial regression approach (Cameron and Trivedi 2013). Since a majority of our engagement variables display a substantial number of zeroes, we employ a more flexible model - Zero-Inflated Negative Binomial or ZINB (Lambert 1992), which converges to Zero-Inflated Poisson (ZIP) if the dispersion parameter is close to zero (Feng 2021). For dependent variables that do not display any zeroes (e.g., Instagram likes), we deploy the Hurdle model (Mullahy 1986), which can also manage zeroes. We detail the model specifications and discuss the results in the e-companion. The findings for Instagram and Facebook are presented in Tables EC.6 and EC.7, respectively. We perform several robustness tests, including the standard Negative Binomial, in Section EC.1.4.5; they all validate our findings.

Our empirical analysis indicates that user engagement on an image post changes as the number of secondary features changes. Building on the theoretical support, we introduce a quadratic term for secondary features in the regression model. As postulated, the number of secondary features demonstrates a nonlinear relationship with likes on Instagram and shares and comments on Facebook, with different patterns observed in each case. We do not find any statistically significant relationship between secondary features and comments on Instagram, which we explain more in Section EC.1.4.6. Following the approach suggested by Hayes (2017), we estimate user engagement values (e.g., the number of likes on Instagram) from their respective regression equations by averaging all other variables and multiplying them with their corresponding coefficients as shown in Table 2. We exclude comments on Instagram in our optimization framework as our main independent variables (linear and quadratic terms for secondary features) were statistically insignificant. We present the following empirical insights.

Empirical Insight 1: *The relationship between user engagement on a post and the number of secondary features in the post is nonlinear and may differ between engagement types and platforms. In particular, the empirical analysis reveals that the relationships between the number of secondary features and likes on Instagram and comments and shares on Facebook are quadratic.*

Empirical Insight 2: *Posts having too many features do not generate high engagement levels on Instagram. By contrast, posts with many features on Facebook receive high engagement levels. This association follows a quadratic pattern, with the point at which engagement switches varying between Facebook and Instagram and the corresponding engagement types.*

Our results also reveal several important insights related to social media user engagement. First, we observe that post interval—the time between two consecutive image posts—is negatively associated with engagement. This finding highlights the importance of frequent posting on social media platforms. Second, posts on different platforms exhibit different patterns regarding text-based captions. On Instagram, posts with more textual information in their captions are associated with a higher number of likes and comments. By contrast, Facebook posts with less text in their captions are associated with a higher number of shares and comments. Finally, we observe that posts that have images with high levels of saturation or color strength experience lower levels of engagement. These three findings answer our Q1 (raised in Section 1.3). We also find insights on the features that answer Q2 in Section 1.3.

Our empirical analysis suggests the following functional form for the relationship between user engagement x_{lmk} and the number of primary features s_{lmk} and the number of secondary features f_{lmk} in post k . Thus, engagement type m with post k on platform l can be expressed as $x_{lmk} = b_{lm} + \beta_{lm}s_{lmk} + \zeta_{lm}f_{lmk} + \eta_{lm}f_{lmk}^2$, where b_{lm} is a constant, β_{lm} is the coefficient for primary features, and ζ_{lm} and η_{lm} are coefficients for secondary features (linear and quadratic, respectively). These coefficients are estimated via empirical analysis for a particular planning horizon and provided in Table 2. This functional relationship is used in the optimization model in the next section.

Table 2 Parameters Obtained from Empirical Analysis

Coefficients	Instagram ($l = 1$)	Facebook ($l = 2$)	
	Likes ($m = 1$)	Shares ($m = 2$)	Comments ($m = 3$)
b_{lm}	7.4952	2.7067	3.3796
β_{lm}	-0.0348	-0.0500	0.0055
ζ_{lm}	0.0945	-0.1012	-0.1224
η_{lm}	-0.0063	0.0098	0.0106
u_{1l}	10	19	19
u_{2l}	14	18	18

4. Problem Formulation

This section develops an optimization model using the insights derived in the previous section. Specifically, we use the functional relationship, $x_{lmk} = b_{lm} + \beta_{lm}s_{lmk} + \zeta_{lm}f_{lmk} + \eta_{lm}f_{lmk}^2$, which describes user engagement in our model. In Section 4.1, we develop a plan for a single platform with a single engagement type. In Section 4.2, we present the structural insights for different platforms. We present the generalized case for multiple platforms and engagement types in Section 4.3.

4.1. Single Platform Single Engagement Type

In this subsection, we present the single platform and single engagement type case of the generalized problem. The firm's objective is to maximize user engagement during a planning horizon comprising K periods, where $k = 1, 2, \dots, K$ and each period represents a day. Similar to the approach adopted in Mallipeddi et al. (2022), we consider a planning horizon of $K = 14$ days for ease of exposition. The firm operates under a limited budget that covers the costs associated with developing posts and social media analytics. We assume that the firm publishes one post per period. This enables us to consider K posts (usually one per day) to be developed and published during the planning horizon. Social media analytics operations see the firm gather data on posts and corresponding information, perform feature extraction from posts, and estimate the association between features and engagement. Next, the firm decides on social media planning using these insights according to its budget. Because primary features for a product have to be advertised, the firm needs to decide how many secondary features should be included in a post.

For a single platform, we represent the features of an image by defining two decision variables: s_k (respectively, f_k) denotes the number of primary (respectively, secondary) features to be included in the post k . Our empirical analysis reveals that the number of features is finite; s_k and f_k have the upper bounds u_1 and u_2 , which are estimated from our data and specific to each platform. Among primary features, we denote q_k ($q_k \leq s_k$) as the mandatory primary features particular to a product(s) to be displayed in the image of a post. We maintain this relationship throughout our analysis. Finally, we also mention here that for multiple platforms l and engagement types m , x_{lmk} denotes user engagement (e.g., for Instagram, the number of likes) for post k . We allocate a budget of B to cover the cost of (i) developing posts and (ii) utilizing social media analytics. Recall that the firm performs social media analytics on posts published during the previous planning horizon. Our objective is to help firms develop a social media marketing strategy by deciding the number of image features on posts during the planning horizon. In this way, social media analytics contributes to developing posts and corresponding user engagement for the current planning horizon.

In summary, firms use the insights from social media to deliver more engaging posts to their users. Accordingly, we update the parameters in Table EC.35 (in the e-companion) for each planning horizon and run our optimization model with updated parameters for each period with the goal of periodically analyzing social media data across platforms, deriving empirical insights on the relationship between user engagement types and image features in the posts, and designing image posts for the next planning horizon. It follows that no two posts are the same during a particular

planning horizon for the same platform, which is consistent with practitioner suggestions to avoid identical and repetitive posts (Abimbola 2025). Next, we discuss three important attributes of our model: (i) the intensity of engagement across a week, (ii) the cost of social media analytics, and (iii) the cost of creating image posts.

4.1.1. Intensity of Engagement. The intensity of engagement represents how users interact or engage on social media platforms on different days. For example, an article on Sprout Social (2025b) states that weekdays generate a relatively higher engagement level (or intensity) than posts on weekends. Although our data do not reveal any statistically significant differences in user engagement between days, it is not surprising that users access social media platforms at times convenient to them. To generalize, we introduce intensity of engagement via weight w_k , for user engagement for post k , which depends on the day the post is scheduled. We set $\sum_{k=1}^7 w_k = 1$ for one week during our planning horizon, where w_k assumes lower values during the weekend. Note that weights w_k may differ according to the industry. For instance, Kumar et al. (2019) find that users do not engage substantially with social media shopping sites during weekends. We assume that user engagement is lower on weekends, and thus, we use different weights w_k for different days. Our model can be easily extended with different weights for different days.

4.1.2. Social Media Analytics Cost. Firms can perform social media analytics in-house or outsource it to a third-party vendor. As discussed, even if a firm outsources the analytics operations, it still needs a team to coordinate with the vendor(s) regularly. Wu et al. (2020) comment that many firms manage social media analytics in-house for better control and ease of decision-making. The in-house team monitors social media activities and helps the content designers develop effective posts for the next planning horizon. The team (i) collects data from multiple platforms on multiple firms that include images, text information, and user reactions, (ii) extracts features from the images, and (iii) analyzes their impact on user engagement.

We refer to the cost of maintaining a team, C_f , as a fixed cost amortized for the planning horizon. There is a variable cost eU per planning horizon, where U is the upper bound on f_k (or U_l for f_{lk}), quantifying the team's effort (including computational power) required to perform social media analytics and e represents the per unit analytics cost for extracting and analyzing the number of secondary features. The secondary features variable f_k is bounded by U for a planning horizon. Thus, the costs of feature extraction and analysis comprise a fixed cost C_f and a variable cost eU per planning horizon. Estimating e is challenging, with a firm needing resources for data collection,

data structuring, and running computational algorithms on the data. These operations are costly and time-consuming, demanding that the firm devote more effort U for advanced analytics for understanding user engagement, which consumes the budget for developing posts, $C_c = eU$. We perform computational experiments on a testbed of various parametric values of e to observe user engagement outcomes at different costs. Without loss of generality, our framework applies to both in-house and outsourced analytics operations.

4.1.3. Image Post Development Cost. Similar to analytics operations, the firm can also have an in-house design team(s) with the required expertise to develop posts or can subcontract this activity to another firm. The team may include designers and photographers requiring office space. As such, each additional feature in an image requires effort and resources beyond leasing or purchasing the objects associated with the features. We denote a linear and identical cost per feature to be included in a post as a . Thus, for a planning horizon, the total costs associated with creating posts can be represented as $a \sum_{k=1}^K (s_k + f_k)$, with each post k having $(s_k + f_k)$ features. The same approach extends to the generalized model, which assumes each image of a post to have $(s_{lk} + f_{lk})$ features where l represents each platform (we present it later).

We present our model for social media management for a single platform or SMM_s with one engagement type. The objective of SMM_s is to maximize the weighted user engagement Π_s , where the weight w_k is introduced to capture the intensity of user engagement on day k . As indicated, we omit the indices l and m in SMM_s .

Problem SMM_s :

$$\mathbf{Max} \ \Pi_s = \sum_{k=1}^K w_k x_k$$

Subject to:

$$a \sum_{k=1}^K (s_k + f_k) + eU + C_f \leq B \quad (1)$$

$$x_k = b + \beta s_k + \zeta f_k + \eta f_k^2, \ \forall k \quad (2)$$

$$s_k \leq u_1, \ \forall k \quad (3)$$

$$f_k \leq U, \ \forall k \quad (4)$$

$$U \leq u_2, \ \forall k \quad (5)$$

$$s_k \geq q_k, \ \forall k \quad (6)$$

$$x_k, s_k, f_k, U: \text{ Integer variable } (\geq 0), \ \forall k \quad (7)$$

In SMM_s , through constraint (1), we impose the budget constraint with three cost components. The first component ($a \sum_{k=1}^K (s_k + f_k)$) is the cost of developing K image posts, where each post k includes $(s_k + f_k)$ features. The second component (eU) refers to the social media analytics cost, which is a variable cost. The third component (C_f) represents the fixed cost of social media analytics operations. Constraint set (2), which is obtained by our empirical analysis, illustrates the functional relationship between user engagement and the image features to be included in the post. Note that the parameters b , β , ζ , and η are estimated from our empirical analysis (see Table EC.35 in the e-companion for these values for different platforms). However, this constraint introduces non-linearity via a quadratic term. Constraint sets (3) and (4) restrict the total number of simple and secondary features below their upper bound values, with U representing the decision variable. We obtain the upper bound values from our empirical data. In constraint set (5), the decision variable U is bounded by u_2 , which is obtained from the data used in the empirical analysis. Constraint set (6) provides lower bounds set by the firm regarding the number of primary features for post k . Note that q_k represents the mandatory primary feature(s) the firm wants to advertise in period k , such as a bed or chairs for a furniture firm. Because this is easier to extract, it pertains to the primary features category. In constraint set (7), we define the non-negativity constraints.

4.2. Solving Single Platform Problem SMM_s for a Platform

We now present structural insights for two platforms, Instagram and Facebook. We use the subscripts l and m for platforms and engagement types, respectively. We illustrate those attributes in the following subsections and include the proofs in this manuscript's e-companion.

4.2.1. Solving the Single Platform Problem SMM_s for Instagram. Instagram's user engagement attributes are given under Case 1, where $\beta_{lm} < 0$, $\zeta_{lm} > 0$, $\eta_{lm} < 0$. We now provide the solution approach by linearizing Problem SMM_s and utilizing a polynomial-time algorithm to solve it. Any platform pertaining to this case can use this approach.

Case 1: $\beta_{lm} < 0$, $\zeta_{lm} > 0$, and $\eta_{lm} < 0$. An example of such a platform in our case study is Instagram, as demonstrated in Table 2, where $l = 1$ and $m = 1$. The following results for the platform satisfy the condition in Case 1.

We let $q_0 = \sum_{k=1}^K q_k$ and $f_0 = \sum_{k=1}^K f_k$. Because $\beta_{lm} < 0$, in Problem SMM_s , for maximum user engagement x_k , we must have $s_k = q_k$, $\forall k$. Additionally, in this case, we demonstrate the optimal number of secondary features f_k^* to maximize user engagement for the period k . We have the following lemmas. Also, note that we do not consider budget constraints to generate these structural results. We state that in the lemmas by mentioning constraints (2) from Problem SMM_s .

LEMMA 1. In Problem SMM_s , maximum user engagement is obtained by setting $s_k = q_k, \forall k$ and $\sum_{k=1}^K s_k = q_o$.

LEMMA 2. In Problem SMM_s with only constraint (2) under condition Case 1 for platform l and engagement type m for a given value of s_k , the maximum user engagement for period k is x_k^* , when either $f_k^* = \lfloor \frac{\zeta_{lm}}{2|\eta_{lm}|} \rfloor$ or $f_k^* = \lfloor \frac{\zeta_{lm}}{2|\eta_{lm}|} \rfloor + 1$ whichever provides the maximum x_k^* .

LEMMA 3. As $\zeta_{lm} > 0$ and $\eta_{lm} < 0$, in Problem SMM_s for platform l and engagement type m , $U_0 = \min\{u_2, f_k^*\}$ and $U \leq U_0$.

LEMMA 4. In Problem SMM_s under condition Case 1, the maximum, $f_0 = \lfloor \frac{(B-C_f-eU-aq_0)}{a} \rfloor$, where $f_0 = \sum_k f_k$ and B represents the budget.

As the lemmas make it apparent, under Case 1, we observe a concave relationship between engagement and secondary features given that $\zeta_{lm} > 0$ and $\eta_{lm} < 0$. This results in optimal user engagement at f_k^* for a sufficient budget. These findings enable the following managerial insights.

Observation 1: For a social media platform where the association between engagement and secondary features is concave, there exists an optimal user engagement level. For example, in our case, the maximum number of likes on Instagram occurs when the number of secondary features on a post is $f_k^* = 7$ or 8 .

This insight is interesting for firms because they can derive the most engagement (likes) from Instagram posts using a certain number of features. However, it is also important to note that a less-than-sufficient budget can restrict the firm from generating such engagement. Firms need to prioritize the development of posts in such a situation, as we shall demonstrate. Now, we linearize the problem SMM_s and describe the formulation. Before we introduce the model, we highlight the linearization process and the structural insights.

Here, we set $b + \zeta f_k + \eta f_k^2 = \sum_{j=0}^{f_k^*} g_j y_{k,j}$, where $g_j = b + \zeta j + \eta j^2, j = 0, 1, \dots, f_k^*$ as defined in Table EC.36. Note that g_j is the contribution of the number of secondary features to the user engagement and $y_{k,j}$ is a boolean variable to determine the number of secondary features to be included in the image.

THEOREM 1. Problem $LSMM_s$ (Case 1) under condition Case 1 is equivalent to the linear version of Problem SMM_s .

Observation 2: The user engagement estimated by $LSMM_s$ (Case 1) is equivalent to that estimated by SMM_s .

The intuition behind Observation 2 can be explained as follows. We linearize constraint set (2) in SMM_s via constraint sets (9) and (10) in $LSMM_s(Case1)$. We create the variable $y_{k,j}$ to denote each secondary feature in the model. The parameter g_j represents the impact on user engagement based on the number of secondary features used. This way, we replace the nonlinearity with linear constraints in $LSMM_s(Case1)$. In Table EC.36, we enumerate the Instagram likes (Case 1) values for each secondary feature f_k , which we derive from our empirical results, finding that the highest level of engagement x_k^* occurs at $f_k^* = 7$ or 8, which is consistent with Managerial Insight 1. Linearizing all formulations for both single and multiple platforms removes the non-linearity associated with our formulations and produces simple and efficient models.

Problem $LSMM_s(Case1)$:

$$\mathbf{Max} \quad \Pi_s = \sum_{k=1}^K w_k x_k$$

Subject to:

$$a \sum_{k=1}^K s_k + a \sum_{k=1}^K \sum_{j=0}^{f_k^*} j y_{k,j} + eU + C_f \leq B \quad (8)$$

$$x_k = \beta s_k + \sum_{j=0}^{f_k^*} g_j y_{k,j}, \quad \forall k \quad (9)$$

$$\sum_{j=0}^{f_k^*} y_{k,j} = 1, \quad \forall k \quad (10)$$

$$s_k \leq u_1, \quad \forall k \quad (11)$$

$$\sum_{j=0}^{f_k^*} j y_{k,j} \leq U, \quad \forall k \quad (12)$$

$$U \leq f_k^*, \quad (13)$$

$$s_k \geq q_k, \quad \forall k \quad (14)$$

$$x_k, s_k, U : \text{Integer variable } (\geq 0), \quad \forall k \quad (15)$$

$$y_{k,j} \in \{0, 1\} \quad \forall k; \quad \forall j \quad (16)$$

4.2.2. Solving the Single Platform Problem SMM_s for Facebook (Shares). The attributes of Facebook for user engagement (Shares) appear under Case 2, where $\beta_{lm} < 0$, $\zeta_{lm} < 0$, and $\eta_{lm} > 0$. We now provide the solution approach by linearizing Problem SMM_s . Any platform pertaining to this case can use the approach presented in this subsection.

Case 2: $\beta_{lm} < 0, \zeta_{lm} < 0, \eta_{lm} > 0$. An example of such a platform is Facebook, where the engagement type is the number of shares. Readers can refer to Table 2, where $l = 2$ and $m = 2$. We have the following results for the Platform satisfying condition in Case 2.

As in Case 1, we must have $s_k = q_k, \forall k$ and $\sum_{k=1}^K s_k = q_o$ (Lemma 1). The next result provides the optimal number of secondary features f_k^* that minimizes user engagement in period k .

LEMMA 5. *In Problem SMM_s with only constraint (2) under condition Case 2 for platform l and engagement type m for a given value of s_k , the minimum user engagement x_k^* occurs in period k , when either $f_k^* = \lfloor \frac{|\zeta_{lm}|}{2\eta_{lm}} \rfloor$ or $f_k^* = \lfloor \frac{|\zeta_{lm}|}{2\eta_{lm}} \rfloor + 1$ whichever provides the minimum x_k^* .*

4.2.3. Solving the Single Platform Problem SMM_s for Facebook (Comments). The attributes of Facebook for user engagement (comments) appear under Case 3, where $\beta_{lm} > 0, \zeta_{lm} < 0$, and $\eta_{lm} > 0$. Any platform similar to this case can utilize the approach presented in this subsection.

Case 3: $\beta_{lm} > 0, \zeta_{lm} < 0$, and $\eta_{lm} > 0$. An example of such a platform from our case study is Facebook (comments), as demonstrated by Table 2, where $l = 2$ and $m = 3$. We have the following results for the Platform satisfying the condition in Case 3. The next result provides the optimal number of secondary features f_k^* that minimizes user engagement in period k .

LEMMA 6. *In Problem SMM_s with only constraint (2) under condition Case 3 for platform l and engagement type m for a given value of s_k , the minimum user engagement x_k^* occurs in period k , when either $f_k^* = \lfloor \frac{|\zeta_{lm}|}{2\eta_{lm}} \rfloor$ or $f_k^* = \lfloor \frac{|\zeta_{lm}|}{2\eta_{lm}} \rfloor + 1$ whichever provides minimum x_k^* .*

Unlike Case 1, Cases 2 and 3 demonstrate convex patterns and exhibit the minimum engagements that occur at $f_k = 5$ and 6 for Facebook shares and comments, respectively. Because all other structural properties remain the same, we can modify $LSSM_s(Case1)$ by replacing f_k^* with u_2 , the upper bound of secondary features on Facebook. The upper bound u_2 remains the same for shares and comments because these engagement types pertain to the same platform. The specifics of $LSMM_s(Cases2 - 3)$ are described formally in Section EC.2.

Observation 3: *The relationship between user engagement as measured by the number of shares (respectively, comments) on Facebook and secondary features reveals a convex pattern. User engagement on Facebook posts is associated with more secondary features, leading to more shares (and comments).*

Table 3 Relative Weights w_k

w_1	w_2	w_3	w_4	w_5	w_6	w_7
10	12	16	14	16	14	10

4.2.4. Numerical Analysis. In this subsection, we conduct a numerical analysis by running $LSMM_s(Case1)$ for Instagram likes and $LSMM_s(Cases2-3)$ for Facebook shares and comments separately. We then extend the analysis by combining shares and comments for Facebook and developing Problem $LSMM_f$. We set the feature development cost a at 20 and vary e between 20 and 90, with increments of 10. This changing e enables us to observe the impact of the analytics costs and budget on feature use and engagement levels. The number of mandatory primary features q_k is generated randomly from the set of integer numbers between 1 and 6, fixed for all platforms. For ease of exposition, we select the values for w_{lk} from a sample of posts from Instagram as depicted in Figure EC.10, with the planning horizon beginning on a Sunday and ending on a Saturday over a two-week period. While the pattern suggests that Instagram posts experience higher user activities during weekdays, we find similar user behavior on Facebook as well, where engagement on weekends is relatively lower than on weekdays (Lang 2025). Thus, there is an overwhelming majority of the opinion that weekends are usually associated with lower user engagement (e.g. Hutchinson 2023). Thus, we keep w_{lk} similar across the platforms. However, our optimization framework can easily incorporate different weights for different platforms and our primary insights will continue to hold. We present a case here. Table 3 shows the relative weights w_k , where $k = 1, 2, \dots, 7$. We repeat the same values for $k = 8, 9, \dots, 14$ and fix them across platforms and engagement types for comparison.

Table EC.37 (in the e-companion) presents the results for Instagram. It is apparent that the total engagement (likes) on Instagram reduces as e increases because the budget allocation tightens for the development of post activities. The highest total engagement of 714,161 (x_k^*) occurs when e is set at the lowest ($e = 20$). However, as e increases, fewer secondary features (f_k^*) are assigned on posts to be published on the days, especially the days characterized by lower engagement intensity. On average, the firm achieves maximum user engagement x_k^* by including the optimal number of secondary features ($f_k^* = 7$ or 8) on posts (as specified by Lemma 2).

We present the results for Facebook shares and comments in Tables EC.38 and EC.39, respectively (in the e-companion). Unlike the relationship between likes and secondary features on Instagram, on Facebook, the two engagement types demonstrate a convex relationship with secondary features. Table EC.36 shows that $f_k^* = 5$ minimizes shares. The firm utilizes $f_k = u_2$ (or, 18 for Facebook)

secondary features on posts to be published on days characterized by relatively high engagement intensity under a budget constraint. We observe a similar pattern for Facebook comments in Table EC.36, where the minimum engagement occurs at $f_k^* = 6$. The results confirm our theoretical insights for the convex relationship between secondary features and Facebook shares (respectively, comments). In order to differentiate the weights on Facebook posts from Instagram posts, we assign higher weights to Thursdays and Fridays for Facebook, following the article by Lang (2025) (Table EC.41). Table EC.42 presents the results for Facebook with these new weights. Our insights continue to be similar to these new weights.

Now, we run $LSSM_f$ for a platform with multiple engagement types and the main parameter setting and present the results in Table EC.40 (in the e-companion). Because the engagement types are derived from the same platform, the number of features (both primary and secondary), the effort U , and the upper limits for both feature types remain the same. The engagement levels are x_{1k} and x_{2k} for shares and comments. We assign Facebook shares and comments two different relative weights—0.7 (μ_2) and 0.3 (μ_3)—to include the impact of different engagement types. We discuss them in detail in the e-companion, along with an analysis with no engagement weights.

4.3. Multiple Platforms and Multiple Engagement Types: Generalized Problem

Firms want their presence across multiple social media platforms to reach the maximum number of users. Furthermore, users retrieve information from multiple platforms (Zhou et al. 2015). This demands that firms have a comprehensive social media marketing management strategy that considers multiple platforms to achieve maximum user engagement via organic posts. This requires that the following vital aspects be incorporated into the generalized model:

- Modalities of engagement can differ within and across platforms. As discussed, engagement type signals the degree of interest of a user. For instance, commenting indicates greater interest in a post than liking that post (Aichner and Jacob 2015). It also depends on the types of content. For instance, on Facebook, interactive features of a post influence comments, whereas sensory and visual aspects of a post affect sharing (Kim and Yang 2017).
- Social media platforms are different in terms of context. For instance, Instagram is considered a more creative outlet (Unnava and Aravindakshan 2021). These differing roles also impact how users react to posts. Thus, we develop our frameworks aligned with the differences in platforms to help firms efficiently adapt their image post design and development strategies.
- Our empirical analysis reveals that posts on Facebook and Instagram exhibit different engagement levels for different numbers of secondary image features.

- Achieving high overall engagement on all platforms requires a standardized approach to incorporating the aforementioned aspects of different platforms within a multi-platform framework.

We construct z_{lm} based on the user base v_l for each platform l and the weight μ_{lm} of an engagement type m of a platform l , following the model developed by Aichner and Jacob (2015). The authors describe two metrics: social media impact factor (SMIF) and social media use (SMU). They estimate the SMIF of every platform l by dividing the number of active users of platform l by the total number of active users across platforms V , where $V = \sum_{l=1}^L v_l$. We use v_l/V to represent the SMIF and estimate it from a firm's social media accounts. Next, the SMU of each platform is measured by computing user reactions based on engagement types. For each platform, Aichner and Jacob (2015) compare different engagement types and assign weights accordingly. For example, on Facebook, one share is ten times more valuable than one like and twice as valuable as a comment. We capture the engagement weights within each platform, display z_{lm} using v_l and μ_m , and present the weights for Facebook and Instagram in Table EC.44.

Problem SMM_G :

$$\mathbf{Max} \Pi_g = \sum_{l=1}^L \sum_{m=1}^M \sum_{k=1}^K z_{lm} w_{lk} x_{lmk}$$

Subject to:

$$a \sum_{l=1}^L \sum_{k=1}^K (s_{lk} + f_{lk}) + e \sum_{l=1}^L U_l + LC_f \leq B \quad (17)$$

$$x_{lmk} = b_{lm} + \beta_{lm} s_{lk} + \zeta_{lm} f_{lk} + \eta_{lm} f_{lk}^2, \forall k; \forall l; \forall m \quad (18)$$

$$s_{lk} \leq u_{1l}, \forall k; \forall l \quad (19)$$

$$f_{lk} \leq U_l, \forall k; \forall l \quad (20)$$

$$U \leq u_{2l}, \forall k, \forall l \quad (21)$$

$$s_{lk} \geq q_{lk}, \forall k; \forall l \quad (22)$$

$$x_{lmk}, s_{lk}, f_{lk}, U_l : \text{Integer variable } (\geq 0), \forall l; \forall m; \forall k \quad (23)$$

In Problem SMM_G , constraint set (17) limits the three cost parameters under the total budget, similar to the single platform and single engagement type case but differing by including the platform (l) notations. In constraint set (18), we associate user engagement with our empirical results for each platform l and user engagement m . However, the features are independent of m . Constraint sets (19) and (20) restrict the simple and secondary features according to their upper

bound and the decision variable U_l . In constraint set (21), we restrict the decision variable U_l under the upper bound of secondary features u_{2l} for each platform l . We set the primary feature(s) criterion using constraint set (22). In constraint set (23), we define the non-negativity constraints. We extend the generalized formulation to two scenarios in the following two subsections.

4.3.1. Two Platforms. Problem SMM_G is applicable to any number of platforms and engagement types. However, the model is nonlinear. We linearize constraint set (18) and present its linear version $LSMM_G$ in Section EC.2 (e-companion). We linearize the nonlinear constraint set (18) using constraint sets (EC.21) and (EC.22). There are substantial differences between the linearized general model $LSMM_g$ and the single platform combined formulation $LSMM_f$. The binary variable $y_{l,k,j}$ depends on the platform l because the firm can vary the number of features on posts across platforms. Depending on the convex or concave relationships between secondary features and engagement, the firm can use either the upper bound on the secondary features or the optimal secondary features (represented by J^l). For example, J^l take the values of $f_{l=1,k}^*$ for Instagram and $u_{2,l=2}$ for Facebook. The enumerated values g_{lmj} depend on both platform and engagement type.

Using a set of computational experiments, we discuss the results of $LSMM_G$ for Instagram and Facebook. We use a scaled user base for the platforms and vary them to observe the overall change in the total engagement. We also vary the unit feature cost a between 20 and 50 with an increment of 10. For ease of exposition, we kept q_k similar across two platforms and set a budget of \$10,000. As discussed, this is within the range of firms' social media budgets. Tables EC.45 - EC.49 (in the e-companion) present the results. The columns represent the engagement levels in Instagram (likes) and Facebook (weighted shares and comments). We observe that likes on Instagram posts reduce as either e or a increases. Additionally, an increase in the number of Instagram users generates more engagement, causing the Facebook engagement level to fall. Given the increase in a , overall engagement also reduces because firms cannot utilize secondary features to boost engagement. We observe similar patterns across all scenarios, as tabulated in Tables EC.45 - EC.49.

5. Algorithms for Social Media Engagement Maximization

This section devises efficient algorithms for single platforms. The two algorithms proposed in this section build on the cases and structural insights presented earlier. We further demonstrate the robustness of our algorithms by conducting a computational experiment using the data from Instagram and Facebook and present their applications.

5.1. Algorithm for a Single Platform: Case 1

We devise the optimal and easy-to-implement algorithm FAAS(Case 1) to allocate image features in social media posts for a single platform. We use the insights given in Lemmas (1–4) to develop the heuristic. Given the budget for Single Platform: Case 1, the algorithm prioritizes feature allocation on days with higher levels of engagement intensity. That is, the algorithm maximizes social media engagement during the planning horizon. The complexity of FAAS(Case 1) is $O(K f_{max}^*)$. Note that Instagram falls under Case 1 in our study. We now formally present the algorithm.

Algorithm 1 FAAS(Case 1)

Input: Coefficients: b, β, ζ , and η ; Cost parameters: a, e, C_f ; Budget B ; q_k, U (Upper bound on secondary features); $f_k^*, g_j, j = 1, 2, \dots, f_k^*$; K days; weights, $w_k, k = 1, 2, \dots, K$.

Begin

Set objective value, $\Pi_s^* = 0$; $U = f_{max}^*, s_k = q_k$; arrange weights in non-increasing order: $w_{i_1} \geq w_{i_2} \geq \dots \geq w_{i_K}$, tie is broken arbitrarily, where i_k indicates the day for the weight w_{i_k} .
Find the budget, B' available for secondary features, $B' = B - a \sum_{k=1}^K s_k - eU - C_f$

While ($U \geq 0$) **do**

While ($B' \geq 0$) **do**

For $k = 1$ to K **do**

 Allocate maximum no. of secondary features r on the post for day i_k such that

$B' = B' - r * a \geq 0$ and $r \leq U$. Register $i_k = r$ and $\Pi_s = w_{i_k} x_{i_k}$, where $x_{i_k} = \beta s_k + g_{j_k}$

 If $\Pi_s^* < \Pi_s$ then $\Pi_s^* = \Pi_s$ and $i_k^* = i_k, k = 1, 2, \dots, K$

 If $B' < a$ then set $B' = -a$.

End(for k)

 Set $B' = -a$

End(while B')

Set $U = U - 1$ and $B' = B - \sum_{k=1}^K s_k - eU - C_f$

End(while U)

End

Output: K posts with $i_k^*, k = 1, 2, \dots, K$ and total engagement for the planning horizon, Π_s^* .

THEOREM 2. *Algorithm FAAS(Case 1) solves Problem LSMM_s(Case 1) in polynomial time. The complexity of the algorithm is $O(K f_{max}^*)$.*

5.2. Algorithm for a Single Platform: Case 2

We now present the algorithm FAAS(Case 2), based on Lemma 5, for allocating image features for a single platform. FAAS(Case 2) closely resembles FAAS(Case 1) in terms of the approach.

However, in Case 2, x_k is convex of f_k for a fixed value of s_k , and we set $f_k \leq u_2$ as the upper bound of secondary features for Case 2.

Algorithm 2 FAAS(Case 2)

Input: Coefficients: b, β, ζ , and η ; Cost parameters: a, e, C_f ; Budget B ; q_k, U ;

$\mu_m, f_k^{max}, g_{jm}, j = 1, 2, \dots, f_k^{max}$; K number of days; weights, $w_k, k = 1, 2, \dots, K$.

Begin

Set $f'_k = \lceil \frac{|\zeta|}{\eta} \rceil$. Set objective value, $\Pi_s^{max} = 0$; $U = u_2, s_k = q_k$; arrange the weights in the non-increasing order: $w_{i_1} \geq w_{i_2} \geq \dots \geq w_{i_K}$, tie is broken arbitrarily, where i_k indicates the day for w_{i_k} .

Find the budget, B' available for secondary features, $B' = B - a \sum_{k=1}^K s_k - eU - C_f$

While ($U \geq 0$) **do**

While ($B' \geq 0$) **do**

For $k = 1$ to K **do**

 Allocate maximum no. of secondary features $r \geq f'_k$ on the post for day i_k such that

$B' = B' - r * a \geq 0$ and $r \leq U$.

 Register $i_k = r$ and $\Pi_s = w_{i_k} \sum_{m=1}^M \mu_m x_{ikm}$, where $x_{ikm} = \beta_m s_k + g_{jkm}$.

 If $\Pi_s^{max} < \Pi_s$ then $\Pi_s^{max} = \Pi_s$ and $i_k^{max} = i_k, k = 1, 2, \dots, K$

 If $B' < a$ then set $B' = -a$.

End(for k)

 Set $B' = -a$

End(while B')

Set $U = U - 1$ and $B' = B - \sum_{k=1}^K s_k - eU - C_f$

End(while U)

End

Output: K posts with $i_k^{max}, k = 1, 2, \dots, K$ and total engagement for the planning horizon, Π_s^{max} .

THEOREM 3. *The algorithm FAAS(Case 2) solves Problem LSMM_f in polynomial time. The algorithm's complexity is $O(Kf_{max})$.*

We extend our analysis in Section EC.2.3 (in the e-companion) to include four platforms, using the data from Instagram, Facebook, and two simulated platforms that resemble two real-world social media platforms. The analysis of the four platforms is designed to reduce the complexity associated with multiple social media platforms and develop a simple framework for generating higher user engagement. We propose an algorithm—Feature Allocation Algorithm Multiple (FAAM)—that can be extended for any number of platforms. Table EC.59 presents the performance of FAAM, demonstrating its robustness. We also discuss the limitations of this heuristic (e-companion).

6. Results and Managerial Insights

This section first discusses the managerial insight derived from the numerical analysis done in Section 4.2.4 and then describes the proposed algorithms and structural insights into several manager-relevant questions. We discuss the performance of the algorithms developed for individual platforms and describe situations where they are more effective. To derive useful insights for managerially relevant questions, we run extensive computational experiments in the next two subsections using our algorithms and insights. First, we explain why firms should focus beyond platform user bases when making budget decisions. Second, we discuss the importance of advanced image features for driving user engagement in multi-platform settings. Finally, we discuss a setting where firms must consider using duplicate features in posts.

6.1. Insights Gained from Section 4.2.4

In Section 4.2.4, we primarily examine how user engagement varies under budget limitations. We also explore this change in user engagement on high- and low-engagement intensity days. The findings from our analysis lead to the following actionable managerial insight.

Managerial Insight 1: *A limited budget forces firms to be tactical in terms of utilizing their resources to gain maximum user engagement on social media platforms. While an optimal (or maximum) number of secondary features is desirable for Instagram (and Facebook), a limited budget enforces a policy where a firm must spend more resources on social media posts, which are to be published on days characterized by relatively high-intensity engagement for achieving maximum engagement. Firms must prioritize posts to be published on these days.*

The primary takeaway from this insight is that firms should carefully decide how to utilize their budget to increase user engagement on social media platforms. In particular, focusing too much on days characterized by lower intensity of engagement might engender lower overall engagement. The firm must use its budget strategically to specify the features on posts for maximum engagement.

6.2. Effectiveness of the Proposed Algorithms

We run two experiments to assess FAAS algorithms. The parameters used in the algorithms resemble those used in Section 4.3.1. The results for Instagram appear in Table EC.50 (in the e-companion). We find that the optimal engagement matches the output of FAAS(Case 1) at $U = 7$ or 8, given e , with the minimum engagement revealed to be 681,961 at $f_k = 0$, approximately 4.5% lower than the optimal value. For Facebook, we observe the highest possible engagement using FAAS(Cases 2–3) in Table EC.51 (in the e-companion). Differing from the example of Instagram, the firm

obtains more benefits under FAAS(Cases 2–3) at $U = 0$ than $U \in [1, 11]$ in our context. However, the maximum engagement with all secondary features stands at 302,284, which is 16.2% higher than the engagement at $f_k = 0$.

6.3. Social Media User Bases and User Engagement

Firms rely on the number of users to design their strategies for different social media platforms (Cucu 2022). However, having a clear focus on utilizing the user base solely for social media management is essential, as the outcomes and implications of the analysis may differ substantially. It may be possible that considering only the user base when developing a social media strategy may not be the best option. To address this issue, we compare our optimization framework to a setting in which budget allocation for each platform is based solely on the user base. For this comparison, we develop a test bed by varying the key parameters: budget, weights, user base ratio of the two platforms, and unit cost e . We present them in Table EC.52, indicating a total of 192 instances.

For ease of presentation, we report the difference in terms of increase in engagement in our approach from the user base-focused approach in Table EC.53. We observe that, in our context, our proposed approach exhibits a higher overall engagement than the strategy focused solely on the user base; the user base-focused approach may reduce the engagement level by up to 14.8% in our context. In addition, engagement reduces as both the budget and the user base on Instagram grow, highlighting the compounding effect of broader reach and a higher spend on Instagram compared to Facebook. Instagram is more popular among young adults and more often firms target that demographic, indicating a higher weightage to Instagram. These results highlight that the approach focusing solely on a platform's user base may produce sub-optimal performance in most realistic instances. This answers Q3 (presented in Section 1.3).

Managerial Insight 2: *It is important to define the social media management clearly in terms of only using the user base of different platforms to allocate the budget as it may often lead to sub-optimal engagement outcomes, as seen in our setting, which may result in up to approximately 15% less overall engagement than our proposed approach.*

6.4. Importance of Secondary Features for Social Media Engagement

It costs more to incorporate secondary features into images because of additional analytics and development efforts. Although they improve user engagement differently across different platforms, analyzing the level of visible improvements is worthwhile. To analyze the benefits of using secondary features, we follow the budget allocation policy described in Section EC.2.3 and conduct a test. This

enables us to find an allocation parameter θ_l for each platform l , where $\sum_{l=1}^L \theta_l = 1$. Table EC.58 in the e-companion presents these parameter values, with platform 1 referring to Instagram and platform 2 referring to Facebook.

To answer Q4 and demonstrate our heuristic's performance, we increase ζ_{lm} by 5% and reduce η_{lm} by 5% to generate two scenarios for each platform. Therefore, we generate $2 \times 2 \times 2 \times 2 = 16$ combinations or scenarios. This way, we create a testbed of 448 runs to capture diverse scenarios. We present the results for the two platforms in Table EC.54 (in the e-companion). After applying the allocation formula, we note the maximum engagement levels with Instagram (likes) and Facebook (shares and comments) from the 2-1 combination, respectively. Without any secondary features, the maximum engagement levels are taken from the 1-1 combination. We observe that engagement levels increase between 4.93% and 16.21% with the addition of secondary features under the allocated budget. Given the growth in users on Instagram and Facebook (Sprout Social 2025a), the increase is significant.

Managerial Insight 3: *Under a fixed budget, it is always beneficial to use secondary features to substantially improve engagement levels. In our context, we observe an increase in user engagement of about 5% and 16% with secondary features to the posts on Instagram and Facebook, respectively.*

6.5. Optimization Framework without Duplicate Image features

In this subsection, we answer Q5 by considering a case where a social media marketing team of a firm may not want to keep duplicate features or features of the same type on the developed images. For instance, they may want to keep only one feature (e.g., a couch) in an image instead of showing two or more of the same feature. Using relevant features in advertising images is an interesting debate in practice. It is a common practice for social media content creators to feature the same feature or product in their advertising images for various reasons, including highlighting important items (Mialki 2023), maintaining visual consistency (Klahre and Arpaia 2024), brand-building (Campaign Monitor 2021), and artistic choices (Soegaard 2024). For instance, as Mialki (2023) comments, social media content creators often use repetition to emphasize the context of the advertisement and guide users to the appropriate features.

We analyze this setting in our optimization framework (EC.2.4) by conducting a new analysis without these duplicate features and comparing it with our primary analysis, where we keep all the features of image posts. We show that for a majority of the scenarios, the framework where duplicates are considered exhibits a higher level of engagement (Table EC.62). This additional analysis presents some interesting insights. Secondary features, as seen in the model with duplicates,

exhibit similar patterns in the model without duplicates for both platforms. Our optimization model also exhibits the reduction in engagement levels as a and e increase, a similar pattern observed in our primary analysis. However, given the industry practice of using duplicate features for more appropriateness in advertisements, we maintain the framework with all features as our focal model.

Managerial Insight 4: *Analyzing social media user engagement in the context of utilizing duplicate image features is important for firms' objectives, more importantly if the budget is limited. In our context, social media image posts with duplicate features exhibit higher engagement levels than posts without duplicate features in a majority of scenarios. The inclusion of duplicate features aligns with industry practices and enhances advertisement relevance. Therefore, marketers should prioritize models that retain all features to maximize user engagement.*

7. Discussion and Conclusion

Our study proposes a data-driven optimization framework to maximize user engagement for firms that publish image posts across social media platforms. These platforms facilitate real-time reactions to posts, enabling firms to monitor and collect user feedback information, which has enormous operational value for those firms (e.g., Kumar et al. 2018, Lee et al. 2018), helping them produce and deliver more relevant advertising posts. Irrelevant ads on social media fail to capture users' interest and are less likely to encourage engagement on posts (Forbes 2023). Thus, it is essential to develop a framework for creating engaging social media posts for users. Under this framework, social media analytics can aid in that process (Barkakoty 2025) given the value of data analytics in operations (Feng and Shanthikumar 2022). Our optimization framework proposes a simple and efficient process for utilizing social media analytics to develop relevant image posts for higher levels of user engagement on different platforms with varying characteristics.

We estimate the model parameters and the functional form of user engagement and image features using empirical analysis. Among these features, we classify the secondary features that form a story around the primary advertising items and profoundly influence user actions. We develop insights into (i) identifying the relationship between user engagement and image features across platforms and engagement types, (ii) achieving maximum engagement under budget limitations, and (iii) offering easy-to-implement and manager-relevant approaches to developing social media posts. We demonstrate the effectiveness of our algorithms in both single- and multi-platform settings. The generic structure of our framework enables extended applications in different industries and across social media contexts where image posts crucially contribute to advertising outcomes.

7.1. Theoretical Contributions

Our research contributes to the growing literature on social media user engagement while also considering resource allocation, offering insights into how companies can use social media analytics to guide the development of such posts. We also compare our study and the closely related papers in Table EC.1, which summarizes various findings in the literature. Figure EC.1 presents our contributions visually. We articulate our theoretical contributions in the following two subsections.

7.1.1. Drivers of User Engagement on Social Media FGC Posts. We extend the existing literature on images and social media engagement, which has largely emphasized the effects of the mere presence of images and low-level imagery information or peripheral attributes such as color and quality, by systematically identifying and examining central image features. The major emphasis on peripheral features in the existing literature is not altogether surprising, given the challenges in extracting and measuring central attributes. Leveraging the advancements in computer vision tasks, we extend the understanding of the antecedents of social media engagement by studying both central and peripheral features, as highlighted in the ELM framework (Petty et al. 1986), thereby providing a more comprehensive perspective on how image content influences user behavior.

Guided by narrative framing theory (Baumgartner 2002), we differentiate between primary and secondary visual features, the latter representing miscellaneous image features that provide additional layers of information and represent visual complexity (Pieters et al. 2010). Prior studies in the traditional marketing literature have explored these aspects, including image features, and have shown mixed evidence of their impact on users. There are conflicting findings on whether simplicity with fewer features (Aitchison 2012) or complexity with more intricate details (Putrevu et al. 2004) better influences user engagement. Our findings reveal that secondary features exert a nonlinear influence on user engagement, thereby underscoring the importance of considering image features in the context of social media engagement.

This expansion approach that uses the alternative theoretical lenses discussed above (Gopal et al. 2024) enables us to confirm that—similar to traditional advertising—these image features are associated with user engagement on social media, and that this association is not uniform. Using the data from two major social media platforms — Instagram and Facebook — we show that this effect varies across these two platforms. It is not surprising since social media platforms differ in user demographics and content presentation formats. Moreover, our study sheds light on how other important aspects of image posts, such as textual captions, image color properties, posting frequency, and the use of specific objects, such as people or plants, impact user engagement.

In addition, our study adds to the growing body of research on FGC in social media, which prior studies have found to be more positively received by users (e.g., Unnava and Aravindakshan 2021). Thus, examining FGC is important as it may enhance the knowledge about consumer preferences and result in future sales (Ren et al. 2023). Also, these FGC posts are under the direct control of firms. The predominant focus on UGC in the literature (e.g., Liu et al. 2021) has left important dimensions of FGC underexplored. As a result, key considerations—such as budget allocation, platform selection, and development costs—have received limited scholarly attention despite their practical relevance. By employing a parsimonious modeling approach and incorporating realistic parameters (Gupta 2018), our optimization framework systematically addresses these key considerations through rigorous scenario testing and developing polynomially solvable algorithms. Across various scenarios considered in our research setting, our approach outperforms commonly employed social media marketing strategies that allocate budgets solely based on the user base of platforms, those that ignore secondary features, and those that use only single, distinct features instead of multiple features that enhance informational richness and context. Together, these insights expand the existing literature on social media advertising strategy by demonstrating the value of data-driven decisions for optimizing cross-platform engagement with image-based posts, modeling several realistic scenarios that generate insights within available resources.

7.1.2. Utilization of Analytics via Deep Learning Algorithms. We also contribute to the growing body of literature on the utilization of analytics, particularly its application in social media advertising, by quantifying different costs and addressing resource allocation decisions across platforms. Studies in the literature either develop or use pre-trained deep learning algorithms to analyze images and are mostly descriptive in nature (e.g., Jaakonmäki et al. 2017, Li and Xie 2020). What is missing in the literature is a systematic work that quantifies the value of different algorithms in social media analytics while also considering the operational trade-offs associated with the costs of analytics and development. Insights from our optimization framework contribute to the literature on social media user engagement and image analysis, which often ignores what delivers the most value given resource limitations, bridging empirical insights with prescriptive analytics and showing that maximizing engagement is a multi-factor, trade-off-driven problem that varies by platforms, suggesting what is more effective under resource limitations.

The increasing popularity of social media platforms as prominent advertising channels calls for a better understanding of the user engagement levels on social image posts across different platforms and how analytics can be utilized (e.g., Gupta et al. 2022, Boesch 2024). While there is a rise in the

availability of large-scale unstructured image data and advanced deep learning algorithms (Himeur et al. 2023) that can aid this process, it is not fully explored in the literature. We have employed two different deep learning algorithms, SSD and Faster RCNN, each characterized by differing levels of efficiency in terms of performance and computational costs, to extract image features and quantify their cost/benefit in understanding the factors contributing to higher user engagement levels on social media posts. Advanced models such as these are increasingly being used to develop new measures (Pan et al. 2019), such as our operationalization of image features. Moreover, we jointly consider analytics and the development of social media posts from a resource allocation perspective, which no other study has systematically addressed. As social media platforms, user preferences across these platforms, and post-related costs, including analytic efforts, continue to evolve, our framework is designed to remain robust and valuable.

7.2. Managerial and Industry Implications

Despite the substantial growth and upward forecast of social media advertising, little attention has been given to issues related to developing image posts using social media analytics to better match user expectations (Overgoor et al. 2022). Our optimization framework, designed to maximize user engagement on social media posts, incorporates key parameter values, with several derived from our rigorous empirical analysis. This offers a social media management framework that enables firms to use social media data to generate insights and apply them to maximize user engagement periodically. Our framework is appropriate in an evolving social media marketing environment that sees firms increase social media spending and invest more in analytics (Kumar and Qiu 2021).

By studying the antecedents of user engagement with social media posts, we estimate the impact of secondary features across different platforms. Our results show that most engaging posts on Instagram and Facebook differ in terms of the number of secondary features. This means that a “one-size-fits-all” strategy does not work. Firms must creatively design posts for different platforms (Markerly 2022). This reflects the comments of IKEA’s head of mobile solutions, Victor Bayata: “So it’s very important for us to understand these [people] behaviors because that is how we start supporting them and bringing relevant communication. Our ambition as a company is all about, again, bringing the customer a unique, relevant, and integrated experience that can allow us to engage” (Jaekel 2015). Relevant social media posts improve user engagement on social media platforms and increase potential future sales (Lang 2025).

Most contemporary companies, regardless of their size, spend substantial resources on social media platforms to gain a competitive advantage. An article by Soderlund (2022) reveals that

74% of companies spend almost one-third of their marketing budget on social media advertising, highlighting the importance of budget allocation. This is important because firms are often guided to prioritize certain days for posts to utilize budget efficiently (Sprout Social 2021). We demonstrate how firms can be tactical in terms of creating social media image posts for different platforms under a limited budget. However, deciding their budget allocation based only on a platform's number of users can produce sub-optimal results: the user base approach can lead to a reduction in engagement by up to 14%, as we observe in our study setting. This finding aligns with the ongoing debate that overall engagement is more important than simply the user base number (Sprout Social 2025a). We incorporate this feature in our multi-platform setting and find that adding secondary features to image posts increases engagement levels by about 5% on Instagram and 16% on Facebook. The increase in engagement has major implications because even a 1% engagement increase produces substantial benefits (Newberry 2025).

We discuss three additional results obtained in our empirical analysis, which we present in Q1 in Section 1.3. First, the post interval or the duration between two consecutive posts is negatively associated with user engagement across all platforms. Given the competition between marketers to command user attention, firms should frequently publish posts on their social media accounts. Second, we find that posts with more textual information on Instagram are associated with higher engagement, as evidenced by the positive relationship between an Instagram post's word count and the number of likes. Meanwhile, on Facebook, it is the opposite; posts' captions with fewer words are associated with more engagement. Finally, we observe how the saturation (the strength of color in a picture) of an image is negatively associated with engagement on Facebook Posts. We also present interesting results with the use of specific features, such as humans or plants, in Q2 in Section 1.3.

7.3. Emerging Trends in Social Media and Future Trends

Given the popularity of social media platforms (Gao et al. 2022), businesses flock there and publish posts to connect with their target customers. In the current decade, there is a consistent increase in social commerce that allows social media users to buy directly on social media. Naceva (2024) comments that the US will likely have 108 million social buyers by 2025. Thus, there will be a growing need for more relevant social media posts for users. Another emerging social media trend is the use of AI and human touch. Firms will strive to produce more relevant and relatable content for their users on social media platforms. In this context, the personalization of advertisements will always gain a lot of traction. In terms of social media platforms, Facebook and Instagram

will continue to remain the top choices for advertisers, while newer platforms such as TikTok will continue to gain more audience (Clark-Keane 2025). However, firms can prioritize one platform over another depending on their target market. For instance, on average, TikTok has more younger users than Meta (Morello 2025). Against this backdrop, our multi-platform social media optimization framework addresses how firms can develop and publish their content across different platforms using the insights from social media analytics while utilizing their budget efficiently.

7.4. Limitations and Future Research

Certain limitations of this study open up exciting future research avenues. We have collected data from Instagram and Facebook. Although our optimization framework approach can be generalized for any platform, future research could examine data from other platforms.

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All URLs below are accessed on October 23, 2025.

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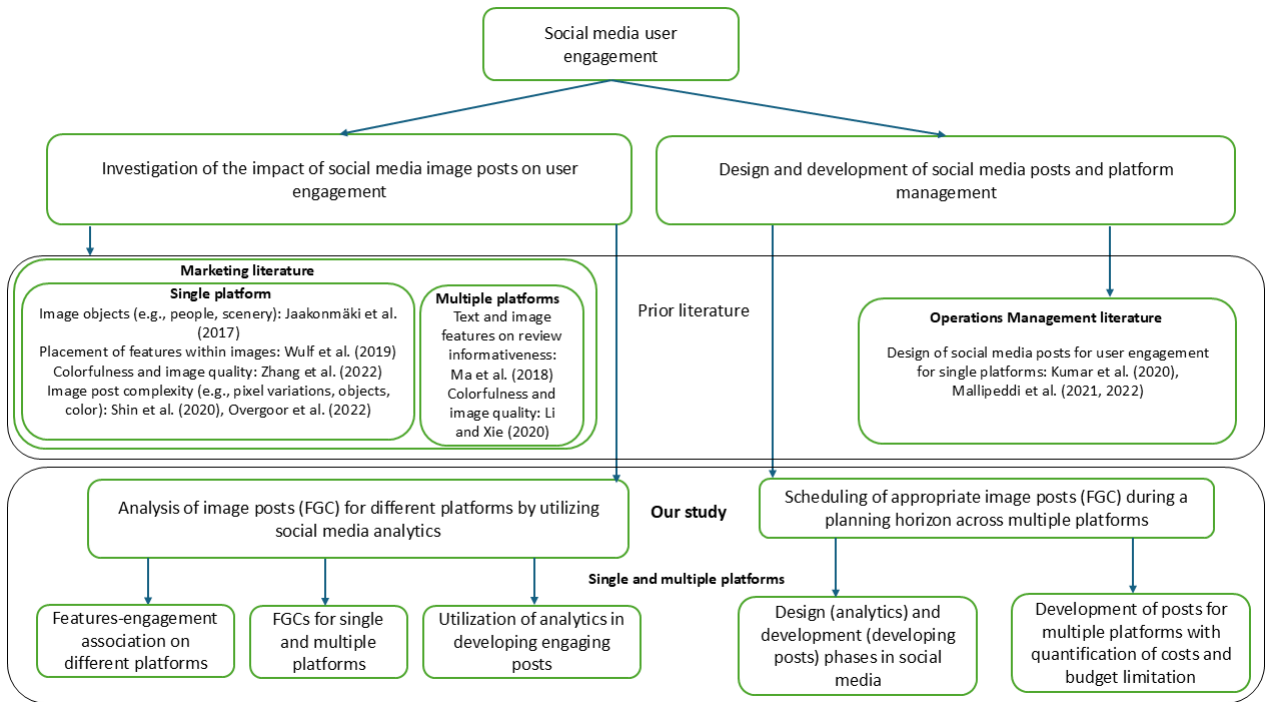
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E-Companion to Know Your Users via Image Analytics Before Developing Posts: Data-Driven Optimization Framework to Enhance Social Media Engagement

This e-companion provides the details of our empirical analysis, proofs of lemmas and theorems, and tables and figures used in the proposed optimization framework.

Figure EC.1 Positioning of Our Study in the Literature



EC.1. Empirical Analysis

In this section, we discuss our empirical analysis. We mainly provide more details on (i) the data collection, (ii) the deep learning approach, (iii) the econometric approach, (iv) a discussion of the findings and how we utilize them in our optimization framework, and (v) the robustness tests.

EC.1.1. Data Collection and Variables

Our dataset comes from the posts published by furniture firms on Facebook and Instagram, accessed through the platforms' application programming interfaces (APIs). We collected a total of 2,733 recent posts from Facebook and Instagram. These APIs only allow certain features to be extracted from social media posts. We select each firm's official social media accounts that periodically publish organic posts for each platform's audience. Baseline data were collected from social media

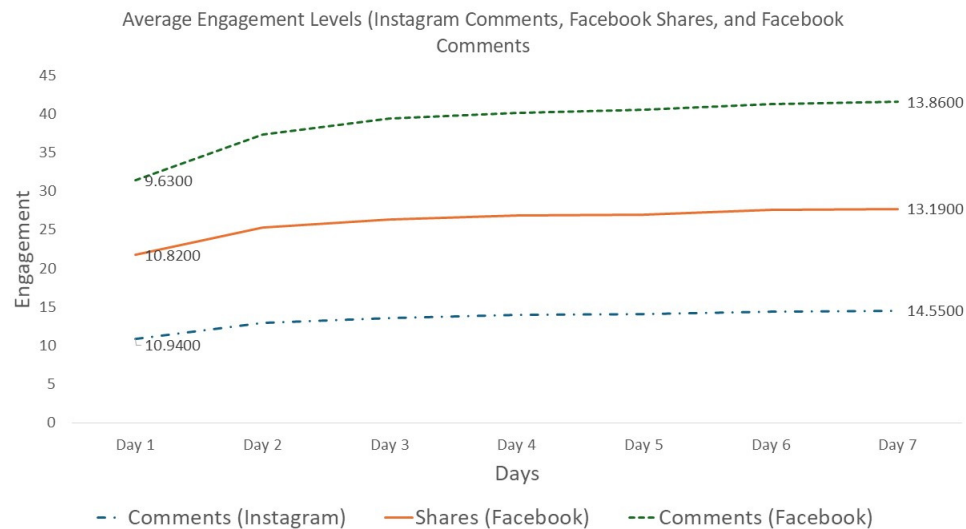
Table EC.1 Social Media Image Analysis in Marketing Literature and Our Study

Authors	Research setting					Methodology	Deep learning use
	Research question	Post type	Image analysis level	Multi-platform	Multi-engagement type		
Jaakonmäki et al. (2017)	Effect of post creator and post features on user engagement	UGC	Peripheral	No	Yes	Empirical (LASSO)	Extract features
Ma et al. (2018)	Information value of user-provided photos and review text on review helpfulness	UGC	Peripheral	No	No	Empirical (machine learning-based)	Extract features
Wulf et al. (2019)	Placement of features within images and how it affects popularity	UGC	Central	No	No	Empirical (linear regression)	Extract features
Li and Xie (2020)	Effect of colorfulness and quality of images on user engagement	UGC	Peripheral	Yes	Yes	Empirical (Zero-Inflated Negative Binomial)	Extract features
Liu et al. (2020)	Analysis of the link between user-created images and brand perception	FGC	Peripheral	No	No	Empirical (survey-based)	Extract features
Shin et al. (2020)	Impact of image complexity and text-image fitness on user engagement	UGC	Central and peripheral	No	Yes	Empirical (Negative Binomial regression)	Extract features
Zhang et al. (2021)	Understanding of the quality of images and how it affects Airbnb property demand	UGC	Peripheral	No	No	Empirical (difference-in-difference)	Extract features
Overgoor et al. (2022)	Effect of design and feature complexity on user engagement on Instagram	FGC and UGC	Central and peripheral	No	No	Empirical (Negative Binomial regression)	Extract features
Our study	Utilization of social media analytics to develop image posts with appropriate number of features to maximize engagement across platforms under budget constraint	FGC	Central and peripheral	Yes	Yes	Empirical and analytical modeling (Hurdle, Zero-Inflated Negative Binomial, and optimization)	Extract features and quantify engagement and costs

platforms, with additional information on the firms manually gleaned from their social media accounts and official websites. We built scrapers in Python that interact with the social media APIs to collect posts with images. It also collects the number of likes (available for Instagram only), comments, shares (available for Facebook only), captions, and posting dates. Comments are

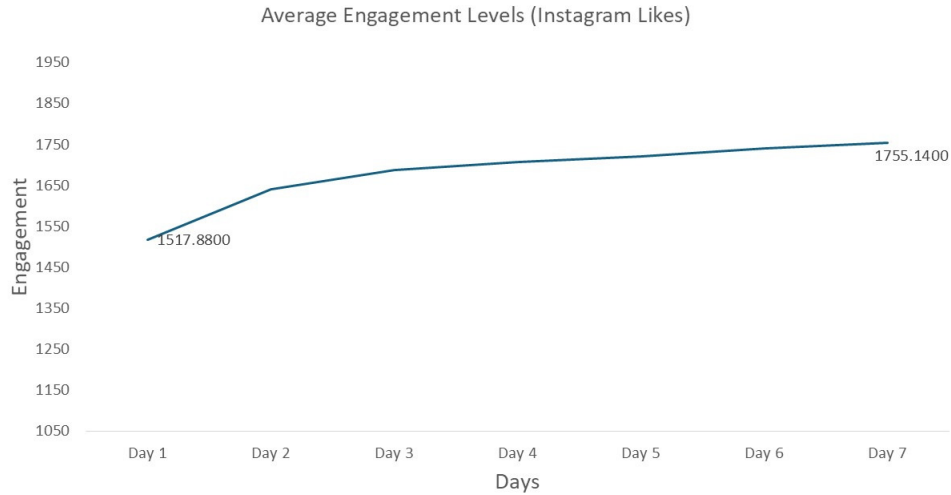
Table EC.2 Comparison between Li and Xie (2020) and Our Study

	Li and Xie (2020)	Our study
Research focus	Effect of image characteristics on user engagement	Utilization of social media analytics to develop image posts with an appropriate number of features for multiple platforms to maximize engagement across platforms under budget constraint
Methodology	Empirical (Zero-Inflated Negative Binomial)	Empirical and analytical modeling (Hurdle, Zero-Inflated Negative Binomial, and optimization)
Key image/post characteristics studied	Mere presence of an image in a post, human face in a post, colorfulness, professionally shot	Image features (primary and secondary), color properties, post interval, presence of specific features (e.g., person or plant)
Image characteristics route (ELM)	Peripheral	Central and peripheral
Post type	UGC	FGC
Platforms	Instagram and Twitter	Instagram and Facebook
Evaluation of deep learning algorithms	No	Yes
Budget and cost considerations	No	Yes
Joint consideration of post design and development	No	Yes

Figure EC.2 Average Engagement Levels (Instagram Comments, Facebook Shares, and Facebook Comments)

common and can be measured across both platforms. The study by Aldous et al. (2019) finds that most engagements happen during the initial couple of hours from posting time, and therefore, day normalization may not be required. Our data reveal the same property we discussed in Section 3.1.1. We describe the main variables used in our empirical analysis in Table EC.3.

Dependent variables: The dependent variables in our study are likes and comments on Instagram and shares and comments on Facebook. These variables exhibit the properties of a count variable with high overdispersion; we discuss this further in EC.1.4.5. Note that, on average, user engagement

Figure EC.3 Average Engagement Levels (Instagram Likes)**Table EC.3 Variables Used in Empirical Analysis**

Variable	Description	Variable type
likes	number of likes (count)	dependent (Instagram)
comments	number of comments (count)	dependent (Instagram and Facebook)
shares	number of shares (count)	dependent (Facebook)
primary features	number of features extracted using the SSD algorithm	independent
secondary features	number of additional features extracted using Faster R-CNN algorithm	independent
time	duration between post date and data collection date	control
post-interval	duration between two consecutive posts	control
revenue	revenue of a firm	control
posts (Instagram only)	total number of posts by a firm on Instagram	control
page active (Facebook only)	length of the official account of a firm on Facebook since opening	control
saturation	saturation value of an image	control
log(wc)	logarithm of word count of a post's caption	control

on Instagram is substantially higher than that on Facebook. We identified these variables from the data we collected from Instagram and Facebook.

Independent variables: The key independent variables in the empirical analysis are the two classes of image features extracted from the social media posts - primary and secondary features. SSD is a simpler algorithm that only extracts prominent primary features from images. We refer to them as primary features. On the other hand, Faster R-CNN extracts additional features in addition to the primary feature(s), given its algorithmic complexity and performance. These additional features are secondary features. It is noteworthy to mention here that we apply deep learning algorithms in our empirical analysis to illustrate the use of social media analytics. Given the high accuracy and several applications of pre-trained deep learning algorithms (e.g, Cheng et al. 2017), we use the pre-trained models, both SSD and Faster R-CNN. The use of pre-trained models is prevalent in academic research, given their high accuracy levels in classification tasks (Yim et al. 2021).

Control variables: We employ several post-level and firm-level control variables to control for the heterogeneity among posts and firms. We run a simple Python program to extract image color properties, including red, green, and blue (RGB) channel values and hue, saturation, and lightness (HSL) values. Given the high correlation among these six color properties, we only use saturation as a control variable in the econometric analysis. Besides extracting these color features, we also collect and analyze the properties of posts' captions using the Linguistic Inquiry and Word Count (LIWC) tool. The LIWC tool ignores irrelevant items such as punctuation and hashtags and preserves the relevant information from the textual data. Post captions are crucial in our analysis since they offer social media users valuable product information. We use log-transformed word count to control for the effect of additional information on user engagement. We also construct a time variable that refers to the duration between posts and data collection dates. This variable is used for two purposes. First, it controls the length of time a post has been on the platform. Second, it also acts as a proxy that controls the impact of the number of followers around the post's publication time. We control for the firm effects using firms' number of posts and revenue, consistent with the literature for controlling post volume that indicates active participation (Audrezet et al. 2020).

It is important to discuss the choice of the industry in our study. We extract social media posts of large furniture firms. We chose the furniture industry to capture a wide range of audiences, similar to many other sectors. Additionally, furniture industries focus more on the products in their posts than human brands and logos, which is more appropriate for our work. More importantly, product image photography is an important display for furniture retailers, as customers give importance to professionally shot furniture images and the beautiful room scene graphics displayed on images. In this regard, our work can be extended to several other industries.

EC.1.2. Applications of Deep Learning Algorithms

We use deep learning methods to extract features or objects from images that we use as variables in our econometric model to assess the feature-engagement relationship and develop the optimization model to maximize user engagement. For that purpose, we deploy deep learning algorithms using TensorFlow in Python (All codes are available upon request). TensorFlow is an end-to-end open-source platform developed by Google that has built-in functionalities for training and testing deep learning algorithms. Since we used pre-trained models with a labeled dataset and applied them to the social media data we collected, our computational tools were sufficient to run the models. Social media analytics help firms identify the antecedents of user engagement, especially via deep

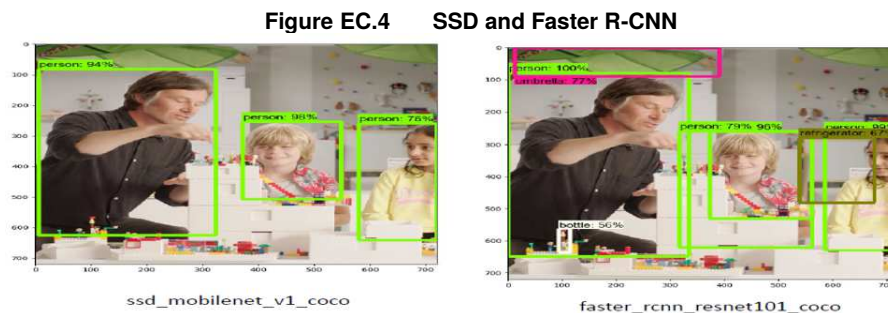
learning methods, since they help identify the features without large-scale experiments and inherent biases (Shin et al. 2020).

Deep learning methods vary based on complexity and performance in detecting objects. Depending on the applications, several deep learning methods are available for use. Convolutional neural networks, or CNNs, are popular deep learning algorithms for object detection with various degrees of complexity and accuracy. CNNs are primarily built with multilayer perceptrons, a basic neural network structure with inputs, activation functions, and outputs, with fully connected layers. In other words, each neuron within a layer is connected to all neurons in the next layer. CNNs learn to optimize the network by analyzing the hierarchical patterns in the data, in this case, an image. It first assembles and analyzes simpler patterns and then advanced patterns through the layers. The higher the layers, the more computationally complex the algorithm becomes, with better accuracy but lower memory efficiency. SSD uses one such layer and quickly detects prominent objects in an image. It is significantly faster in speed, but offers lower performance in terms of detecting features.

Within the family of CNNs, Region-based CNN (R-CNN) methods take image input and identify the features via bounding boxes or region proposals through selective search. Once the region proposals are identified, R-CNN warps the input to a standard size for processing, passes it to a CNN-based feature extractor that extracts features, and finally uses a support vector machine (SVM) algorithm to classify the features into the corresponding classes. This process is slow because it requires the training of three different models: (i) the CNN model to generate the features, (ii) the classifier that predicts the class objects, and (iii) the regression model to reduce bounding box errors for classification (Yadav and Binay 2017). In the family of R-CNNs, the Fast R-CNN model is developed to expedite the process by jointly training the CNN, classifier, and bounding box regressors and replacing the SVM classifier with a SoftMax for object classification. However, one bottleneck with the Fast R-CNN was that the region proposals were created using the slow selective search process. The Faster R-CNN overcomes the bottleneck by using a single CNN to carry out both region proposals and classification (Ren et al. 2015).

Compared to the other CNN methods for image analysis, the Faster R-CNN method enables the end-to-end learning of all layers. Faster R-CNN uses a region proposal network (RPN) instead of slow selective search methods. For our optimization model, we identify features from the social media data and analyze the appropriate number of features to be used in images. The Faster R-CNN is the appropriate method in our context to identify the additional features from an image and the primary features extracted by SSD. Compared to Faster R-CNN, SSD is a simpler model

that does not require regional proposal generation or feature resampling. The difference in the performance between SSD and Faster R-CNN allows us to quantify social media analytics to gather more user information efficiently. Faster R-CNN is 5-6 times slower than SSD processing an image. Ideally, more advanced analytics employ a large amount of data, more computational tools, and better technology teams, which adds to the social media analytics costs. We use the two deep learning methods to illustrate the difference. In Figure EC.4, we demonstrate these two algorithms' differences in feature detection. Both algorithms are pre-trained on the COCO dataset. Faster R-CNN detects more features. We use these two algorithms on all the social media posts we collected and identify the simple and secondary features in the econometric method and optimization model.



The use of deep learning algorithms to extract image features and their relevance in social media marketing warrants a brief discussion. We utilize the image recognition task of deep learning in the optimization framework. Image recognition deals with identifying objects in an image and classifying them into categories. It is an ongoing research area in computer vision applications. Significant improvements have been made in image recognition algorithms, both in terms of speed and accuracy, over time. As noted by researchers at Brown University, a substantial part of this improvement is the constant dialogue between AI systems and neuroscience to make a better synergy (Pikul 2023). According to them, “The synergies go both ways: neuroscience gives us good inspiration for improving AI. But the fact that we do improve AI from those neuroscience mechanisms is also a way to validate the discoveries made in neuroscience and to identify key mechanisms of general intelligence, visual intelligence, and more.” This can be seen from the developments in the field. Meta developed an image recognition algorithm called YOLO, or you only look once, which identifies objects quickly. This motivates us to use deep learning algorithms to classify image features into primary or easily identifiable and secondary or miscellaneous features. To identify the primary features, we apply the SSD or single-shot detector algorithm. SSD is faster as it uses a single neuron to identify the distinct and non-overlapping features. SSD is extensively

used in self-driving cars, surveillance systems, and robotics (Baeldung 2023). We use Faster RCNN to extract the primary features and the secondary features. This way, we incorporate deep learning algorithms within social media analytics to analyze images and extract the features.

EC.1.3. Costs Associated with Analytics

Given an increase in the advancements in social media analytics tools, the creation of content has become more efficient, both in terms of expenditures and production time. For instance, an experiment by Reisenbichler et al. (2021) shows a reduction of 90% in production time and associated labor costs in producing artificial intelligence or AI-generated textual content after being edited by a human. However, social media analytics will continue to require a substantial budget in the future, mainly due to the following three reasons: (1) the rising costs of training the models, (2) finding appropriate and sizable data to train models, and (3) creating content with originality.

First, we discuss the rising costs of developing and using AIs. For example, according to a news report, the CEO of OpenAI, Sam Altman, is attempting to raise around \$7 trillion to improve the performance of the system (Business Insider 2024). Companies invest heavily in AI systems as they require powerful processors and complex software; these tools drive the costs of AI (Boesch 2024). For example, Netflix spends a substantial amount of its \$1.5 billion annual expenditure on technology in AI (Reilly 2024). It is also reported that a custom AI model can cost \$10 million USD, and may increase with more sophistication in the system. Second, it is essential to acknowledge the resource-intensive nature of creating advanced and customized analytics processes, as they necessitate a substantial amount of data (Bommasani et al., 2023). Companies often find it difficult to access a sizable amount of appropriate data to build their models (BCG 2024). Additionally, building more sophisticated systems often requires expensive data and storage processes (Agrawal 2023). Finally, we must discuss the implications of originality in creating AI-generated social media posts. While there are sufficient benefits in efficiency and consistency, firms are raising significant concerns about AI-generated posts (Dowling 2023). For instance, Google does not allow publishing content generated solely by a machine. As AI evolves, it will become more useful in developing content rapidly to enhance the social media experience. However, firms and users will also search for originality and human touch, requiring more creative, highly skilled, and highly paid content creators. This need for originality and advancements in AI will keep costs substantial.

Although an improvement in efficiency is likely to be seen in many applications due to AI, the costs of developing and using AI are expected to rise in the next few decades, with an increased focus on more resources, sophistication, and data requirements. Against this backdrop, our research question

Table EC.4 Summary Statistics of Instagram Variables

Variable	Obs	Mean	Std. Dev.	Min	Max
likes	542	2888.7600	3243.9750	96	19815
comments	542	44.4658	61.4368	0	353
time	542	164.3358	119.2445	1	449
saturation	542	58.1697	49.0072	0	255
log(wc)	542	3.3761	0.7042	1.0980	5.6970
primary features	542	1.5572	1.4692	0	10
secondary features	542	3.8450	2.9365	0	14
post-interval	542	2.2103	1.9504	0	33
revenue	542	3.2576	1.6654	1.4500	5.3300
posts	542	2120.2920	495.0938	1227	2514

Table EC.5 Summary Statistics of Facebook Variables

Variable	Obs	Mean	Std. Dev.	Min	Max
shares	2,191	13.6020	26.6060	0	388
comments	2,191	35.8380	63.0630	0	573
time	2,191	352.3580	246.7230	1	1057.2900
saturation	2,191	56.9330	43.3854	0	255
log(wc)	2,191	3.4888	0.6033	0	5.2200
primary features	2,191	1.7122	1.6960	0	19
secondary features	2,191	3.8621	3.0050	0	18
post-interval	2,191	1.2590	1.2582	0	8.0818
revenue	2,191	3.7400	2.9200	1.4500	9.1200
page active	2,191	3965.4200	289.3200	3499	4344

and insights will continue to be relevant as firms are spending resources (Sprout Social 2021) on social media advertising to develop image posts for enhanced user engagement and potential future sales (Lang 2025). Our framework models the competing costs of social media analytics and developing posts. Our framework can be adjusted to cover different costs of social media analytics. Social media managers can use our framework as a guide to consider social media analytics in developing relevant posts for users on different platforms.

EC.1.4. Econometric Approach

This section describes our econometric approach in which we examine the relationship between secondary features and user engagement. The summary statistics of variables are presented in Tables EC.4 and EC.5 for Instagram and Facebook, respectively.

EC.1.4.1. Model Specification. Our dependent variables represent count data and are heavily skewed. According to the literature, count models such as the Poisson or Negative Binomial are suitable in our context. All four engagement metrics we analyze exhibit high dispersion, where their variances exceed their mean values. Thus, we model the engagement variables using a Negative Binomial model, which is less restrictive than the Poisson regression, as it allows for more general

forms of overdispersion and underdispersion (Cameron and Trivedi 2013). In our context, most of the engagement variables contain zeroes, which is common across many social media engagement metrics. To account for this, we employ a more flexible specification - the Zero-Inflated Negative Binomial (ZINB) model. Since Instagram likes do not display any zeroes, we instead apply a hurdle model for Instagram likes. We present the results using the standard Negative Binomial as robustness tests later, and show that the results are qualitatively similar between the models. We present our regression equation: $x_{lmk} = \alpha_{lm} + \beta_{lm}s_{lmk} + \zeta_{lm}f_{lmk} + \eta_{lm}f_{lmk}^2 + \theta V_{lmk} + \gamma Z + \nu Y_l + \epsilon_{lmk}$.

As discussed earlier, the key independent variables used in the analysis are s_{lmk} and f_{lmk} for the post k on platform l and engagement type m . Note that the features are engagement-independent. However, their coefficients differ across platforms and engagement types. The variable s_{lmk} comprises the primary features of an image, conveying some product information about the advertisement to users. On the other hand, f_{lmk} complements s_{lmk} and forms a narrative for users by creating a story. Users obtain their information through these features. As hypothesized, the association between secondary features f_{lmk} and engagement is likely to be nonlinear and may differ across platforms. Thus, we construct the quadratic variable for f_{lmk} to test the relationship.

The coefficient of s_{lmk} represents the linear impact of primary features on engagement, while the coefficient of f_{lmk} captures the linear impact of secondary features on user engagement. The coefficient of the quadratic term for f_{lmk} highlights the nature of the nonlinear association between secondary features and user engagement. V_{lmk} denotes the post-specific controls (time, log-transformed word count of a post's caption, saturation value of the image of a post, and post-interval) while Z denotes the control variables for firm effects. Variable such as revenue for a firm is usually fixed in a single period, and they can be updated while running the analysis in future periods. Revenue refers to the market positioning of a firm. Since estimating the revenue generated from each platform is difficult, we consider the firm's revenue fixed across platforms. On the other hand, the number of posts may vary across platforms but not across the engagements within a platform. We denote such platform-specific control variables as Y_l .

EC.1.4.2. Results. We report the empirical results for Instagram in Table EC.6 with three models for likes and comments - the control-only models with primary features (1 and 4), the main effects models with secondary features (2 and 5), and the full models with the quadratic terms of secondary features (3 and 6). Note that we report the p values within parentheses for all the tables. The model fits (log likelihood values) substantially improve from Model 1 to Model 2 and from Model 4 to Model 5, highlighting the importance of including secondary features in the regression model. The

Table EC.6 Regression Analyses of Likes (Hurdle) and Comments (ZINB) on Instagram Data

	(1) likes	(2) likes	(3) likes	(4) comments	(5) comments	(6) comments
time	-0.0007** (0.0190)	-0.0009*** (0.0060)	-0.0009*** (0.0040)	-0.0021*** (0.0000)	-0.0021*** (0.0000)	-0.0021*** (0.0000)
saturation	-0.0002 (0.7360)	-0.0001 (0.8290)	-0.0001 (0.9360)	0.0009 (0.3210)	0.0009 (0.3250)	0.0008 (0.3330)
log(wc)	0.6683*** (0.0000)	0.6839*** (0.0000)	0.6982*** (0.0000)	0.7054*** (0.0000)	0.6952*** (0.0000)	0.6817*** (0.0000)
primary features	-0.0321 (0.1460)	-0.0360* (0.0920)	-0.0348 (0.1000)	-0.0576** (0.0250)	-0.0549** (0.0270)	-0.0543** (0.0240)
post-interval	-0.0711*** (0.0000)	-0.0693*** (0.0000)	-0.0682*** (0.0000)	-0.0352* (0.0290)	-0.0361** (0.0090)	-0.0369** (0.0080)
revenue	0.3428*** (0.0000)	0.3473*** (0.0000)	0.3472*** (0.0000)	0.4262*** (0.0000)	0.4252*** (0.0000)	0.4271*** (0.0000)
posts	0.0012*** (0.0000)	0.0012*** (0.0000)	0.0013*** (0.0000)	0.0017*** (0.0000)	0.0017*** (0.0000)	0.0017*** (0.0000)
secondary features		0.0307** (0.0240)	0.0945*** (0.0080)		-0.0109 (0.4820)	-0.0539 (0.1880)
secondary features squared term			-0.0063** (0.0460)			0.0042 (0.2580)
constant	2.0747*** (0.0000)	1.8230*** (0.0000)	1.6409*** (0.0000)	-3.6792*** (0.0000)	-3.5744*** (0.0000)	-3.4482*** (0.0000)
Inflate secondary features				-0.9921* (0.0770)	-0.8245 (0.1400)	-1.0084 (0.1560)
log likelihood	-4645.4704	-4641.8303	-4639.7014	-2335.1490	-2334.7810	-2334.0170
Observations	542	542	542	542	542	542

p values in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

coefficient of secondary features (Model 2) is positive and statistically significant for likes (0.0307, $p = 0.0240$). This result suggests that a one-unit increase in secondary features is associated with a 3% increase in likes. On the other hand, the effect of secondary features on comments (Model 5) is negative but not statistically significant (-0.0109, $p = 0.4820$). The primary features variable (Model 5) is negatively associated with comments (-0.0549, $p = 0.0270$), indicating that having more primary features is associated with fewer comments on Instagram. Next, we discuss the quadratic term of the secondary features variable.

We include the quadratic term for secondary features in Models 3 and 6 for likes and comments, respectively. Not surprisingly, these two models have the best fit with the quadratic terms for secondary features included. The quadratic term is negative and significant (-0.0063, $p = 0.0460$) for

Table EC.7 Regression Analyses of Shares and Comments on Facebook Data (ZINB)

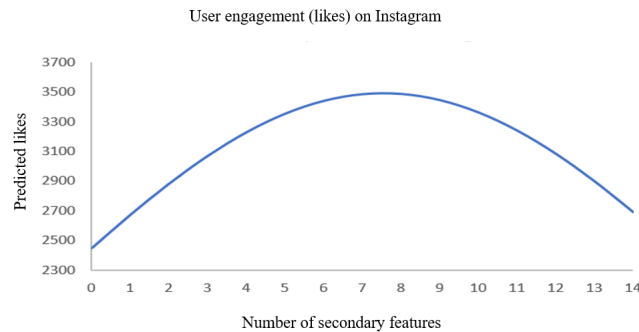
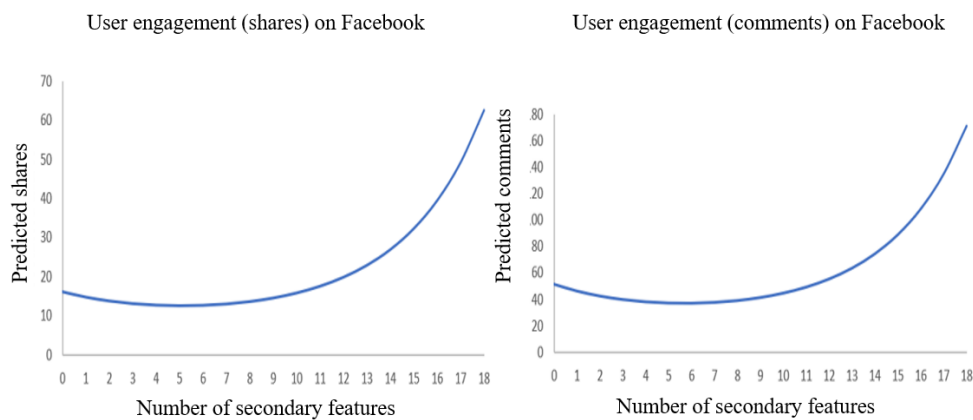
	(1) shares	(2) shares	(3) shares	(4) comments	(5) comments	(6) comments
time	2.9500×10^{-6} (0.9910)	1.6500×10^{-6} (0.9950)	-3.1200×10^{-5} (0.8980)	-0.0022*** (0.0000)	-0.0022*** (0.0000)	-0.0022*** (0.0000)
saturation	-0.0020** (0.0190)	-0.0020** (0.0190)	-0.0019** (0.0200)	-0.0018** (0.0110)	-0.0018** (0.0130)	-0.0016** (0.0220)
log(wc)	-0.3158*** (0.0030)	-0.3148*** (0.0030)	-0.3311*** (0.0040)	-0.7325*** (0.0000)	-0.7360*** (0.0060)	-0.7718*** (0.0000)
primary features	-0.0349 (0.4790)	-0.0359 (0.4390)	-0.0500 (0.2130)	0.0111 (0.7550)	0.0143 (0.6830)	0.0055 (0.8550)
post-interval	-0.0820** (0.0350)	-0.0815** (0.0330)	-0.0797** (0.0320)	-0.0817*** (0.0090)	-0.0834*** (0.0070)	-0.0800*** (0.0080)
revenue	0.1158*** (0.0000)	0.1157*** (0.0000)	0.1089*** (0.0000)	0.3010*** (0.0000)	0.3017*** (0.0000)	0.2956*** (0.0000)
page active	-0.0007*** (0.0000)	-0.0007*** (0.0000)	-0.0008*** (0.0000)	0.0004*** (0.0080)	0.0004*** (0.0060)	0.0003** (0.0190)
secondary features		0.0022 (0.8890)	-0.1012** (0.0200)		-0.0075 (0.5470)	-0.1224*** (0.0000)
secondary features squared term			0.0098** (0.0200)			0.0106*** (0.0010)
constant	6.3125*** (0.0000)	6.3130*** (0.0000)	6.7429*** (0.0000)	4.0835*** (0.0000)	4.0902*** (0.0000)	4.6861*** (0.0000)
Inflate secondary features	-1.2984 (0.1290)	-1.3941* (0.0700)	-1.1057** (0.0470)	-1.6572 (0.2640)	-1.5380 (0.2360)	-1.3717* (0.0700)
log likelihood	-7635.9460	-7635.9160	-7626.1090	-9158.4150	-9158.0180	-9144.9040
Observations	2,191	2,191	2,191	2,191	2,191	2,191

p values in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

likes. The linear and quadratic terms are positive and negative, respectively, suggesting a concave pattern in the relationship between secondary features and engagement (likes). In other words, likes initially increase with secondary features and then decrease after a certain point. We present the discussion on the changing behavior of the engagement types later in Section EC.1.6 in the e-companion. The quadratic term is not significant for Instagram comments (Model 6, 0.0042, $p = 0.2580$).

Turning our attention to the results for Facebook posts, in Columns 2 and 5 of Table EC.7, we find that the linear term of secondary features has no significant impact on either shares or comments, indicating that the linear forms may not capture the actual association. The coefficients for primary features have no association with shares or comments. The full models in columns 3 and 6 in Table EC.7 show that the quadratic term is both positive and significant for shares (0.0098, $p = 0.0200$) and comments (0.0106, $p = 0.0010$). The number of shares and comments on a Facebook

Figure EC.5 Instagram Analysis**Figure EC.6 Facebook Analysis**

post reduces initially as the number of secondary features increases, and starts to increase after certain thresholds. We present these patterns in Figures EC.5 and EC.6, which visually show the concave and convex patterns.

EC.1.4.3. User perception of the image features. Users are exposed to a substantial amount of information each time they use a social media platform. A post needs to stand out from the crowd to attract a user's attention before any engagement occurs. Thus, firms must develop relevant posts that are useful to users. However, this notion is also not free from challenges as users have limited cognitive ability to process an advertisement's content (Pieters et al. 2010). As Cacioppo and Petty (1982) remark, some users think more than other users. In other words, some social media users may need more motivation to evaluate an advertisement (a post in our context). Thus, an important question arises in the use of social networks, as Petty et al. (1986) discussed in traditional advertising: When will users evaluate posts of social network images and process the information? We can rephrase this question in our study context: Do social media users see the image features?

Table EC.8 Goodness of Fit Measures between Models without Features and Our Full Models

	Instagram (Likes)		Facebook (Shares)		Facebook (Comments)	
	No features	Model with features	No features	Model with features	No features	Model with features
Goodness of fit	-4646.5510	-4639.7014	-7639.7170	-7626.1090	-9158.7640	-9144.9040

Table EC.9 Empirical Models without Image Features

Variables	Instagram likes	Facebook shares	Facebook comments
time	-0.0007** (0.0200)	-4.7600×10^{-5} (0.8560)	-0.0022*** (0.0000)
saturation	-0.0002 (0.8010)	-0.0019** (0.0370)	-0.0018** (0.0130)
log(wc)	0.6624*** (0.0000)	-0.3174*** (0.0030)	-0.7322*** (0.0000)
post-interval	-0.0708*** (0.0000)	-0.0757* (0.0740)	-0.0828** (0.0100)
revenue	0.3424*** (0.0000)	0.1162*** (0.0000)	0.3008*** (0.0000)
posts (page active for Facebook)	0.0012*** (0.0000)	-0.0007*** (0.0000)	0.0004*** (0.0080)
Observations	542	2,191	2,191
Log likelihood	-4646.5510	-7639.7170	-9158.7640
p values in parentheses			
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$			

We follow the discussions from the traditional advertising literature where Petty et al. (1986) emphasize that users have high and low involvement levels with advertising posts and central routes may induce high involvement compared to low involvement via peripheral routes. High involvement enables users to exert more cognitive effort for evaluation, and their attitude becomes a function of information processing. Wojdyski and Bang (2016) show that users pay more attention to relevant advertisements. In our context, we study the impact of image features on user engagement. Users process an advertising image with various objects, which can influence their attitude and, in turn, their level of engagement with social media posts. These objects in an advertising image influence users via the central route of the ELM framework. To understand the effects on user engagement in our study context, we run econometric models without primary and secondary feature variables and compare the goodness-of-fit measures of our models with and without features for all three platforms in Table EC.8 and the coefficients in Table EC.9.

The goodness-of-fit results and the statistically significant coefficients of the linear and quadratic terms for secondary features collectively demonstrate the importance of these features in influencing user engagement. We can conclude that image features impact user attitudes in terms of their social

media engagement behavior. The influence is visible across different social media platforms and types of engagement.

It is also important to discuss whether the deep learning algorithms actually reflect human perceptions of measures, since we utilize these algorithms to extract image features. There has been a perception that these advanced analytic tools are inferior to human performance (Dai and Tayir 2022). However, the recognition performance of deep learning algorithms has improved over time, given more availability of training data and better generalizations (Goodfellow et al. 2013). He et al. (2015) show that these algorithms can match or even surpass human performance in several image recognition tasks. Dodge and Karam (2017) find that deep learning algorithms' recognition performance is at par with humans on good quality images. While these detections reflect human evaluation in general, which is an ongoing research area, significant improvements are being made in image recognition through the continuous dialogue between artificial intelligence (AI) and neuroscience communities. Shin et al. (2020) show that human coders rank images similarly to those done by deep learning algorithms. This series of evidence motivates us to use deep learning algorithms to classify image features into primary or easily identifiable and secondary or miscellaneous features.

EC.1.4.4. Discussion of Empirical Findings. Our empirical analysis reveals that high engagement occurs on Instagram posts that have fewer secondary features. On the other hand, posts on Facebook with higher engagement have more secondary features. We also discuss some additional findings that are interesting in our context. The positive and significant coefficient of the log word count (0.6982, $p = 0.0000$) in Column 3 of Table EC.6 suggests that Instagram posts with more information in the caption exhibit more likes. Another important finding is the negative association of post-interval with both likes (-0.0682 , $p = 0.0000$) and comments (-0.0369 , $p = 0.0080$) on Instagram. These two results suggest synergistic effects between user engagement and how frequently a firm should post, as infrequent postings may hurt both likes and comments. For Facebook, the impact of log word count is negative and significant for both shares (-0.3311 , $p = 0.0040$) and comments (-0.7718 , $p = 0.0000$) as reported in Columns 3 and 6, respectively, in Table EC.7.

We also present the predicted outcomes to show how the shape of the curve, especially the value of the outcome variable at every point of the independent variable, is affected. We present them in Tables EC.10, EC.11, and EC.12 for Instagram likes, Facebook shares, and Facebook comments, respectively. The effect sizes are statistically significant at all values of secondary features for Instagram likes and Facebook comments and most of the values of secondary features for Facebook shares, further supporting our primary empirical insights. Figures EC.7, EC.8, and EC.9 clearly

Table EC.10 Margins: Instagram Likes

secondary features	margin	std.err.	z	p value	95% confidence	interval
0	2445.6490	211.2799	11.5800	0.0000	2031.5480	2859.7500
1	2671.2600	166.0791	16.0800	0.0000	2345.7510	2996.7690
2	2881.3590	141.7061	20.3300	0.0000	2603.6210	3159.0980
3	3069.2910	144.6875	21.2100	0.0000	2785.7080	3352.8730
4	3228.7760	165.0118	19.5700	0.0000	2905.3590	3552.1930
5	3354.2620	188.5683	17.7900	0.0000	2984.6750	3723.8490
6	3441.2440	208.6782	16.4900	0.0000	3032.2420	3850.2450
7	3486.5270	226.1855	15.4100	0.0000	3043.2120	3929.8430
8	3488.4300	247.5958	14.0900	0.0000	3003.1510	3973.7090
9	3446.8810	281.8018	12.2300	0.0000	2894.5590	3999.2020
10	3363.4250	334.7284	10.0500	0.0000	2707.3690	4019.4810
11	3241.1300	406.0063	7.9800	0.0000	2445.3730	4036.8880
12	3084.3990	490.3779	6.2900	0.0000	2123.2760	4045.5220
13	2898.7040	580.6525	4.9900	0.0000	1760.6460	4036.7620
14	2690.2730	669.5648	4.0200	0.0000	1377.9500	4002.5960

indicate the nonlinear pattern of engagement across the two platforms and engagement types with higher secondary features.

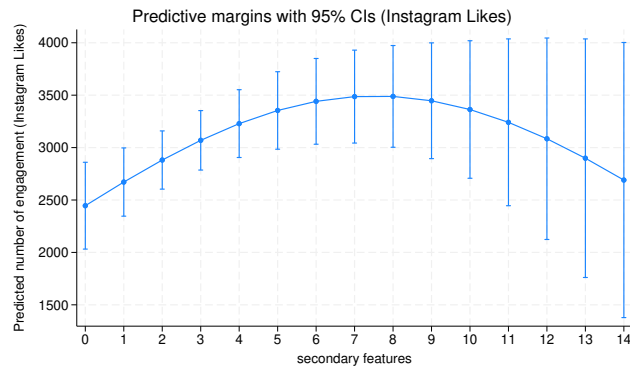
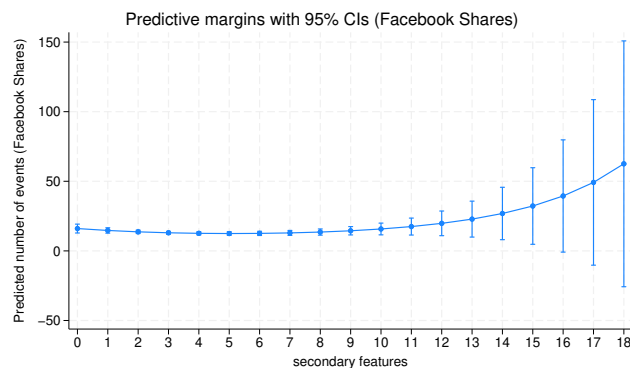
Figure EC.7 Predicted Margins (Instagram Likes)**Figure EC.8 Predicted Margins (Facebook Shares)**

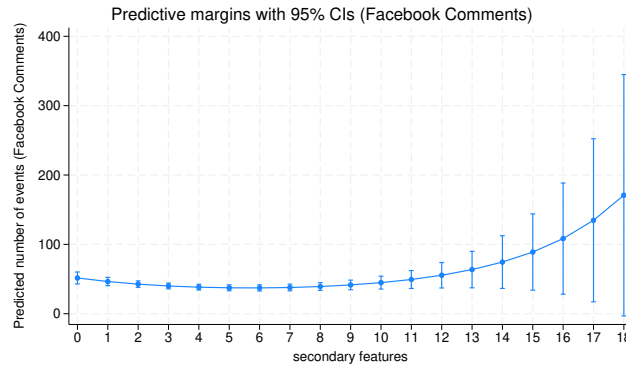
Table EC.11 Margins: Facebook Shares

secondary features	margin	std.err.	z	p value	95% confidence	interval
0	15.9276	1.6080	9.9100	0.0000	12.7759	19.0793
1	14.6918	0.9913	14.8200	0.0000	12.7488	16.6348
2	13.7080	0.6353	21.5800	0.0000	12.4629	14.9532
3	13.0204	0.5358	24.3000	0.0000	11.9703	14.0705
4	12.6080	0.5948	21.2000	0.0000	11.4421	13.7738
5	12.4501	0.6926	17.9800	0.0000	11.0926	13.8076
6	12.5384	0.7987	15.7000	0.0000	10.9729	14.1038
7	12.8782	0.9375	13.7400	0.0000	11.0407	14.7157
8	13.4901	1.1631	11.6000	0.0000	11.2104	15.7698
9	14.4120	1.5459	9.3200	0.0000	11.3822	17.4419
10	15.7030	2.1633	7.2600	0.0000	11.4630	19.9430
11	17.4497	3.1098	5.6100	0.0000	11.3545	23.5449
12	19.7761	4.5174	4.3800	0.0000	10.9222	28.6301
13	22.8582	6.5850	3.4700	0.0010	9.9519	35.7646
14	26.9458	9.6171	2.8000	0.0050	8.0967	45.7950
15	32.3958	14.0825	2.3000	0.0210	4.7945	59.9971
16	39.7223	20.7086	1.9200	0.0550	-0.8658	80.3104
17	49.6738	30.6341	1.6200	0.1050	-10.3679	109.7155
18	63.3532	45.6615	1.3900	0.1650	-26.1416	152.8480

Table EC.12 Margins: Facebook Comments

secondary features	margin	std.err.	z	p value	95% confidence	interval
0	50.1553	4.3807	11.4500	0.0000	41.5694	58.7412
1	46.3807	3.0921	15.0000	0.0000	40.3203	52.4411
2	42.7288	2.3910	17.8700	0.0000	38.0425	47.4150
3	39.9442	2.1394	18.6700	0.0000	35.7509	44.1374
4	38.0743	2.1193	17.9700	0.0000	33.9205	42.2281
5	37.0503	2.1857	16.9500	0.0000	32.7664	41.3342
6	36.8189	2.2997	16.0100	0.0000	32.3116	41.3261
7	37.3681	2.4983	14.9600	0.0000	32.4716	42.2647
8	38.7341	2.8760	13.4700	0.0000	33.0973	44.3710
9	41.0062	3.5692	11.4900	0.0000	34.0107	48.0016
10	44.3373	4.7382	9.3600	0.0000	35.0506	53.6241
11	48.9614	6.6058	7.4100	0.0000	36.0142	61.9086
12	55.2208	9.8834	5.5900	0.0000	35.8496	74.5919
13	63.6086	13.4217	4.7400	0.0000	37.3026	89.9146
14	74.8330	19.4359	3.8500	0.0000	36.7393	112.9267
15	89.9156	28.3253	3.1700	0.0020	34.3991	145.4321
16	110.3422	41.5700	2.6500	0.0080	28.8664	191.8179
17	138.2968	61.5097	2.2500	0.0250	17.7399	258.8537
18	177.0301	91.8842	1.9300	0.0540	-3.0596	357.1198

EC.1.4.5. Robustness Checks. We conducted several robustness checks to ensure the validity of our empirical findings. However, first, we conduct a quantile regression analysis to observe the engagement levels across various quantiles. Next, we also conduct tests using Mixed-Effects and Generalized Negative Binomial (GNB) models and present the main findings that support our empirical results. Finally, we apply the Hurdle model (Instagram comments and Facebook shares

Figure EC.9 Predicted Margins (Facebook Comments)

and comments), the Negative Binomial with clustered-robust errors, and the standard Negative Binomial to test our empirical findings. We elaborate on them next.

Quantile regression is particularly useful for analyzing variables like social media engagement, where the relationship between predictors and the outcome may vary across different points in the distribution (Aldous et al. 2019). It allows researchers to analyze engagement values at different quantiles. An analysis of the data at different quantiles can also help firms in their decision-making (Mendelson and Moon 2018). We are also interested in analyzing the engagement at different quantiles. We select 0.5, 0.7, and 0.9 quantiles for our analysis and present the results in Tables EC.13, EC.14, and EC.15 for Facebook shares, Facebook comments, and Instagram likes. We chose relatively higher quantile values for the engagements since posts at the bottom quantiles may have lower engagement, especially on Facebook. The squared secondary features variable is significant for both shares and comments at 0.7 and 0.9 quantiles, with values almost similar to our main findings. It is not surprising since comments and shares on Facebook posts tend to be low in volume (KWSM 2024). For Instagram, it is statistically significant across all quantiles since Instagram posts attract a substantially higher number of likes. We focus on the mean effect, as it closely follows the practice in the literature and provides ease of interpretation, making it more accessible for managerial relevance.

Social media engagement analysis often requires the consideration of groups in the regression models, which makes Mixed-Effects models appropriate. However, Mixed-Effects models require a reasonably large number of clusters to avoid bias (McNeish and Stapleton 2016). Thus, while conducting the robustness tests using Mixed-Effects models, we treat the firm and the days of the week on which the posts were made to define the grouping structure for our analysis to alleviate the issues regarding small groups. This construction of grouping allows us to create a relatively large number of clusters to run the Mixed-Effects model. We present the results of our main independent

Table EC.13 Quantile Regression Analyses of Facebook Shares

	(1)	(2)	(3)
	0.5	0.7	0.9
time	-0.0002 (0.2950)	-0.0001 (0.2530)	-0.0003 (0.1710)
saturation	-0.0015 (0.1100)	-0.0001 (0.7410)	-0.0002 (0.4830)
log(wc)	-0.8674*** (0.0000)	-0.4741*** (0.0000)	-0.3157*** (0.0000)
primary features	-0.1119*** (0.0000)	-0.1119*** (0.0000)	-0.0627 (0.6980)
post-interval	-0.1082* (0.0500)	-0.0456 (0.1880)	-0.0515** (0.0120)
revenue	0.1720*** (0.0000)	0.1128*** (0.0000)	0.0901*** (0.0000)
page active	-0.0007*** (0.0000)	-0.0009*** (0.0000)	0.0009*** (0.0000)
secondary features	0.0014 (0.9680)	-0.0420 (0.1030)	-0.1587*** (0.0010)
secondary features squared term	0.0014 (0.6400)	0.0030*** (0.0000)	0.0105*** (0.0000)
constant	7.4697*** (0.0000)	7.8646*** (0.0000)	8.2514*** (0.0000)
Observations	2,191	2,191	2,191

p values in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

variables in Table EC.16. We also present the results from the GNB model in Table EC.17, which are qualitatively similar to our main insights. These results are consistent with our main findings, further reinforcing the robustness of our key conclusions. The findings from other robustness tests in tables EC.18, EC.19, and EC.20 also support our main insights.

Recall that the nature of our dependent variables across the platforms is of a count type, which makes the Poisson or Negative Binomial model an appropriate method. We describe the summary statistics of our dependent variables in Table EC.21. We can easily observe the overdispersion of these three count variables; the variances are substantially higher than the means. For Instagram likes, the variance-to-mean ratio is around 3,643, whereas the values for this ratio for Facebook shares and comments are 52 and 110, respectively. In Table EC.22, we present the results of the statistical test for overdispersion. All the dependent variables exhibit significant overdispersion. Overdispersion is a common issue in analyzing social media user engagement using metrics such as

Table EC.14 Quantile Regression Analyses of Facebook Comments

	(1)	(2)	(3)
	0.5	0.7	0.9
time	-0.0023*** (0.0000)	-0.0023*** (0.0000)	-0.0024*** (0.0000)
saturation	-0.0013 (0.1750)	-0.0006 (0.3720)	-0.0008 (0.1110)
log(wc)	-1.4216*** (0.0000)	-1.0814*** (0.0000)	-0.7039*** (0.0000)
primary features	-0.0329 (0.1430)	-0.0064 (0.9040)	-0.0091 (0.7520)
post-interval	-0.0769** (0.0500)	-0.1017** (0.0026)	-0.0546 (0.3750)
revenue	0.2869*** (0.0000)	0.2909*** (0.0000)	0.2883*** (0.0000)
page active	0.0001 (0.8380)	0.0002 (0.7380)	0.0002 (0.1790)
secondary features	-0.0368 (0.4140)	-0.0605 (0.0080)	-0.1283*** (0.0000)
secondary features squared term	0.0020 (0.6440)	0.0058* (0.0390)	0.0094** (0.0040)
constant	7.3500*** (0.0000)	6.0865*** (0.0000)	5.4000*** (0.0000)
Observations	2,191	2,191	2,191

p values in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

likes, comments, or shares. Researchers use a Negative Binomial approach to model user engagement (Rietveld et al. 2020). Studies show that modeling it under Poisson may lead to misleading conclusions (Gardner et al. 1995), which can be avoided by using a Negative Binomial. We also find that the model fit (log likelihood) for Instagram likes is -374236.5900, which is substantially lower than our main model (Hurdle) fit value of -4639.7014. The results of all robustness tests (see Table EC.31 for a summary of the findings) are consistent with our main results, reinforcing the suitability of a Negative Binomial assumption over the Poisson model for our analysis. Thus, we have selected the models under the Negative Binomial assumption over Poisson, which is also suggested in the literature (Cameron and Trivedi 2013).

Additionally, we also conduct our econometric analyses using the standard Negative Binomial method. In Tables EC.18, EC.19, and EC.20, we present the results for the standard Negative Binomial, which are qualitatively similar to our main insights.

Table EC.15 Quantile Regression Analyses of Instagram Likes

	(1) 0.5	(2) 0.7	(3) 0.9
time	-0.0005*** (0.0010)	-0.0003*** (0.0000)	-0.0006*** (0.0000)
saturation	-0.0012*** (0.0000)	-0.0008*** (0.0070)	0.0004*** (0.0000)
log(wc)	0.6011*** (0.0000)	0.8481*** (0.0000)	0.7550*** (0.0000)
primary features	-0.0256*** (0.0000)	-0.0631*** (0.0000)	-0.0449*** (0.0000)
post-interval	-0.0612*** (0.0000)	-0.0978*** (0.0000)	-0.0533*** (0.0000)
revenue	0.4019*** (0.0000)	0.3642*** (0.0000)	0.2768*** (0.0000)
posts	.0011*** (0.0000)	.0013*** (0.0000)	.0014*** (0.0000)
secondary features	0.1345*** (0.0000)	0.0967*** (0.0080)	0.0649*** (0.0000)
secondary features squared term	0.1345*** (0.0000)	-0.0072*** (0.0000)	-0.0649*** (0.0000)
constant	1.5843*** (0.0000)	1.1802*** (0.0000)	1.8638*** (0.0000)
Observations	542	542	542

p values in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$ **Table EC.16 Mixed-Effects Model Results for Primary Independent Variables**

Variables	Instagram likes	Facebook shares	Facebook comments
secondary features	0.0750*** (0.0000)	-0.0803* (0.0700)	-0.1046*** (0.0060)
secondary features squared term	-0.0045*** (0.0000)	0.0074* (0.0880)	0.0089*** (0.0080)
Observations	542	2,191	2,191

p values in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$ **Table EC.17 Generalized Negative Binomial Model Results for Primary Independent Variables**

Variables	Instagram likes	Facebook shares	Facebook comments
secondary features	0.1022*** (0.0050)	-0.0860** (0.0350)	-0.1179*** (0.0000)
secondary features squared term	-0.0068** (0.0350)	0.0082* (0.0360)	0.0109*** (0.0010)
Observations	542	2,191	2,191

p values in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table EC.18 Robustness Tests for Instagram Likes (Negative Binomial, Hurdle model, and Negative Binomial (Clustered robust))

Variables	Negative Binomial	Hurdle model	Negative Binomial (Clustered-robust)
time	-0.0009*** (0.0040)	-0.0009*** (0.0040)	-0.0009 (0.1050)
saturation	-0.0001 (0.9500)	-0.0001 (0.9360)	-0.0001 (0.9410)
log(wc)	0.6980*** (0.0000)	0.6982*** (0.0000)	0.6982*** (0.0000)
post-interval	-0.0680*** (0.0080)	-0.0682*** (0.0000)	-0.0682** (0.0340)
revenue	0.3470*** (0.0000)	0.3472*** (0.0000)	0.3472*** (0.0040)
posts	0.0012*** (0.0000)	0.0016*** (0.0000)	0.0016*** (0.0020)
primary features	-0.0340 (0.1240)	-0.0348 (0.1000)	-0.0348** (0.0330)
secondary features	0.0940*** (0.0080)	0.0945*** (0.0040)	0.0945*** (0.0260)
secondary features squared term	-0.0062** (0.0460)	-0.0063** (0.0350)	-0.0063** (0.0350)
Observations	542	542	542
p values in parentheses			
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$			

EC.1.4.6. Tests for Instagram Comments. We perform several robustness tests to confirm our primary finding on the relationship between secondary features and Instagram comments. In Table EC.24, we present and compare the results. Each of these tests indicates that the quadratic terms for secondary features are statistically insignificant, thereby validating our primary finding.

One plausible reason behind the insignificance of the quadratic term for secondary features on Instagram comments may relate to the type of engagement. Commenting is a different form of engagement than liking since it requires more effort from the users. On the other hand, the coefficients for both the linear and quadratic terms of secondary features are statistically significant for Facebook. The number of comments on a Facebook post reduces initially as the number of secondary features increases and then it gradually increases after certain thresholds. In contrast to secondary features, we find that the primary features variable is negatively associated with comments, suggesting that posts on Instagram with too many primary features do not receive higher comments, on average. However, the impact of primary features on Facebook comments is not statistically significant. The differences in the engagement types also exhibit the different mechanisms of user interactions across different platforms.

Table EC.19 Robustness Tests for Facebook Shares (Negative Binomial, Hurdle model, Negative Binomial (Clustered-robust), and ZINB)

Variables	Negative Binomial	Hurdle model	Negative Binomial (Clustered-robust)	ZINB
time	-0.0001 (0.8920)	0.0001 (0.3960)	-0.0001 (0.9250)	-3.1200×10^{-5} (0.8980)
saturation	-0.0019** (0.0190)	-0.0011 (0.0990)	-0.0019*** (0.0000)	-0.0019** (0.0200)
log(wc)	-0.3322*** (0.0030)	-0.0209 (0.6880)	-0.3322** (0.0140)	-0.3311*** (0.0040)
post-interval	-0.0079** (0.0310)	-0.0376 (0.1310)	-0.0797* (0.0530)	-0.0797** (0.0320)
revenue	0.1090*** (0.0000)	0.0564*** (0.0000)	0.1094** (0.0030)	0.1089*** (0.0000)
page active	-0.0007*** (0.0000)	-0.0009*** (0.0000)	-0.0008*** (0.0010)	-0.0008*** (0.0000)
primary features	-0.0500 (0.2090)	-0.0424*** (0.0040)	-0.0502 (0.1170)	-0.0500 (0.2130)
secondary features	-0.0988** (0.0280)	-0.1159*** (0.0000)	-0.0988* (0.0820)	-0.1012** (0.0290)
secondary features squared term	0.0090** (0.0200)	0.0110*** (0.0000)	0.0097* (0.0570)	0.0098** (0.0200)
Observations	2,191	2,191	2,191	2,191

p values in parentheses

 $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

EC.1.4.7. Selection of the Regression Models. We have selected our empirical model from both theoretical perspectives and empirical results. The narrative framing theory emphasizes the importance of a narrative around the primary items, which motivates us to test the nonlinear relationship between user engagement and secondary features. We also analyze it from the modeling perspective by analyzing the goodness of fit. As an example, we test five different models to check the goodness of fit, as presented in Table EC.25 and present the appropriate empirical model for our optimization framework. Note that a high log likelihood value represents a better fit. We incrementally add the variables and check the goodness of fit. Model (5), with both linear and quadratic terms for both primary features and secondary features variables, does not add substantial value compared to Model (4), which does not have the quadratic term for primary features. It shows that Model (4) is reasonable in our context. We discuss the result for the squared term for the primary features

Table EC.20 Robustness Tests for Facebook Comments (Negative Binomial, Hurdle model, Negative Binomial (Clustered-robust), and ZINB)

Variables	Negative Binomial	Hurdle model	Negative Binomial (Clustered-robust)	ZINB
time	-0.0021*** (0.0000)	-0.0021*** (0.0000)	-0.0022*** (0.0000)	-0.0022*** (0.0000)
saturation	-0.0016** (0.0180)	-0.0009 (0.1300)	-0.0016*** (0.0000)	-0.0016** (0.0220)
log(wc)	-0.7760*** (0.0000)	-0.4814*** (0.0000)	-0.7760* (0.0700)	-0.7718*** (0.0000)
post-interval	-0.0800*** (0.0080)	-0.0477** (0.0310)	-0.0804*** (0.0060)	-0.0800*** (0.0080)
revenue	0.2970*** (0.0000)	0.2512*** (0.0000)	0.2973*** (0.0000)	0.2956*** (0.0000)
page active	0.0003** (0.0210)	0.0003*** (0.0020)	0.0003 (0.4530)	0.0003** (0.0190)
primary features	0.0036 (0.9050)	0.0169 (0.2480)	0.0036 (0.8830)	0.0055 (0.8550)
secondary features	-0.1140*** (0.0000)	-0.1195*** (0.0000)	-0.1147*** (0.0060)	-0.1224*** (0.0000)
secondary features squared term	0.0100*** (0.0010)	0.0104*** (0.0000)	0.0101*** (0.0050)	0.0106*** (0.0000)
Observations	2,191	2,191	2,191	2,191

p values in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table EC.21 Summary Statistics of Primary Dependent Variables

Platform (engagement type)	Obs	Mean	Std. dev.	Min	Max
Instagram (Likes)	542	2888.7600	3243.9750	96	19815
Facebook (Shares)	2,191	13.6029	26.6070	0	388
Facebook (Comments)	2,191	35.8384	63.0630	0	573

Table EC.22 Presence of Overdispersion

Platform (engagement type)	Coefficient	Std. error	t	P value	95% confidence interval
Instagram (Likes)	0.4411	0.0627	7.0400	0.0000	0.3180 0.5643
Facebook (Shares)	2.2921	0.7096	3.2300	0.0010	0.9006 3.6837
Facebook (Comments)	1.2331	0.1503	8.2000	0.0000	0.9383 1.5279

Table EC.23 Poisson Results

Variables	Instagram likes	Facebook shares	Facebook comments
time	-0.0002 (0.4750)	0.0001 (0.06820)	-0.0020*** (0.0000)
saturation	-0.0001 (0.9370)	-0.0015* (0.0940)	-0.0015 (0.9410)
log(wc)	0.3699*** (0.0000)	-0.2419*** (0.0000)	-0.3561*** (0.0000)
post-interval	-0.0528* (0.0960)	-0.0611 (0.1850)	-0.0033 (0.9160)
revenue	0.3193*** (0.0000)	0.0835*** (0.0000)	0.2339*** (0.0000)
posts (page active for Facebook)	0.0010*** (0.0000)	-0.0009*** (0.0000)	0.0002* (0.0710)
primary features	-0.0155 (0.5600)	-0.7160 (0.2780)	0.0200 (0.4620)
secondary features	0.0626* (0.0870)	-0.1016*** (0.0060)	-0.0777** (0.0100)
secondary features squared term	-0.0041 (0.1770)	0.0080*** (0.0040)	-0.0066** (0.0100)
Observations	542	2,191	2,191
p values in parentheses			
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$			

in Table EC.26. The coefficient for the squared term for primary features is not statistically significant for Instagram likes. This supports our main empirical model, where we test the nonlinear relationship between secondary features and engagement.

EC.1.4.8. Additional Checks for Endogeneity Concerns. Our empirical analysis may be subject to endogeneity due to the observational nature of our data. The decision to include additional features in a post could be a deliberate choice by the post developers, potentially correlating with other control variables. To address endogeneity concerns, particularly those related to selection bias, we conduct additional tests to validate the robustness of our main findings. First, we follow a Propensity Score Matching process or PSM (Rosenbaum and Rubin 1984) by categorizing posts with high and low features (a binary variable based on the median value of secondary features) and use a matched sample in the second stage of the regression to correct any selection bias. Second, we use the Heckman model to further validate the robustness of our findings by addressing selection due to unobservable factors.

- **Propensity Score Matching (PSM):** We create a binary variable (high_features) to classify whether a post has high or low secondary features for each platform by considering the median number of secondary features. A value of 1 is assigned if the number of secondary features

Table EC.24 Robustness Tests for Instagram Comments

Variables	Negative Binomial	Negative Binomial (Clustered-robust)	Hurdle	ZINB
time	-0.0021*** (0.0000)	-0.0021** (0.0280)	-0.0022*** (0.0000)	-0.0021*** (0.0000)
saturation	0.0008 (0.3250)	0.0008 (0.2810)	0.0009 (0.2410)	0.0008 (0.2670)
log(wc)	0.6817*** (0.0000)	0.6817*** (0.0000)	0.6609*** (0.0000)	0.6817*** (0.0000)
post-interval	-0.0369*** (0.0080)	-0.0369*** (0.0000)	-0.0344** (0.0360)	-0.0369** (0.0210)
revenue	0.4271*** (0.0000)	0.4271*** (0.0000)	0.4391*** (0.0000)	0.4271*** (0.0000)
posts	0.0017*** (0.0000)	0.0017*** (0.0000)	0.0016*** (0.0000)	0.0017*** (0.0000)
primary features	-0.0543** (0.0270)	-0.0543** (0.0213)	-0.0513** (0.0340)	-0.0543** (0.0230)
secondary features	-0.0539 (0.1880)	-0.0539** (0.0270)	-0.0625* (0.0930)	-0.0539 (0.1480)
secondary features squared term	0.0042 (0.2580)	0.0042 (0.1040)	0.0046 (0.1780)	0.0042 (0.2222)
Observations	542	542	542	542
p values in parentheses				
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$				

Table EC.25 Goodness of Fit of Different Models on Instagram Data

Models	Log likelihood value
Without any feature variables (1)	-4646.5481
With primary features only (2)	-4645.4705
With both primary and secondary features variables (linear terms) (3)	-4641.8303
With linear terms for primary and secondary features and a quadratic term for secondary features (4)	-4639.7014
With linear and quadratic terms for primary and secondary features (5)	-4639.5808

is higher than the median and 0 otherwise. Thus, we perform (i) a logistic regression (logit) model to estimate the probability of treatment (high features) and (ii) a matching process in which each treated unit is matched with one control unit using the nearest neighbor process. We then execute our regression analysis on the matched sample and present the results in Table EC.27. The results show that our main insights continue to hold. We also present the

Table EC.26 Results with Quadratic Term of Primary Features for Instagram Likes

Platform(engagement type)	Variables	Coefficient
Instagram(Likes)	primary features	-0.0542 (0.2710)
	primary features squared term	0.0036 (0.5980)
	secondary features	0.0963*** (0.0080)
	secondary features squared term	-0.0064** (0.0420)
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$		

Table EC.27 PSM Results

Variables	Instagram likes	Facebook shares	Facebook comments
secondary features	0.0934** (0.0200)	-0.1701*** (0.0040)	-0.1376*** (0.0000)
secondary features squared term	-0.0070** (0.0410)	0.0158*** (0.0040)	0.0122*** (0.0000)
Observations	542	2,191	2,191
p values in parentheses * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$			

Table EC.28 Differences in Mean Before and After Matching (Instagram Likes)

Variable	Mean Treated	Control	t-test %bias	t	$p > t $	V (T)/ V(C)
time	178.0900	157.9800	16.9000	1.9200	0.0550	1.0300
saturation	56.1600	56.692	-1.1000	-0.1300	0.9000	0.8100
log(wc)	3.3015	3.2754	3.7000	0.4300	0.6650	0.8300
post-interval	2.2129	2.3384	-6.5000	-1.0100	0.3110	1.0800
revenue	3.2458	3.1503	5.7000	0.6800	0.4990	1.0600
posts	2098.8000	2056.2000	8.6000	0.9200	0.3590	0.9300

Table EC.29 Differences in Mean Before and After Matching (Facebook Shares and Comments)

Variable	Mean Treated	Control	t-test %bias	t	$p > t $	V (T)/ V(C)
time	365.3200	368.2900	-1.2000	-0.2700	0.7850	1.0800
saturation	58.4450	57.9080	1.2000	0.2800	0.7830	0.9200
log(wc)	3.4488	3.4519	-0.5000	-0.1200	0.9080	0.8500
post-interval	1.2677	1.1943	5.8000	1.3500	0.1760	1.1200
revenue	3.8919	3.7610	4.5000	1.0400	0.3000	1.0700
page_active	3968.2000	3986.2000	-6.2000	-1.4000	0.1610	1.0400

difference in means in Table EC.28 and Table EC.29. We see that, except for the variable time for Instagram likes, no other control variable shows any statistically significant difference.

- **Heckman Model:** We also perform the Heckman model (Certo et al. 2016) to observe and correct if there is any selection bias in our data corresponding to the secondary features and

Table EC.30 Heckman Model Results with Inverse Mills Ratio (IMR)

Variables	Instagram likes	Facebook shares	Facebook comments
IMR	-1.1753 (0.7710)	-13.4264 (0.2080)	-29.0849*** (0.0020)
secondary features	0.0943*** (0.0090)	-0.1048** (0.0260)	-0.1322*** (0.0000)
secondary features squared term	-0.0063** (0.0470)	0.0101** (0.0180)	0.0113*** (0.0000)
Observations	542	2,191	2,191
p values in parentheses			
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$			

engagement types due to unobserved factors. We use a Probit model in the first stage with the higher than median variable as the dependent variable to estimate the Inverse Mills Ratio (IMR). IMR captures the selection bias. In the second stage, we use this IMR as a covariate in the Negative Binomial model, which helps control for any selection bias. We present the results in Table EC.30. Note that there is no selection issue for Instagram likes and Facebook shares (a statistically non-significant coefficient for IMR). However, a statistically significant IMR indicates that selection bias exists, which we observe for Facebook comments, and the Heckman correction has adjusted for it. The adjusted coefficients are qualitatively similar to our main results.

While the two tests validate the robustness of our findings by showing qualitatively similar results, in the absence of a randomized experiment, we cannot entirely dismiss the possibility that selection is influenced by unobserved variables. Therefore, we suggest that the readers interpret our results with this limitation in mind. However, in our context, since our focus is on maximizing engagement by varying the image features, endogeneity may have less relevance. We summarize the coefficients of our main independent variables across all the robustness tests in Table EC.31. They are all qualitatively similar to our main insights.

EC.1.4.9. Clustering. We analyze the image features present on social media image posts. These features are used to advertise products, which can also be considered as clusters of products. To analyze this clustering aspect of features, we adopt two approaches. First, we identify specific features from these images. Often, these images can be grouped into distinct clusters based on the presence of certain features (e.g., the presence of people). Such analyses (e.g., the presence of a human face in images in Li and Xie (2020)) exist in the literature. We group images into two distinct clusters based on the presence (1) or absence (0) of persons and plants in the primary features

Table EC.31 Summary of Key Variables across Robustness Tests

Variable	Model	Instagram likes	Facebook shares	Facebook comments
Secondary features	Generalized NB	0.1022***	-0.0860**	-0.1179***
	ZINB		-0.1012**	-0.1224***
	Mixed-Effects	0.0750***	-0.0803*	-0.1046**
	Hurdle	0.0945***	-0.1159***	-0.1195***
	Negative Binomial (clustered-robust)	0.0945**	-0.0988*	-0.1147***
	Negative Binomial	0.0940***	-0.0988**	-0.1140***
	PSM	0.0934**	-0.1701***	-0.1376***
	Heckman Model	0.0943***	-0.1048**	-0.1322***
Squared secondary features	Generalized NB	-0.0068**	0.0082**	0.0109***
	ZINB		0.0098**	0.0106***
	Mixed-Effects	-0.0045***	0.0074*	0.0089***
	Hurdle	-0.0063**	0.0110***	0.0104***
	Negative Binomial (clustered-robust)	-0.0063**	0.0097*	0.0101***
	Negative Binomial	-0.0062**	0.0090**	0.0100***
	PSM	-0.0070**	0.0158***	0.0122***
	Heckman Model	-0.0063**	0.0101**	0.0113***

p values in parentheses
 * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

set. Note that two different binary variables are created for these two features. We chose these two features as they are quite regular in the furniture image posts. Next, we incorporate another set of clustering approaches – whether a post can be classified into the living room, bedroom, or other categories. This is particularly relevant in our context, as we analyze data from furniture companies.

In Table EC.32, we present the results of our analysis when image posts are identified with the presence of persons and plants in them. While the coefficients for the secondary features and its quadratic term are similar in nature to our main results, the coefficients for person and plant across the engagement types and platforms are interesting. The coefficient for person (-0.5136, $p=0.0000$) is negative and significant for Instagram likes. Thus, posts with people receive fewer likes on average on Instagram in our case context. We do not observe such a negative association on Facebook. While we do not observe any significant impact of these two variables on Facebook comments, posts with plants receive fewer shares (-0.1778, $p=0.0570$) than others, on average. These results further emphasize the importance of analyzing separate platforms individually, as the current analysis also reveals variations in their user preferences.

Table EC.33 presents the results of our three-cluster approach, where we assign a cluster ID 1, 2, or 3 to a post depending on whether it shows a living room, bedroom, or other, respectively. Our main insights continue to hold even with the addition of the cluster variable. The coefficients associated with the clusters are not statistically significant for Instagram likes or Facebook comments. This is not surprising, as users may not have a particular interest in any specific type of furniture post. However, the coefficients are statistically significant for Facebook shares. Cluster 2 has a negative coefficient (-0.3441, $p = 0.0010$), and Cluster 3 has a positive coefficient (0.1643, $p = 0.0660$)

Table EC.32 Cluster with Person and Plants

Variables	Instagram likes	Facebook shares	Facebook comments
secondary features	0.1090*** (0.0030)	-0.0786* (0.0720)	-0.1293*** (0.0000)
secondary features squared term	-0.0070** (0.0280)	0.0084** (0.0320)	0.0110*** (0.0000)
person	-0.5136*** (0.0000)	-0.0458 (0.7760)	0.0461 (0.7050)
plant	-0.0319 (0.7020)	-0.1778* (0.0570)	0.0440 (0.5620)
Observations	542	2,191	2,191
p values in parentheses * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$			

Table EC.33 Three Cluster Approach

Variables	Instagram likes	Facebook shares	Facebook comments
secondary features	0.0967*** (0.0060)	-0.0981** (0.0290)	-0.1238*** (0.0000)
secondary features squared term	-0.0063** (0.0430)	0.0091** (0.0240)	0.0107*** (0.0010)
cluster 2	0.1450 (0.3770)	-0.3441*** (0.0010)	-0.0777 (0.4900)
cluster 3	-0.0128 (0.9120)	0.1643* (0.0660)	-0.0860 (0.3010)
Observations	542	2,191	2,191
p values in parentheses * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$			

relative to Cluster 1. These two coefficients indicate that users share other types of posts more than living room or bedroom types.

EC.1.4.10. Dimensionality Reduction via Removal of Common Features. We run a separate empirical analysis after reducing the dimensionality of the feature space by removing the three most commonly occurring objects from both primary features and secondary features. These three objects are chairs, persons, and plants. Removing these features is an interesting extension of our current framework, as it helps eliminate the most frequently repeating features in the dataset. It follows the dimensionality reduction approach mostly followed in the text analysis methods, where the most frequent words are often ignored. Table EC.34 presents the results. We see that the results for Facebook shares and comments qualitatively align with our main findings. However, for Instagram, the coefficient for the squared term of secondary features is no longer statistically significant, while the coefficient for the linear term is still statistically significant. This result is not surprising. Originally, the association between secondary features and likes showed a concave pattern. However, removing

Table EC.34 Econometric Analysis after Dimension Reduction via Features Removal

Variables	Instagram likes	Facebook shares	Facebook comments
secondary features	0.0756* (0.0680)	-0.2133** (0.0010)	-0.2216*** (0.0000)
secondary features squared term	-0.0037 (0.3980)	0.0179*** (0.0040)	0.0166*** (0.0020)
Observations	542	2,191	2,191
p values in parentheses * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$			

observations has disproportionately affected the quadratic factor. For Facebook, the removal did not affect the convex pattern substantially.

EC.1.5. Analyzing Differences Across Platforms on Engagement Levels

In this subsection, we observe how the engagement levels are different across platforms by conducting two analyses. First, we (1) conduct an analysis of the engagement levels on similar posts across platforms and (2) perform a platform-level analysis where we code the two platforms as a binary variable and observe its impact on a common engagement type (comments).

First, we observe the correlation between similar engagement types across the platforms by considering the same posts. This analysis allows us to observe how users engage with posts across different platforms. We find that the correlation between Facebook comments and Instagram comments is 0.8543. It suggests that those posts resonate similarly with users who comment on Facebook and Instagram. To understand the phenomenon better, we also extracted posts from the platform (formerly Twitter) and analyze the identical posts across platforms, focusing on Facebook shares, X retweets, Instagram likes, and X favorites. Since X retweets and favorites are quite similar in nature to Facebook shares and Instagram likes, this comparison provides meaningful insights into engagement patterns across platforms. We find that the correlation values between X retweets and Facebook shares is 0.3268 and X favorites and Instagram likes is 0.3577. It suggests that users may not engage similarly with the same posts across different platforms. This analysis highlights how users may engage similarly or differently with the same posts across various platforms, further underscoring the value of our engagement maximization approach across platforms.

Next, we proceed by analyzing a platform model for comments (Instagram and Facebook) using a binary variable coded as 0 for Instagram and 1 for Facebook. Applying the standard Negative Binomial model, we find that the platform variable is negative and statistically significant (-0.3019, $p = 0.0010$). This result indicates that, on average, the number of comments is lower for Facebook posts compared to Instagram posts. These findings suggest differences in user engagement behavior,

emphasizing that users generally comment more on Instagram posts than on Facebook posts. We also qualitatively found more engagement on Instagram. In the four-platform analysis (EC.2.3), we incorporate user exposure for each platform by leveraging their engagement activity across platforms.

EC.1.6. Discussion of Empirical Results for Optimization Framework

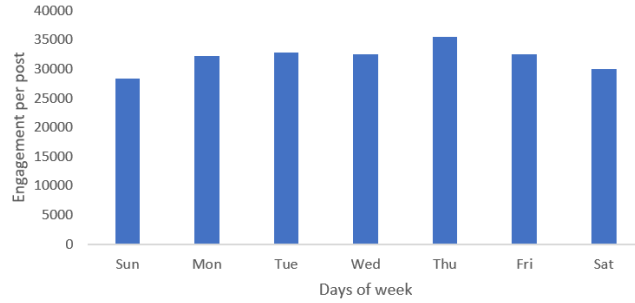
Table EC.35 Parameter Estimation Using Empirical Analysis

Coefficients for	Instagram ($l = 1$)	Facebook ($l = 2$)	
	Likes ($m = 1$)	Shares ($m = 2$)	Comments ($m = 3$)
Constant (α_{lm})	1.6409***	6.7429***	4.6861***
primary features (β_{lm})	-0.0348	-0.0500	0.0055
secondary features (ζ_{lm})	0.0945***	-0.1012**	-0.1224***
Squared secondary features (η_{lm})	-0.0063**	0.0098**	0.0106***
User engagement (x_{lmk})	$x_{1,1,k}$	$x_{2,2,k}$	$x_{2,3,k}$
p values in parentheses			
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$			

In this subsection, we analyze the empirical results for each Instagram and Facebook engagement type and present the insights to be used to develop our optimization framework. We present the empirical summaries in Table EC.35 and discuss their implications. The estimation results from the Instagram analysis (Table EC.35) show that the quadratic impact of the secondary features is negative and significant. The curve bends when $f_{1,1,k} = \lfloor \frac{0.0945}{2|0.0063|} \rfloor + 1 = 8$. Thus, the number of likes on Instagram reduces after the number of secondary features crosses 8 in an image, on average. The relationship between user engagement and secondary features is non-monotonic as the switching point is within the range of secondary features (Table EC.4). The equation for the Instagram likes can be written as follows: $\hat{x}_{1,1,k} = 1.6409 - 0.0348s_{1,1,k} + 0.0945f_{1,1,k} - 0.0063f_{1,1,k}^2 + \theta_1V_{1,1,k} + \gamma_1Z + \nu_1Y_1$.

The coefficient of $f_{1,i,k}$ shows the rate of change in likes on Instagram while other variables are fixed at their means. We can assess the rate of change in the dependent variable $\hat{x}_{1,1,k}$ as we change $f_{1,1,k}$. However, the quadratic relationship tells us that the change is not the same at different levels and the direction of change switches after using 8 secondary features. We present the engagement values near the switching point by following Hayes (2017), where other variables except for primary features and secondary features are fixed at their respective means and added to α_{lm} to construct b_{lm} . For example, at $f_{1,1,k} = 7$, the expected value of $\hat{x}_{1,1,k} = 7.4410 + 0.0945 * 7 - 0.0063 * 7^2 = 7.7956$. Similarly, $\hat{x}_{1,1,k}$ obtains 7.7961 and 7.7842 when $f_{1,1,k}$ is set to 8 and 9, respectively. We show the coefficient values in Table 2. Thus, we can observe a downward trend in likes after the number of

Figure EC.10 Instagram: Engagement Intensity per Post for Different Days of the Week
Interactions per post (Instagram)



secondary features exceeds 8. The number of likes increases initially as the number of secondary features increases and starts to reduce after $f_{1,1,k} = 8$.

Table EC.35 shows that the coefficients of the quadratic term are all positive and significant for Facebook shares (0.0098, $p = 0.0200$) and comments (0.0106, $p = 0.0010$). The curves are convex. The curves bend for Facebook shares and comments when the number of secondary features exceeds 6 and 7, respectively. Compared to likes, comments and shares involve a higher level of engagement from the users, allowing them to comment or share to their networks more on the images, particularly the features. Users can express their perspectives on the features of the images. For example, the user can inquire about a couch or a desk displayed in an image. After setting all variables at their mean, we present the equations for Facebook below, except for primary and secondary features.

- Facebook shares: $\hat{x}_{2,2,k} = 2.7067 - 0.0500s_{2,2,k} - 0.1012f_{2,2,k} + 0.0098f_{2,s,k}^2$.
- Facebook comments: $\hat{x}_{2,3,k} = 3.3796 + 0.0055s_{2,3,k} - 0.1224f_{2,3,k} + 0.0106f_{2,3,k}^2$.

EC.2. Optimization Models and Tables

In this section, we present the proofs of our lemmas and theorems with the supporting tables. We report the results of our computational experiments in tabular formats and describe the relevant extensions of our primary models. In Table EC.36, we tabulate the engagement levels at every value of secondary features following Hayes (2017). We do not record the engagement levels beyond $j = 8$ for Instagram likes, as the engagement continues to decrease thereafter.

EC.2.1. Proofs of Lemmas and Theorems

Proof of Lemma 1: It is straightforward since $\beta_{lm} < 0$ for Case 1. Thus, firms keep the minimum primary features required, which is q_k . Q.E.D.

Proof of Lemma 2: Note that $x_k = b + \beta s_k + \zeta f_k + \eta f_k^2$. By taking derivative of x_k with respect to f_k and set equal to zero, we have $\frac{d(x_k)}{d(f_k)} = \zeta + 2\eta f_k = 0$. As f_k is an integer and the second derivative is negative, the result follows. Q.E.D.

Table EC.36 g_j Values for Instagram Likes, Facebook Shares, and Facebook Comments

j	g_j (Instagram Likes)	g_j (Facebook Shares)	g_j (Facebook Comments)
0	7.4952	2.7067	3.3796
1	7.5835	2.6154	3.2678
2	7.6592	2.5437	3.1771
3	7.7224	2.4917	3.1075
4	7.7730	2.4594	3.0590
5	7.8112	2.4468	3.0316
6	7.8368	2.4538	3.0253
7	7.8498	2.4806	3.0401
8	7.8504	2.5270	3.0760
9		2.5931	3.1330
10		2.6789	3.2111
11		2.7844	3.3103
12		2.9095	3.4306
13		3.0543	3.5720
14		3.2189	3.7345
15		3.4031	3.9181
16		3.6069	4.1228
17		3.8305	4.3486
18		4.0737	4.5956

According to Lemma 2, the maximum user engagement x_k^* occurs at either $f_k^* = \lfloor \frac{\zeta_{lm}}{2|\eta_{lm}|} \rfloor = 7$ or $f_k^* = \lfloor \frac{\zeta_{lm}}{2|\eta_{lm}|} \rfloor + 1 = 8$ whichever provides maximum x_k^* . Here $f_k^* = 8$. Note that U is the upper bound on f_k quantifying the effort required to extract secondary features. The following lemma provides an upper bound for U .

Proof of Lemma 3: The result follows from Lemma 2 as x_k is concave function in f_k . Q.E.D.

Proof of Lemma 4: The budget constraint (1) in Problem SMM_s can be expressed as follows:

$$aq_0 + af_0 + eU + C_f \leq B. \text{ This implies that the maximum, } f_0 = \lfloor \frac{(B - C_f - eU - aq_0)}{a} \rfloor. \text{ Q.E.D.}$$

Proof of Theorem 1: Since x_k is integer, constraints (9) and (10) linearize the nonlinear constraint (2) in Problem SMM_s using boolean variables $y_{k,j}$. Since we linearize f_k as $f_k = \sum_{j=0}^{f_k^*} jy_{k,j}$, constraint (4) is rewritten as constraint (12) in $LSMM_s(Case1)$. Similar change is made in constraint (1) that becomes constraint (8) to place $\sum_{j=0}^{f_k^*} jy_{k,j}$ in place of f_k . We use our regression results to compute x_k at different values of f_k . Constraints (11) and (14) remain the same in both problems. Constraint (13) to $LSMM_s(Case1)$ is added since optimal solution x_k^* to $LSMM_s(Case1)$ cannot be more than f_k^* . Suppose $x_k^* = f_k^* + z$ is optimal solution in $LSMM_s(Case1)$ for some k , where z is a positive integer. We set $x_k^* = f_k^*$ in $LSMM_s(Case1)$. This change does not affect any constraint in $LSMM_s(Case1)$. But it increases the objective function value of $LSMM_s(Case1)$ since the optimal value is f_k^* for the concave function $x_k = b + \beta s_k + \zeta f_k + \eta f_k^2$ in Problem SMM_s for a fixed value of s_k (Lemma 2). This contradicts with the fact that $x_k^* = f_k^* + z$ is optimal. Thus, the optimal solution x_k^* to $LSMM_s(Case1)$ cannot be more than f_k^* . Similarly, this statement is also valid for Problem SMM_s .

It has to be noted that each feasible integer solution to Problem $LSMM_s$ is also feasible to Problem $SMM_s(Case1)$ with the same objective function value. Since the optimal solution x_k^* to both problems cannot be more than f_k^* , the optimal solution to Problem $LSMM_s(Case1)$ is also optimal to Problem $SMM_s(Case1)$. Thus, Problem $SMM_s(Case1)$ is equivalent to solving the linear version of Problem $LSMM_s(Case1)$. Q.E.D.

Proof of Lemma 5: Note that $x_k = b + \beta s_k + \zeta f_k + \eta f_k^2$. By taking derivative of x_k with respect to f_k and set equal to zero, we have $\frac{d(x_k)}{d(f_k)} = -|\zeta| + 2\eta f_k = 0$. As f_k is an integer and the second derivative is positive, the result follows. Q.E.D.

For Facebook (shares), we use the following equation: $x_k = b + \beta s_k + \zeta f_k + \eta f_k^2$, where b and η are positive and β and ζ are negative. Using the values from Table 2, we can rewrite the equation as: $x_k = 2.7067 - 0.0500s_k - 0.1012f_k + 0.0098f_k^2$. By definition of the problem, $f_k \leq u_2 = 18$. According to Lemma 5, the minimum user engagement x_k^* occurs at period k (for Facebook (shares)), when either $f_k^* = \lfloor \frac{|\zeta_{lm}|}{2\eta_{lm}} \rfloor = 5$ or $f_k^* = \lfloor \frac{|\zeta_{lm}|}{2\eta_{lm}} \rfloor + 1 = 6$ whichever provides minimum x_k^* .

Proof of Lemma 6: Note that $x_k = b + \beta s_k + \zeta f_k + \eta f_k^2$. By taking derivative of x_k with respect to f_k and set equal to zero, we have $\frac{d(x_k)}{d(f_k)} = -|\zeta| + 2\eta f_k = 0$. As f_k is an integer and the second derivative is positive, the result follows. Q.E.D.

In the case of Facebook (comments), we use the following equation: $x_k = b + \beta s_k + \zeta f_k + \eta f_k^2$, where b , η and β are positive, and ζ is negative. Using the values from Table 2, we can rewrite the equation as: $x_k = 3.3796 + 0.0055s_k - 0.1224f_k + 0.0106f_k^2$. By definition of the problem $f_k \leq u_2 = 18$. According to Lemma 6, the minimum user engagement x_k^* occurs at period k (for Facebook (comments)), when either $f_k^* = \lfloor \frac{|\zeta_{lm}|}{2\eta_{lm}} \rfloor = 5$ or $f_k^* = \lfloor \frac{|\zeta_{lm}|}{2\eta_{lm}} \rfloor + 1 = 6$ whichever provides minimum x_k^* . Also, since $\beta_{lm} > 0$, s_k will be determined as a part of our optimization.

In the main document, we have provided the formulation $LSMM_s(Case1)$ for one platform. Next, we describe the Problem $LSMM_s(Cases2-3)$ for another platform, which is Facebook in our study context. It has similarities with $LSMM_s(Case1)$. However, β , g_j , u_1 , and u_2 are different for Facebook and Instagram. β and g_j are different for Facebook shares and comments. The impact of secondary features on user engagement is linearized in this formulation using constraint EC.2. The primary features and secondary features follow their upper limits, which are different on Facebook compared to Instagram. Under $LSMM_s(Cases2-3)$, the relationship between secondary features and user engagement (both comments and shares) on Facebook is convex, resulting in an initial decrease and then an increase in user engagement as the number of secondary features increases.

Table EC.37 Computational Results for Instagram Likes (Regression Parameters from Table 2 Multiplied by 5000)

Unit effort cost e	20			30			40			50			60			70			80			90		
Content	s_k	f_k	x_k	s_k	f_k	x_k	s_k	f_k	x_k	s_k	f_k	x_k	s_k	f_k	x_k	s_k	f_k	x_k	s_k	f_k	x_k	s_k	f_k	x_k
1	3	7	38728	3	7	38728	3	7	38728	3	7	38728	3	7	38728	3	7	38728	3	6	38678	3	6	38678
2	4	7	38554	4	7	38554	4	7	38554	4	7	38554	4	7	38554	4	7	38554	4	7	38554	4	6	38504
3	2	7	38902	2	7	38902	2	7	38902	2	7	38902	2	7	38902	2	7	38902	2	7	38902	2	7	38902
4	2	7	38902	2	7	38902	2	7	38902	2	7	38902	2	7	38902	2	7	38902	2	7	38902	2	7	38902
5	2	7	38902	2	7	38902	2	7	38902	2	7	38902	2	7	38902	2	7	38902	2	7	38902	2	7	38902
6	1	7	39076	1	7	39076	1	7	39076	1	7	39076	1	7	39076	1	7	39076	1	7	39076	1	7	39076
7	2	8	38902	2	8	38902	2	7	38902	2	7	38902	2	7	38902	2	7	38902	2	7	38902	2	6	38852
8	2	7	38902	2	7	38902	2	7	38902	2	7	38902	2	7	38902	2	7	38902	2	7	38902	2	6	38852
9	4	7	38554	4	7	38554	4	7	38554	4	7	38554	4	7	38554	4	7	38554	4	7	38554	4	6	38554
10	6	8	38206	6	8	38206	6	7	38206	6	7	38206	6	7	38206	6	7	38206	6	7	38206	6	7	38206
11	1	8	39076	1	8	39076	1	7	39076	1	7	39076	1	7	39076	1	7	39076	1	7	39076	1	7	39076
12	2	7	38902	2	7	38902	2	7	38902	2	7	38902	2	7	38902	2	7	38902	2	7	38902	2	7	38902
13	2	8	38902	2	8	38902	2	7	38902	2	7	38902	2	7	38902	2	7	38902	2	7	38902	2	7	38902
14	2	8	38902	2	8	38902	2	7	38902	2	7	38902	2	7	38902	2	7	38902	2	7	38902	2	6	38852
Leftover budget	280			200			220			140			60			50			0			10		
Engagement (weighted)	714161			714161			714161			714161			714161			714161			714111			713901		
U	8			8			8			8			8			7			7			7		

The decision variable U differs for each platform. Therefore, as constraint EC.5 demonstrates, the number of secondary features to be used on Facebook posts is also bounded by U .

Problem $LSMM_s(Cases2-3)$:

$$\mathbf{Max} \quad \Pi_s = \sum_{k=1}^K w_k x_k$$

Subject to:

$$a \sum_{k=1}^K s_k + a \sum_{k=1}^K \sum_{j=0}^{u_2} j y_{k,j} + eU + C_f \leq B \quad (\text{EC.1})$$

$$x_k = \beta s_k + \sum_{j=0}^{u_2} g_j y_{k,j}, \quad \forall k \quad (\text{EC.2})$$

$$\sum_{j=0}^{u_2} y_{k,j} = 1, \quad \forall k \quad (\text{EC.3})$$

$$s_k \leq u_1, \quad \forall k \quad (\text{EC.4})$$

$$\sum_{j=0}^{u_2} j y_{k,j} \leq U, \quad \forall k \quad (\text{EC.5})$$

$$U \leq u_2, \quad \forall k \quad (\text{EC.6})$$

$$s_k \geq q_k, \quad \forall k \quad (\text{EC.7})$$

$$x_k, s_k, U : \text{Integer variable } (\geq 0), \quad \forall k \quad (\text{EC.8})$$

$$y_{k,j} \in \{0, 1\} \quad \forall k; \quad \forall j \quad (\text{EC.9})$$

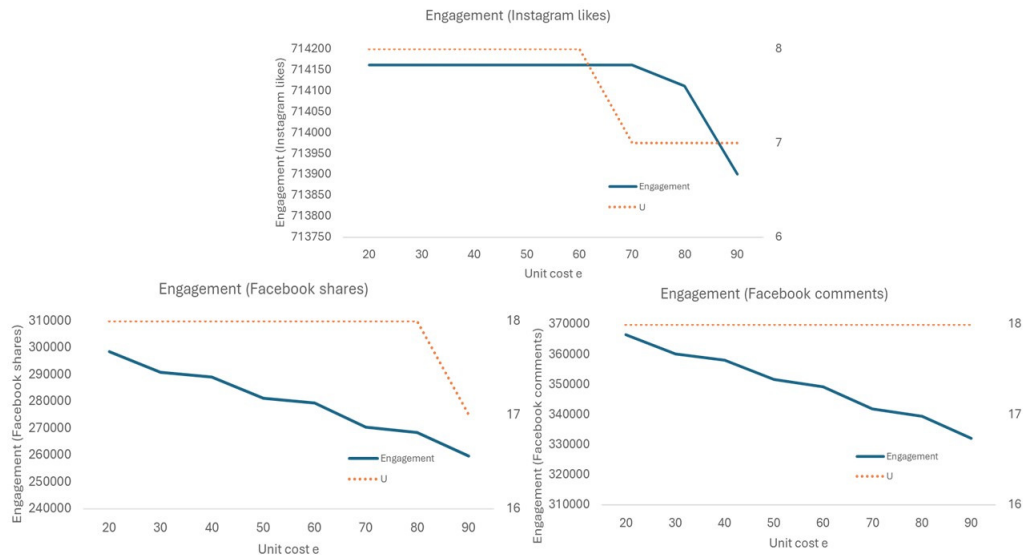
Next, we formally present Problem $LSMM_f$. It represents the combined Facebook shares and comments, where $m = 2$ refers to shares and $m = 3$ denotes comments. This results in a new

Table EC.38 Computational Results for Facebook Shares (Regression Parameters from Table 2 Multiplied by 5000)

Unit effort cost e	20	30	40	50	60	70	80	90
Content	s_k f_k x_k	s_k f_k x_k	s_k f_k x_k	s_k f_k x_k	s_k f_k x_k	s_k f_k x_k	s_k f_k x_k	s_k f_k x_k
1	3 0 12800	3 0 12800	3 0 12800	3 0 12800	3 0 12800	3 0 12800	3 0 12800	3 0 12800
2	4 0 12550	4 0 12550	4 0 12550	4 0 12550	4 0 12550	4 0 12550	4 0 12550	4 0 12550
3	2 18 19850	2 18 19850	2 18 19850	2 18 19850	2 17 18650	2 18 19850	2 18 19850	2 17 18650
4	2 17 18650	2 0 13050	2 17 18650	2 0 13050	2 0 13050	2 0 13050	2 0 13050	2 0 13050
5	2 18 19850	2 18 19850	2 18 19850	2 18 19850	2 18 19850	2 0 13050	2 0 13050	2 17 18650
6	1 0 13300	1 0 13300	1 0 13300	1 0 13300	1 0 13300	1 0 13300	1 0 13300	1 0 13300
7	2 0 13050	2 0 13050	2 0 13050	2 0 13050	2 0 13050	2 0 13050	2 0 13050	2 0 13050
8	2 0 13050	2 0 13050	2 0 13050	2 0 13050	2 0 13050	2 0 13050	2 0 13050	2 0 13050
9	4 0 12550	4 0 12550	4 0 12550	4 0 12550	4 0 12550	4 0 12550	4 0 12550	4 0 12550
10	6 18 18850	6 18 18850	6 18 18850	6 18 18850	6 18 18850	6 18 18850	6 17 17650	6 0 12050
11	1 18 20100	1 0 13300	1 0 13300	1 0 13300	1 0 13300	1 0 13300	1 0 13300	1 0 13300
12	2 18 19850	2 18 19850	2 18 19850	2 18 19850	2 18 19850	2 18 19850	2 18 19850	2 14 15600
13	2 0 13050	2 18 19850	2 0 13050	2 0 13050	2 0 13050	2 0 13050	2 0 13050	2 0 13050
14	2 0 13050	2 0 13050	2 0 13050	2 0 13050	2 0 13050	2 0 13050	2 0 13050	2 0 13050
Leftover budget	0	160	0	160	0	160	0	10
Engagement (weighted)	298650	290810	289130	281290	279370	270410	268490	259770
U	18	18	18	18	18	18	18	17

Table EC.39 Computational Results for Facebook Comments (Regression Parameters from Table 2 Multiplied by 5000)

Unit effort cost e	20	30	40	50	60	70	80	90
Content	s_k f_k x_k	s_k f_k x_k	s_k f_k x_k	s_k f_k x_k	s_k f_k x_k	s_k f_k x_k	s_k f_k x_k	s_k f_k x_k
1	3 0 16982	3 0 16982	3 0 16982	3 0 16982	3 0 16982	3 0 16982	3 0 16982	3 0 16982
2	4 0 17010	4 0 17010	4 0 17010	4 0 17010	4 0 17010	4 0 17010	4 0 17010	4 0 17010
3	2 18 23055	10 18 23275	2 18 23055	2 18 23055	2 18 23055	10 18 23275	2 18 23055	10 18 23275
4	2 17 21805	2 0 16955	2 17 21805	2 0 16955	2 0 16955	2 0 16955	2 0 16955	2 0 16955
5	2 18 23055	2 18 23055	2 18 23055	10 18 23275	2 18 23055	2 18 23055	2 17 21805	2 0 16955
6	1 18 23027	1 18 23027	1 0 16927	1 0 16927	1 0 16927	1 0 16927	1 0 16927	1 0 16927
7	2 0 16955	2 0 16955	2 0 16955	2 0 16955	2 0 16955	2 0 16955	2 0 16955	2 0 16955
8	2 0 16955	2 0 16955	2 0 16955	2 0 16955	2 0 16955	2 0 16955	2 0 16955	2 0 16955
9	4 0 17010	4 0 17010	4 0 17010	4 0 17010	4 0 17010	4 0 17010	4 0 17010	4 0 17010
10	6 18 23165	6 18 23165	6 18 23165	6 18 23165	6 18 23165	6 18 23165	6 18 23165	6 18 23165
11	1 0 16927	1 0 16927	1 0 16927	1 0 16927	1 0 16927	1 0 16927	1 0 16927	1 0 16927
12	2 18 23055	2 18 23055	2 18 23055	2 18 23055	2 17 21805	2 0 16955	2 0 16955	2 0 16955
13	2 0 16955	2 0 16955	2 0 16955	2 0 16955	2 0 16955	2 0 16955	2 0 16955	2 0 16955
14	2 0 16955	2 0 16955	2 0 16955	2 0 16955	2 0 16955	2 0 16955	2 0 16955	2 0 16955
Leftover budget	0	0	0	0	0	0	0	0
Engagement (weighted)	366599	360161	358059	351621	349269	341861	339509	332101
U	18	18	18	18	18	18	18	18

Figure EC.11 Engagement Levels against Unit Cost e and Effort U 

parameter, μ_m , which assigns weights to each engagement type. We use 0.7 and 0.3 for shares and comments, respectively, giving higher weight to shares than comments. Recall that different platforms have different engagement types, which differ based on their value to a firm. In this context, firms usually prefer sharing more than comments, as highlighted in the literature (e.g., Aichner and Jacob 2015). The model can be easily extended for other values of μ_m , where different platforms have different engagement types. β and g_{mj} differ for each engagement type.

In the model, we do not use any platform-specific notation for parameters and variables since the engagement types pertain to the same platform. The total engagement is the sum of engagements from all the engagement types of the platform. Constraints also reflect the setting. We do not have separate upper limits on the primary and secondary features, as well as the required primary features on a post.

Problem $LSMM_f$:

$$\mathbf{Max} \quad \Pi_f = \sum_{m=2}^3 \sum_{k=1}^K \mu_m w_k x_{mk}$$

Subject to:

$$a \sum_{k=1}^K s_k + a \sum_{k=1}^K \sum_{j=0}^{u_2} j y_{k,j} + eU + C_f \leq B \quad (\text{EC.10})$$

$$x_{mk} = \beta_m s_k + \sum_{j=0}^{u_2} g_{mj} y_{k,j}, \quad \forall k; \quad \forall m \quad (\text{EC.11})$$

$$\sum_{j=0}^{u_2} y_{k,j} = 1, \quad \forall k \quad (\text{EC.12})$$

$$s_k \leq u_1, \quad \forall k \quad (\text{EC.13})$$

$$\sum_{j=0}^{u_2} j y_{k,j} \leq U, \quad \forall k \quad (\text{EC.14})$$

$$U \leq u_2, \quad \forall k \quad (\text{EC.15})$$

$$s_k \geq q_k, \quad \forall k \quad (\text{EC.16})$$

$$x_{mk}, s_k, U : \text{Integer variable } (\geq 0), \quad \forall k; \quad \forall m \quad (\text{EC.17})$$

$$y_{k,j} \in \{0, 1\} \quad \forall k; \quad \forall j \quad (\text{EC.18})$$

It is important to discuss the relative weights of different engagement types within the same platform. While platforms are difficult to compare because of different demographics and their

Table EC.40 Computational Results for Facebook Shares and Comments (Regression Parameters from Table 2 Multiplied by 5000)

Unit effort cost c	20	30	40	50	60	70	80	90
Content	s_k f_k x_{1k} x_{2k}	s_k f_k x_{1k} x_{2k}	s_k f_k x_{1k} x_{2k}	s_k f_k x_{1k} x_{2k}	s_k f_k x_{1k} x_{2k}	s_k f_k x_{1k} x_{2k}	s_k f_k x_{1k} x_{2k}	s_k f_k x_{1k} x_{2k}
1	3 0 12800 16982	3 0 12800 16982	3 0 12800 16982	3 0 12800 16982	3 0 12800 16982	3 0 12800 16982	3 0 12800 16982	3 0 12800 16982
2	4 0 12550 17010	4 0 12550 17010	4 0 12550 17010	4 0 12550 17010	4 0 12550 17010	4 0 12550 17010	4 0 12550 17010	4 0 12550 17010
3	2 18 19850 23055	2 18 19850 23055	2 18 19850 23055	2 18 19850 23055	2 18 19850 23055	2 18 19850 23055	2 18 19850 23055	2 18 19850 23055
4	2 17 18650 21805	2 0 13050 16955	2 0 13050 16955	2 0 13050 16955	2 0 13050 16955	2 0 13050 16955	2 0 13050 16955	2 0 13050 16955
5	2 18 19850 23055	2 18 19850 23055	2 18 19850 23055	2 18 19850 23055	2 18 19850 23055	2 18 19850 23055	2 17 18650 21805	2 18 19850 23055
6	1 0 13300 16927	1 0 13300 16927	1 0 13300 16927	1 0 13300 16927	1 0 13300 16927	1 0 13300 16927	1 0 13300 16927	1 0 13300 16927
7	2 0 13050 16955	2 0 13050 16955	2 0 13050 16955	2 0 13050 16955	2 0 13050 16955	2 0 13050 16955	2 0 13050 16955	2 0 13050 16955
8	2 0 13050 16955	2 0 13050 16955	2 0 13050 16955	2 0 13050 16955	2 0 13050 16955	2 0 13050 16955	2 0 13050 16955	2 0 13050 16955
9	4 0 12550 17010	4 0 12550 17010	4 0 12550 17010	4 0 12550 17010	4 0 12550 17010	4 0 12550 17010	4 0 12550 17010	4 0 12550 17010
10	6 18 18850 23165	6 18 18850 23165	6 18 18850 23165	6 18 18850 23165	6 18 18850 23165	6 18 18850 23165	6 18 18850 23165	6 18 18850 23165
11	1 18 20100 23027	1 0 13300 16927	1 17 18900 21777	1 0 13300 16927	1 0 13300 16927	1 0 13300 16927	1 0 13300 16927	1 0 13300 16927
12	2 18 19850 23055	2 18 19850 23055	2 18 19850 23055	2 18 19850 23055	2 17 18650 21805	2 18 19850 23055	2 0 13050 16955	2 0 13050 16955
13	2 0 13050 16955	2 18 19850 23055	2 0 13050 16955	2 0 13050 16955	2 0 13050 16955	2 0 13050 16955	2 0 13050 16955	2 0 13050 16955
14	2 0 13050 16955	2 0 13050 16955	2 0 13050 16955	2 0 13050 16955	2 0 13050 16955	2 0 13050 16955	2 0 13050 16955	2 0 13050 16955
Leftover budget	0	160	0	160	0	160	0	160
Engagement (weighted)	319035	311510	309809	302284	300340	291740	289796	281196
U	18	18	18	18	18	18	18	18

Table EC.41 Relative Weights w_k for Facebook (Based on the Insights from Lang (2025))

w_1	w_2	w_3	w_4	w_5	w_6	w_7
10	12	14	14	16	16	10

Table EC.42 Computational Results for Facebook Shares and Comments with Different Weights (Regression Parameters from Table 2 Multiplied by 5000)

Unit effort cost c	20	30	40	50	60	70	80	90
Content	s_k f_k x_{1k} x_{2k}	s_k f_k x_{1k} x_{2k}	s_k f_k x_{1k} x_{2k}	s_k f_k x_{1k} x_{2k}	s_k f_k x_{1k} x_{2k}	s_k f_k x_{1k} x_{2k}	s_k f_k x_{1k} x_{2k}	s_k f_k x_{1k} x_{2k}
1	3 0 12800 16982	3 0 12800 16982	3 0 12800 16982	3 0 12800 16982	3 0 12800 16982	3 0 12800 16982	3 0 12800 16982	3 0 12800 16982
2	4 0 12550 17010	4 0 12550 17010	4 0 12550 17010	4 0 12550 17010	4 0 12550 17010	4 0 12550 17010	4 0 12550 17010	4 0 12550 17010
3	2 17 18650 21805	2 0 13050 16955	2 0 13050 16955	2 0 13050 16955	2 0 13050 16955	2 0 13050 16955	2 0 13050 16955	2 0 13050 16955
4	2 0 13050 16955	2 0 13050 16955	2 0 13050 16955	2 0 13050 16955	2 0 13050 16955	2 0 13050 16955	2 0 13050 16955	2 0 13050 16955
5	2 18 19850 23055	2 18 19850 23055	2 18 19850 23055	2 18 19850 23055	2 18 19850 23055	2 18 19850 23055	2 18 19850 23055	2 18 19850 23055
6	1 18 20100 23027	1 18 20100 23027	1 18 20100 23027	1 18 20100 23027	1 18 20100 23027	1 0 13300 16927	1 0 13300 16927	1 0 13300 16927
7	2 0 13050 16955	2 0 13050 16955	2 0 13050 16955	2 0 13050 16955	2 0 13050 16955	2 0 13050 16955	2 0 13050 16955	2 0 13050 16955
8	2 0 13050 16955	2 0 13050 16955	2 0 13050 16955	2 0 13050 16955	2 0 13050 16955	2 0 13050 16955	2 0 13050 16955	2 0 13050 16955
9	4 0 12550 17010	4 0 12550 17010	4 0 12550 17010	4 0 12550 17010	4 0 12550 17010	4 0 12550 17010	4 0 12550 17010	4 0 12550 17010
10	6 0 12050 17065	6 0 12050 17065	6 0 12050 17065	6 0 12050 17065	6 0 12050 17065	6 0 12050 17065	6 0 12050 17065	6 0 12050 17065
11	1 18 20100 23027	1 18 20100 23027	1 17 18900 21777	1 0 13300 16927	1 0 13300 16927	1 0 13300 16927	1 0 13300 16927	1 0 13300 16927
12	2 18 19850 23055	2 18 19850 23055	2 18 19850 23055	2 18 19850 23055	2 18 19850 23055	2 18 19850 23055	2 18 19850 23055	2 18 19850 23055
13	2 18 19850 23055	2 18 19850 23055	2 18 19850 23055	2 18 19850 23055	2 17 18650 21805	2 18 19850 23055	2 17 18650 21805	2 0 13050 16955
14	2 0 13050 16955	2 0 13050 16955	2 0 13050 16955	2 0 13050 16955	2 0 13050 16955	2 0 13050 16955	2 0 13050 16955	2 0 13050 16955
Leftover budget	0	160	0	160	0	160	0	160
Engagement (weighted)	319201	311676	309975	302450	300506	291906	289962	281362
U	18	18	18	18	18	18	18	18

own unique characteristics, engagement types can be broadly categorized based on the level of user involvement. As highlighted in Aichner and Jacob (2015), different engagement types have different impact factors. For example, when a user shares a post, it can also be seen by the people in that user's network, resulting in increased exposure for the firm. From a user's perspective, sharing a post indicates a higher level of involvement than liking a post since likes are more casual or passive (Shahbaznezhad et al. 2021). Because of the increased exposure to a user's network, sharing is also treated higher than commenting, as the latter, while indicating higher involvement than liking, does not help in higher exposure. Kim and Yang (2017) comment that a share's value is approximately twice as much as a comment. Thus, within Facebook, we assign weights of 0.7 and 0.3 to shares and comments, respectively. The parameter z_{lm} can be changed accordingly. We also present the results for Facebook without considering these weights in Table EC.43. This analysis assumes any engagement types across platforms to be identical in terms of their importance. The overall pattern of allocating features to posts is similar to our main insights. The firm can set the

Table EC.43 Total Engagement on Facebook (Shares + Comments) without Engagement Weights

e	Total engagement	Secondary features on posts (days)
20	665249	[0, 0, 18, 17, 18, 0, 0, 0, 0, 18, 0, 18, 18, 0]
30	650619	[0, 0, 18, 0, 18, 18, 0, 0, 0, 18, 0, 18, 18, 0]
40	647189	[0, 0, 18, 0, 18, 0, 0, 0, 0, 18, 17, 18, 0, 0]
50	632559	[0, 0, 18, 0, 18, 0, 0, 0, 0, 18, 0, 18, 0, 0]
60	628639	[0, 0, 18, 0, 18, 0, 0, 0, 0, 18, 0, 17, 0, 0]
70	611919	[0, 0, 0, 0, 18, 0, 0, 0, 0, 18, 0, 18, 0, 0]
80	607999	[0, 0, 18, 0, 17, 0, 0, 0, 0, 18, 0, 0, 0, 0]
90	591279	[0, 0, 18, 0, 0, 0, 0, 0, 0, 18, 0, 0, 0, 0]

appropriate weights depending on the impact of each engagement type in its business context. The next subsection presents the generalized problem with all platforms and engagement types.

Problem $LSMM_g$:

$$\mathbf{Max} \Pi_g^l = \sum_{l=1}^L \sum_{m=1}^M \sum_{k=1}^K z_{lm} w_{lk} x_{lmk} \quad (\text{EC.19})$$

Subject to:

$$a \sum_{k=1}^K s_{lk} + a \sum_{l=1}^L \sum_{k=1}^K \sum_{j=0}^{J^l} j y_{l,k,j} + \sum_{l=1}^L e U_l + L C_f \leq B \quad (\text{EC.20})$$

$$x_{lmk} = \beta_{lm} s_{lk} + \sum_{l=1}^L \sum_{j=0}^{J^l} g_{lmj} y_{l,k,j}, \quad \forall k; \quad \forall m, \quad \forall l \quad (\text{EC.21})$$

$$\sum_{j=0}^{J^l} y_{l,k,j} = 1, \quad \forall k, \quad \forall l \quad (\text{EC.22})$$

$$s_{lk} \leq u_l, \quad \forall k, \quad \forall l \quad (\text{EC.23})$$

$$\sum_{j=0}^{J^l} j y_{l,k,j} \leq U_l, \quad \forall; \quad \forall l \quad (\text{EC.24})$$

$$U_l \leq J^l, \quad \forall l \quad (\text{EC.25})$$

$$s_{lk} \geq q_{lk}, \quad \forall k, \quad \forall l \quad (\text{EC.26})$$

$$x_{lmk}, s_{lk}, U_l: \text{ Integer variable } (\geq 0), \quad \forall k; \quad \forall l; \quad \forall m \quad (\text{EC.27})$$

$$y_{l,k,j} \in \{0, 1\} \quad \forall k; \quad \forall j; \quad \forall l \quad (\text{EC.28})$$

The generalized formulation $LSMM_g$ is a multi-platform extension of $LSMM_s$ (Cases2 – 3), where each engagement type is denoted by both l and m . Similarly, the other decision variables, s_{lk} , $y_{l,k,j}$ and U_l are represented with their corresponding platforms. $y_{l,k,j}$ represents the inclusion of each secondary feature j in post k on platform l . The linearized version introduces a change in the formulation by removing the nonlinear constraint (18) and replacing it with constraint sets (EC.21),

Table EC.44 Platform and Engagement Weights

Coefficients	Instagram ($l = 1$)	Facebook ($l = 2$)	
	Likes ($m = 1$)	Shares ($m = 2$)	Comments ($m = 3$)
Platform engagement weight	$\frac{v_1}{V}$	$\frac{v_2^2}{V}$	$\frac{v_2^2}{V}$
Engagement weight	$\mu_{11} = 1.0$	$\mu_{22} = 0.7$	$\mu_{23} = 0.3$
(Engagement+ Platform) weight, z_{lm}	$z_{1,1} = \mu_{11}(\frac{v_1}{V})$	$z_{2,2} = \mu_{22}(\frac{v_2^2}{V})$	$z_{2,3} = \mu_{23}(\frac{v_2^2}{V})$

Table EC.45 Instagram and Facebook Results (Instagram User Base (Scaled) = 8000)

a=20				a=30			
e	Instagram (Likes)	Facebook (Shares & Comments)	Total	e	Instagram (Likes)	Facebook (Shares & Comments)	Total
20	314134	211869	526003	20	304232	197224	501456
30	312734	211869	524603	30	306863	192831	499694
40	311218	211869	523087	40	305441	192831	498271
50	309636	211869	521505	50	303947	192831	496778
60	307787	211869	519656	60	306365	188438	494803
70	305832	211869	517700	70	305227	188438	493665
80	307361	208208	515568	80	303663	188438	492100
90	305654	208208	513861	90	303094	186625	489719
a=40				a=50			
e	Instagram (Likes)	Facebook (Shares & Comments)	Total	e	Instagram (Likes)	Facebook (Shares & Comments)	Total
20	306365	178186	484551	20	305867	167935	473803
30	305085	178186	483271	30	304801	167935	472736
40	303947	178186	482134	40	303663	167935	471598
50	303094	177241	480335	50	303094	167935	471029
60	305867	173061	478928	60	306365	162078	468443
70	304729	173061	477790	70	305441	162078	467518
80	303663	173061	476724	80	304481	162078	466558
90	303094	172116	475210	90	303378	162078	465456

Table EC.46 Instagram and Facebook Results (Instagram User Base (Scaled) = 9000)

a=20				a=30			
e	Instagram (Likes)	Facebook (Shares & Comments)	Total	e	Instagram (Likes)	Facebook (Shares & Comments)	Total
20	334801	200718	535518	20	328681	182682	511362
30	333308	200718	534026	30	327051	182682	509733
40	331712	200718	532430	40	325535	182682	508217
50	330007	200718	530725	50	328264	178520	506784
60	328036	200718	528754	60	326521	178520	505040
70	329628	197249	526877	70	325308	178520	503828
80	327582	197249	524831	80	323641	178520	502160
90	325763	197249	523012	90	326331	173664	499995
a=40				a=50			
e	Instagram (Likes)	Facebook (Shares & Comments)	Total	e	Instagram (Likes)	Facebook (Shares & Comments)	Total
20	326521	168808	495329	20	325990	159097	485087
30	325156	168808	493965	30	324853	159097	483950
40	323944	168808	492752	40	323641	159097	482737
50	327278	163952	491231	50	323034	159097	482131
60	325990	163952	489942	60	326521	153547	480068
70	324777	163952	488730	70	325535	153547	479082
80	323641	163952	487593	80	324512	153547	478059
90	323034	163057	486091	90	323337	153547	476885

(EC.22), and (EC.24). The parameter g_{lmj} represents the engagement impact of each secondary feature for different platforms and engagement types, as we discussed earlier. Note that the variables associated with platforms are only denoted by l , not both l and m . Posts with different features can not be created for different engagement types within the same platform.

Table EC.47 Instagram and Facebook Results (Instagram User Base (Scaled) = 10000)

a=20				a=30			
e	Instagram (Likes)	Facebook (Shares & Comments)	Total	e	Instagram (Likes)	Facebook (Shares & Comments)	Total
20	353401	190682	544082	20	346941	173548	520488
30	351826	190682	542507	30	345221	173548	518768
40	350141	190682	540822	40	347941	169594	517534
50	348341	190682	539022	50	346501	169594	516094
60	346261	190682	536942	60	344661	169594	514254
70	347941	187387	535327	70	343381	169594	512974
80	345781	187387	533167	80	341621	169594	511214
90	347541	184092	531632	90	344461	164981	509441
a=40				a=50			
e	Instagram (Likes)	Facebook (Shares & Comments)	Total	e	Instagram (Likes)	Facebook (Shares & Comments)	Total
20	344661	160368	505028	20	344101	151142	495242
30	343221	160368	503588	30	342901	151142	494042
40	346741	155755	502495	40	341621	151142	492762
50	345501	155755	501255	50	340981	151142	492122
60	344101	155755	499855	60	344661	145870	490530
70	342821	155755	498575	70	343621	145870	489490
80	346261	151142	497402	80	342541	145870	488410
90	344941	151142	496082	90	341301	145870	487170

Table EC.48 Instagram and Facebook Results (Instagram User Base (Scaled) = 11000)

a=20				a=30			
e	Instagram (Likes)	Facebook (Shares & Comments)	Total	e	Instagram (Likes)	Facebook (Shares & Comments)	Total
20	370229	181602	551831	20	367233	161518	528751
30	368579	181602	550181	30	365766	161518	527284
40	366814	181602	548416	40	364509	161518	526027
50	364928	181602	546530	50	363001	161518	524519
60	366175	178464	544639	60	361073	161518	522591
70	364509	178464	542973	70	359732	161518	521250
80	365557	175326	540882	80	362498	157125	519622
90	364090	175326	539416	90	360863	157125	517988
a=40				a=50			
e	Instagram (Likes)	Facebook (Shares & Comments)	Total	e	Instagram (Likes)	Facebook (Shares & Comments)	Total
20	361073	152731	513804	20	360486	143945	504431
30	364300	148338	512638	30	359229	143945	503174
40	363252	148338	511590	40	363252	138924	502176
50	361911	148338	510249	50	362246	138924	501170
60	360486	148338	508824	60	361073	138924	499997
70	363881	143945	507825	70	359983	138924	498907
80	362749	143945	506694	80	358852	138924	497776
90	361366	143945	505311	90	373194	123861	497055

Table EC.49 Instagram and Facebook Results (Instagram User Base (Scaled) = 12000)

a=20				a=30			
e	Instagram (Likes)	Facebook (Shares & Comments)	Total	e	Instagram (Likes)	Facebook (Shares & Comments)	Total
20	385528	173347	558875	20	382408	154176	536584
30	383810	173347	557157	30	380881	154176	535057
40	384982	170352	555334	40	379572	154176	533748
50	383041	170352	553392	50	378001	154176	532177
60	384328	167356	551684	60	380226	149983	530209
70	382692	167356	550048	70	379135	149983	529118
80	380662	167356	548019	80	377477	149983	527460
90	382168	164361	546529	90	380008	145789	525797
a=40				a=50			
e	Instagram (Likes)	Facebook (Shares & Comments)	Total	e	Instagram (Likes)	Facebook (Shares & Comments)	Total
220	380444	141595	522040	20	375382	137402	512784
30	379353	141595	520949	30	379135	132609	511744
40	378262	141595	519858	40	378262	132609	510871
50	376910	141595	518505	50	377215	132609	509824
60	380008	137402	517410	60	375993	132609	508602
70	378917	137402	516319	70	374859	132609	507468
80	377739	137402	515140	80	388713	118231	506944
90	376255	137402	513657	90	388615	118231	506846

Proof of Theorem 2: Due to Lemma 1, since $\beta_{lm} < 0$, in Problem SMM_s , in order to have maximum user engagement, f_k^* , we must have $s_k = q_k, \forall k$. Thus, Algorithm FAAS(Case 1) fixes $s_k = q_k, \forall k$ and then it determines the budget, $B' = B - a \sum_{k=1}^K s_k - eU - C_f$ available for assigning secondary features for the days of the planning horizon. As the objective function is to maximize the weighted engagement, $\Pi_s = \sum_{k=1}^K w_k x_k$, it prioritizes assignment according to the weights of the days by arranging the weights in the nonincreasing order, $w_{i_1} \geq w_{i_2} \geq \dots \geq w_{i_K}$. Since U is a decision variable in Problem $LSMM_s$ (Case 1) (which is equivalent to Problem SMM_s as stated in Theorem 1) and $U \leq f_k^*$, the algorithm finds the budget allocation for each value of $U \leq f_k^*$ by assigning the budget according to the weights of the days. It finds the best objective function value among all U values enumerated. Thus, the algorithm finds the optimal feature allocation in order of time $O(K f_{max}^*)$ and the corresponding optimal value of U . This completes the proof. Q.E.D.

Proof of Theorem 3: Facebook “share” is an example of Case 2 engagement type. Note that Problem $LSMM_f$ under Case 2, where $x_k = b + \beta s_k + \zeta f_k + \eta f_k^2$, where b and η are positive and β and ζ are negative. Lemmas 5 and 6 show that x_k is a convex of f_k for a fixed value of s_k . As $\beta_{lm} < 0$ for Case 2, we must have $s_k = q_k, \forall k$. Thus, FAAS(Case 2) fixes $s_k = q_k, \forall k$ and then it determines the budget, $B' = B - a \sum_{k=1}^K s_k - eU - C_f$ available for assigning secondary features for the days of the planning horizon. Since $\zeta < 0$ and $\eta > 0$, we must have $f_k \geq \lceil \frac{|\zeta|}{\eta} \rceil = f'_k$ for a positive contribution of f_k to x_k . As x_k is a convex of f_k , $f_k \leq f_k^{max} = u_2$. Therefore, $f'_k \leq f_k \leq u_2$. As the objective function is to maximize the weighted engagement, $\Pi_s = \sum_{k=1}^K w_k x_k$, it prioritizes assignment according to the weights of the days by arranging the weights in the nonincreasing order, $w_{i_1} \geq w_{i_2} \geq \dots \geq w_{i_K}$. Since U is a decision variable in Problem $LSMM_f$ (which is fundamentally similar to Problem SMM_s with an additional engagement type) and $U \leq u_2$, the algorithm finds the budget allocation for each value of $U \leq u_2$ by assigning the budget according to the weights of the days. It finds the highest objective function value among all U values enumerated. Thus, the algorithm finds the efficient feature allocation in order of time $O(K f_{max})$ and the corresponding best value of U . Q.E.D.

Note that Problem $LSMM_f$ under Case 3, which in our case is Facebook comments, where $x_k = b + \beta s_k + \zeta f_k + \eta f_k^2$, where b, β and η are positive and ζ is negative. It is shown that x_k is a convex function of f_k for a fixed value of s_k . Algorithm FAAS(Case 3) determines budget $B' = B - a \sum_{k=1}^K s_k - eU - C_f$ that is available for assigning secondary features for the days of the planning horizon. Since $\zeta < 0$ and $\eta > 0$, we must have $f_k \geq \lceil \frac{|\zeta|}{\eta} \rceil = f'_k$ for a positive contribution of f_k to x_k . As x_k is a convex of f_k , $f_k \leq u_2$. Therefore, $f'_k \leq f_k \leq u_2$.

Table EC.50 FAAS(Case 1) Results (Instagram Example)

U	Engagement (Instagram)
8	714161 (Optimal)
7	714161
6	713241
5	710481
4	706801
3	702201
2	696681
1	689321
0	681961 (Minimum)

Table EC.51 FAAS(Cases 2-3) Results (Facebook Example)

U	Engagement (Facebook)
18	302284
17	294508
16	287372
15	280748
14	274892
13	277808
12	269408
11	261788
10	255788
9	250568
8	240236
7	235912
6	233704
5	233704
4	235175
3	238488
2	243640
1	251276
0	260108

Table EC.52 Parameter Setting to Compare with User Base-Focused Strategy

Parameters	Changing values
Budget (B)	9000, 10000, 11000, 12000
Weights (w_k)	Fixed and changed
Firm's user Base ratio (Instagram/Facebook)	1, 1.2, 1.4
Unit cost e	20, 30, 40, 50, 60, 70, 80, 90

Therefore, for Facebook “comment,” we have the following equation:

$$x_k = 3.3796 + 0.0055s_k - 0.1224f_k + 0.0106f_k^2, \text{ where } \beta = 0.0055 \text{ and } f_k \leq u_2 = 18.$$

We also find $f'_k = \lceil \frac{|\zeta|}{\eta} \rceil = \lceil \frac{0.1224}{0.0106} \rceil = 12$.

Note that $0.0055s_k < -0.1224f_k + 0.0106f_k^2$, $s_k \geq q_k$ and $f_k \geq f'_k$. Consequently, secondary features will be used before s_k is used for $f_k \geq f'_k$. Moreover, by taking derivative of x_k with respect to f_k , we have $\frac{d(x_k)}{d(f_k)} = -|\zeta| + 2\eta f_k$, i.e., $\frac{d(x_k)}{d(f_k)} = -|\zeta| + 2\eta f_k = -0.1224 + 2 \times 0.0106 f_k$. For $f_k \geq f'_k = 12$, $\frac{d(x_k)}{d(f_k)} \geq 0.132 > \beta = 0.0055$. This implies each secondary feature contributes more than each primary feature to user engagement when $f_k \geq f'_k = 12$. Thus, FAAS(Case 3) finds maximum for Case 3, Facebook “comment.”

Table EC.53 Comparison of Our Approach and User Base-Focused Approach: Reduction (in %) in Overall Engagement

User base ratio	Budget	Changing weights									Constant weights								
		20	30	40	50	60	70	80	90		20	30	40	50	60	70	80	90	
Instagram/Facebook: 1	9000	5.1694	5.0944	5.3170	5.2149	5.4800	5.3777	5.6460	5.5670		3.7947	3.6139	3.7485	3.6446	3.7284	3.6759	3.7084	3.7283	
	10000	5.4134	5.8837	5.6586	6.0904	5.7860	6.2591	5.9361	6.4305		4.2742	4.6947	4.4737	4.9439	4.5359	4.9590	4.5989	4.9494	
	11000	5.2528	5.2086	5.9792	5.7699	6.4049	6.1685	6.7395	6.4322		4.1153	4.0653	4.8157	4.6268	5.2229	4.9881	5.5860	5.2402	
	12000	3.7031	4.3937	4.5511	5.2528	5.4128	6.1264	5.8545	6.2056		2.5823	3.2654	3.4211	4.1153	4.2736	4.9795	5.0143	5.5023	
Instagram/Facebook: 1.2	9000	9.3778	9.0526	9.6081	9.2520	9.8457	9.4293	10.0089	9.6524		7.8531	7.3806	7.8848	7.3941	7.9171	7.4939	7.9502	7.6389	
	10000	9.5097	9.7686	9.7368	9.9547	9.9463	10.2285	10.1561	10.4715		8.3260	8.5327	8.5002	8.7589	8.5233	8.7987	8.5595	8.8387	
	11000	9.0574	9.8130	9.8078	10.3929	10.2275	10.7925	10.5456	11.0414		7.8785	8.6189	8.6010	9.1996	9.0007	9.5583	9.3507	9.7950	
	12000	8.1597	8.4755	9.0574	9.3782	9.9700	10.2472	10.7669	10.8505		6.9904	7.3028	7.8785	8.1958	8.7811	9.0484	9.5596	9.6350	
Instagram/Facebook: 1.4	9000	12.8257	12.5948	13.1237	12.8672	13.3787	13.0242	13.5606	13.3422		11.1555	10.8183	11.3043	10.9117	11.4092	11.1064	11.5163	11.3509	
	10000	13.0205	13.1391	13.3246	13.4000	13.6079	13.7642	13.8847	14.0733		11.8009	11.8522	11.9546	12.0602	12.0639	12.1862	12.1754	12.3147	
	11000	12.4327	13.3287	13.2069	13.9200	13.6233	14.3158	13.9301	14.6090		11.2172	12.0962	11.9616	12.6885	12.3569	13.0403	12.6984	13.2603	
	12000	11.4998	12.3224	12.4327	13.2692	13.3817	14.1790	14.2051	14.8007		10.2943	11.1080	11.2172	12.0446	12.1559	12.9372	12.9597	13.5412	

Table EC.54 Secondary Features and Engagement Levels on Facebook and Instagram

secondary features	Instagram (1)	Instagram (2)	Facebook (1)	Facebook (2)
0	681961	681523	260108	259886
1	689321	690043	251276	251367
2	696681	697347	243640	244606
3	702201	703433	238488	239603
4	706801	708302	235176	236357
5	710481	711953	233704	234868
6	713241	714388	233704	235137
7	714161	715606	235912	237163
8	714161	715606	240236	240947
9	-	-	250568	251148
10	-	-	255788	255908
11	-	-	261788	261814
12	-	-	269408	268867
13	-	-	277808	277065
14	-	-	274892	274032
15	-	-	280748	279627
16	-	-	287372	285833
17	-	-	294508	292651
18	-	-	302284	300080

EC.2.2. Four Platforms

Our generalized model maximizes the total weighted user engagement across multiple platforms under budget constraints. This subsection extends our analysis to four platforms - platforms 1, 2, 3, and 4. We use the data from Instagram and Facebook for platforms 1 and 2, respectively. We randomly generate data for platforms 3 and 4 that are similar to X (formerly Twitter) retweets and Pinterest likes. X and Pinterest are two other popular platforms that firms use to advertise their products. The parameters for platforms 3 and 4 are tabulated in Table EC.55. We enumerate the engagement levels for these platforms for each secondary feature and show them in Table EC.56. The numbers displayed in bold indicate the minimum and maximum engagement levels for platforms 3 and 4. To run our numerical analysis, we set the budget at \$18,000 and assume user bases of 8,000, 10,000, 9,000, and 10,000 for these four platforms, respectively. These values are scaled by

Table EC.55 Parameters for Platform 3 Retweets and Platform 4 Likes

	Platform 3 Retweets	Platform 4 Likes
b_{lm}	3.9600	4.4200
β_{lm}	-0.0398	-0.0460
ζ_{lm}	-0.0155	0.0750
η_{lm}	0.0050	-0.0078
u_{1l}	9	6
u_{2l}	12	9

Table EC.56 Enumerated Engagement Levels for Platform 3 Retweets and Platform 4 Likes

Secondary features	Platform 3 Retweets	Platform 4 Likes
0	3.9202	4.3740
1	3.9097	4.4412
2	3.9092	4.4928
3	3.9187	4.5288
4	3.9382	4.5492
5	3.9677	4.5540
6	4.0072	4.5432
7	4.0567	4.5168
8	4.1162	4.4748
9	4.1857	4.4172
10	4.2652	x
11	4.3547	x
12	4.4542	x

each platform's user base to make cross-platform comparisons comparable. However, the setting can be extended to test on any values for the user bases. We tabulate the results in Table EC.57. The numerical analysis of the four platforms exhibits some interesting patterns. We list them below.

- As e increases, overall engagement across all four platforms reduces, consistent with our results from the case of two platforms.
- Platforms 1 and 4 receive fewer secondary features on their posts during the planning horizon across all levels of e compared to platforms 2 and 3.
- Like engagement on platform 4 shows similar patterns as like on platform 1. Images on the posts for both of these platforms receive fewer secondary features as the budget tightens. Note that the model still prioritizes the combination of both user base and engagement levels. Thus, in spite of having more users, our model starts reducing secondary features on platform 4 posts first and then on platform 1 posts.
- Posts to be published on platform 2 continue to have all secondary features, whereas posts scheduled for platform 3 either receive the maximum or no secondary features. As can be seen, some of these features are used in posts for platform 1.

However, the linearized version $LSMM_G$ of the generalized problem SMM_G for both two and four platforms are still complex problems under the budget constraint. For this purpose, we use the structural insights generated from the single platforms and develop an algorithm for four platforms based on the single platform algorithms.

In this subsection, we develop a heuristic for multiple platforms. First, we focus on the user exposure or conversion rate. The advertising problem in social media shares some similarities with the traditional budget allocation models. However, unlike the traditional advertising model, a user's interest in a social media post can be observed through engagement types such as likes or comments. Most importantly, it is important for a firm to know what percentage of their users actually engage

with their posts. As highlighted in the traditional advertising literature (Fischer 2019), user exposure influences budget allocation significantly. In social media, user exposure can be constructed by using the ratio of engaged users and total users ($\theta_{eng,l}$) for each firm and platform. We discuss them below for four platforms using their log-transformed values. For example, users who frequently engage on the posts on platform 1 of a firm and the total number of users who follow the firm on their platform 1 account are 3.46 and 3.9, respectively.

- Platform 1: $\frac{\text{Users who liked posts}}{\text{TotalUsers}} = \frac{3.46}{3.9} = 0.88$
- Platform 2: $\frac{\text{Users who shared and commented on posts}}{\text{TotalUsers}} = \frac{1.32}{4} = 0.33$
- Platform 3: $\frac{\text{Users who retweeted posts}}{\text{TotalUsers}} = \frac{1.4}{3.95} = 0.35$
- Platform 4: $\frac{\text{Users who liked posts}}{\text{TotalUsers}} = \frac{2.8}{4} = 0.70$

Next, we consider the aspect of platform potential for each platform. A firm's reach to users increases as more users join and stay active on a platform. Thus, the firm's posts can be visible to more users during the planning horizon, raising that platform's value. We determine platform potential ($\theta_{plt,l}$) for each social media platform by observing the total monthly active users (in millions) of that platform (BackLinco 2025). We tabulate the log-transformed values in the third column of Table EC.58. Note that this value is applicable to all firms. Any firm can update all these values as it executes its process of developing social media posts for every planning horizon.

Finally, we introduce a third parameter called features required ($\theta_{features,l}$) to balance the budget based on features actually required for a platform. For example, we show that Instagram users do not prefer more than eight secondary features. On the other hand, Facebook users prefer all features. Thus, we use the ratio $\theta_{features,l} = \frac{f_{max}^*}{u_{max}}$ for platforms (such as Instagram) where users do not like many features. $\theta_{features,l}$ is 1 for platforms such as Facebook, where users prefer all features.

We combine the three ratios ($\theta_{features,l} * \theta_{plt,l} * \theta_{eng,l}$) accordingly for each platform and construct a generalized θ_l for each platform. For the multiple platforms problem, θ_l is the proportion of the budget allocated for each platform l . Thus, we propose FAAM for more than one platform. Given the number of platforms, we divide the total budget using θ_l and run the FAAS algorithms (different cases) until all the platforms are done. For platforms with multiple engagement types, such as Facebook in our case, FAAS(Cases 2-3) is appropriate within FAAM. The algorithm is formally presented next. The complexity of the algorithm is $O(LK f_{max})$.

THEOREM EC.1. *Algorithm, FAAM solves Problem LSMM_g in polynomial time. The complexity of the algorithm is $O(LK f_{max})$.*

Table EC.58 Parameters for Budget Allocation

Platform	$\theta_{eng,l}$	$\theta_{plt,l}$	$\theta_{features,l}$	θ_l
1	0.88	3.16	0.57	1.58
2	0.33	3.46	1.00	1.14
3	0.35	2.59	1.00	0.90
4	0.70	2.67	0.98	0.98

Algorithm 3 FAAM

Input: Platforms $l, l = 1, 2, \dots, L$ and engagement types $m, m = 1, 2, \dots, M$;

Cost parameters: a, e, C_f ; Budget $B, \beta_{lm}, \zeta_{lm}, \eta_{lm}, q_{lk}, U_l$;

f_{lk}^{max} and $f_{lk}^*, g_{jlm}, j = 1, 2, \dots, f_{lk}^{max}$ or f_{lk}^* ; K days; weights, $w_{lk}, k = 1, 2, \dots, K$.

Begin

For each l , **do**

If($\zeta_{lm} > 0$ and $\eta_{lm} < 0$):

Execute FAAS(Case 1) and store the optimal engagement $Y_l = \Pi_s^{*,l}$

If($\zeta_{lm} < 0$ and $\eta_{lm} > 0$):

If($\beta_{lm} < 0$:

Execute FAAS(Case 2) and store the maximum engagement $Y_l = \Pi_s^{max,l}$

If($\beta_{lm} > 0$:

Execute FAAS(Case 3) and store the maximum engagement $Y_l = \Pi_s^{max,l}$

End(for l)

Sum the total engagements: $\Pi_g^{FAAM} = \sum_{l=1}^L Y_l$

End

Output: K posts for l platforms and the total engagement for the planning horizon, Π_g^{FAAM} .

Proof of Theorem EC.1: The algorithm considers l platforms ($L = 4$ in this case) for the allocation of budget B , which depends on θ_l , where $\theta_l = \theta_{features,l} * \theta_{plt,l} * \theta_{eng,l}$. Following the signs of ζ_{lm} and η_{lm} , FAAS(Case 1) ($\zeta_{lm} > 0$ and $\eta_{lm} < 0$) or FAAS(Cases 2-3) ($\zeta_{lm} < 0$ and $\eta_{lm} > 0$) is executed for each l . Since all the FAAS algorithms are solvable in polynomial time, Algorithm FAAM also solves Problem $LSMM_g$ in polynomial time. Thus, the algorithm finds the efficient budget allocation in order of time $O(LK f_{max})$ for L being the number of platforms. This completes the proof. Q.E.D.

The ordering of $LSSM_g$ and FAAM is presented in the following lemma. In the next section, we discuss FAAM's performance.

LEMMA EC.1. *The ordering of the objective values of $LSSM_g$ and FAAM is as follows: $\Pi_g^l \geq \Pi_g^{FAAM}$.*

Proof of Lemma EC.1: It is straightforward that in terms of engagement, $LSSM_g$ gives a tight upper bound to $FAAM$. Q.E.D.

EC.2.3.1. Performance of FAAM. To demonstrate the performance of FAAM, we generate an experimental testbed by varying two key parameters in FAAM that could potentially affect the quality of our solutions. The testbed is developed as follows. The number of runs within each test instance depends on the number of platforms and the number of secondary features for each platform. Our numerical analysis consists of 47 runs within each test scenario. To show the performance of our model for a wide variety of instances and generate managerially relevant insights, we increase ζ_{lm} by 5% and reduce η_{lm} by 5% to generate two scenarios for each platform. Therefore, we generate $2 \times 2 \times 2 \times 2 = 16$ combinations or scenarios. Thus, combining all parameter settings, we create a testbed of 752 runs to ensure that our testbed covers diverse scenarios. We use the same cost and budget parameters as we used in Section EC.2.2. The objective value for FAAM is Π_g^{FAAM} for each instance v , where $v = 1, 2, \dots, 16$. We show the engagement statistics in Table EC.59.

Table EC.59 Performance Analysis of FAAM against $LSSM_g$

Type	Min diff (%)	Max diff (%)	Avg diff (%)
Four platforms	5.4893	5.9036	5.6965

We compare the performance of FAAM against the model performance in Section EC.2.2 since the linear solution Π_g^l is equivalent to Π_g according to Theorem 1. The optimality gap is calculated using the following formula: $\text{Gap} = \frac{\Pi_g^l - \Pi_g^{FAAM,v}}{\Pi_g^l}$ for each v instance. The average optimality gap is the average over the 16 instances. The average gap from the scenarios is approximately 5.6965% from

Table EC.60 Computational Results of FAAM

secondary features	Platform 1 (1)	Platform 1 (2)	Platform 2 (1)	Platform 2 (2)	Platform 3 (1)	Platform 3 (2)	Platform 4 (1)	Platform 4 (2)
18	x	x	81617	81022	x	x	x	x
17	x	x	79517	79016	x	x	x	x
16	x	x	77590	77175	x	x	x	x
15	x	x	75802	75499	x	x	x	x
14	x	x	74221	73989	x	x	x	x
13	x	x	75008	74808	x	x	x	x
12	x	x	72740	72594	x	x	x	x
11	x	x	70683	70690	87689	87390	x	x
10	x	x	69063	69095	87001	86750	x	x
9	x	x	67653	67810	86391	86182	x	x
8	157115	157433	64864	65056	87174	86854	x	x
7	157115	157433	63696	64034	86317	86062	x	x
6	156913	157165	63100	63487	85604	85409	x	x
5	156306	156630	63100	63414	85400	85177	110252	110960
4	155496	155826	63498	63816	84749	84592	110133	110660
3	154484	154755	64392	64693	84318	84217	109626	109994
2	153270	153416	65783	66044	84109	84054	108732	108958
1	151651	151809	67845	67869	84120	84098	107450	107552
0	150031	149935	70229	70169	84352	84352	105781	105781

$LSSM_g$. The minimum and maximum gaps are approximately 5.4893% and 5.9036%, respectively. As can be seen from these numbers, the optimality gap is relatively smaller in all instances. Additionally, our results show that platforms 1 and 4 exhibit relatively higher engagement, as they receive more secondary features compared to $LSSM_g$ under a limited budget, demonstrating the effectiveness of our budget allocation algorithm in balancing all platforms. We also test the performance of the heuristic by comparing it to the solution generated by $LSSM_g$ when the latter is run for each individual platform with the allocated budget. The total engagement level obtained from the heuristic is 3.2561% lower than $LSSM_g$. It shows that the heuristic is reasonably efficient in guiding social media post development.

The heuristic is developed based on the estimable parameters $\theta_{features,l}$, $\theta_{plt,l}$, and $\theta_{eng,l}$ to allocate budget across different platforms. The combination of these three parameters, θ_l , is reasonable and relevant in our context. However, parameters such as platform potential and users are dynamic; they change over time. Thus, using this heuristic requires updating these parameters. Additionally, decision-makers may have additional parameters to allocate budget across different platforms, which may require further adjustment of θ_l .

EC.2.4. Optimization Framework without Duplicate Image features

In this subsection, we extend our framework by considering a case where a social media marketing team of a firm may not want to keep duplicate features on the developed images. This approach differs from the clustering and dimensionality reduction described in EC.1.4.10. Here, we have not employed any clustering or machine learning method to remove duplicates. We estimate the empirical models for Instagram (likes) and Facebook (shares and comments) without the duplicate features extracted by the Faster RCNN by manually removing them from both primary and secondary features. Table EC.61 presents the main empirical results for the two approaches. However, it is important to note that these two models, with and without duplicates, are structurally different. Further, compared to our original model, where all the quadratic terms were statistically significant, the quadratic term for Instagram is not statistically significant in the model without duplicates.

We develop the optimization model for the two platforms based on the empirical results, where duplicates were removed. Note that this utilization of empirical results in the optimization model can be used for the approaches in EC.1.4.10. We compare the engagement levels of models with and without duplicates and present them in Table EC.62. We present the results for various values of a , the development cost per feature and e , the extraction cost per feature. To make a reasonable comparison, we reduce the budget to \$8,000 and keep the scaled user base for both platforms at

Table EC.61 A Comparison of Models with and without Duplicates

Goodness of fit and variables	With duplicates			Without duplicates		
	Instagram (likes)	Facebook (shares)	Facebook (comments)	Instagram (likes)	Facebook (shares)	Facebook (comments)
Primary features	-0.0348 (0.1000)	-0.0500 (0.2130)	0.0055 (0.8550)	-0.0090 (0.8060)	-0.2132*** (0.0000)	-0.0509** (0.0440)
Secondary features	0.0945*** (0.0080)	-0.1012** (0.0200)	-0.1224*** (0.0000)	0.1533** (0.0250)	-0.2434*** (0.0030)	-0.4545*** (0.0000)
Secondary features squared term	-0.0063** (0.0460)	0.0098** (0.0100)	0.0106*** (0.0010)	-0.0116 (0.2870)	0.0182** (0.0470)	0.0448*** (0.0000)
Observations	542	2,191	2,191	542	2,191	2,191

p values in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$ **Table EC.62 Total Engagement Levels with and without Duplicates**

a=20			a=30		
e	Total engagement (without duplicates)	Total engagement (with duplicates)	e	Total engagement (without duplicates)	Total engagement (with duplicates)
20	498755	543322	20	498755	519328
30	498755	540822	30	498755	517294
40	498755	538327	40	498755	515094
50	498755	535742	50	498755	512681
60	498755	533117	60	498755	510201
70	498755	530537	70	498755	507828
80	498755	528042	80	498755	505588
90	498755	525202	90	498755	502975
a=40			a=50		
e	Total engagement (without duplicates)	Total engagement (with duplicates)	e	Total engagement (without duplicates)	Total engagement (with duplicates)
20	498755	503908	20	498605	494042
30	498755	502255	30	498170	492442
40	498755	500415	40	497495	490810
50	498755	498602	50	496875	489410
60	498755	496922	60	495875	487490
70	498755	494962	70	494775	485818
80	498275	492762	80	493730	484418
90	497495	490810	90	492320	482538

10,000. We observe that, for the majority of scenarios, the framework that considers duplicates exhibits a higher level of engagement. However, as a and e both increase, the framework without duplicates obtains better engagement in some cases since it does not require a higher cost for the features.

From the perspective of costs to firms, removing duplicate features may not reduce their spending in social media analytics, which is a significant part of a firm's social media operations cost. On the other hand, the firm can save a part of its budget if it chooses to remove the duplicate features in its advertising images. However, removing duplicates reduces engagement levels in most scenarios that we analyze, as shown in our optimization model results. It may seem a less appropriate option from a marketing point of view.

We also discuss the technical processes related to duplicate elimination. The deep learning algorithms (SSD and Faster RCNN) used in our study utilize both classification (or what the object is) and localization (where the object is). Both algorithms develop multiple bounding boxes to

predict an object or feature. Then, it uses a non-max suppression (NMS) method to delete duplicate bounding boxes for each object. Thus, for each object, there should be only one bounding box. The feature selection and projection are used by default in these algorithms and are a common practice in training (Wulf et al. 2019). Following the results obtained from the deep learning algorithms, we developed a Python code to eliminate the duplicate features extracted by Faster RCNN from each of the social media advertising images. We utilized this process to construct the variable secondary features for the analysis without duplicate features.

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See references list in the main paper. The additional references are listed below.

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